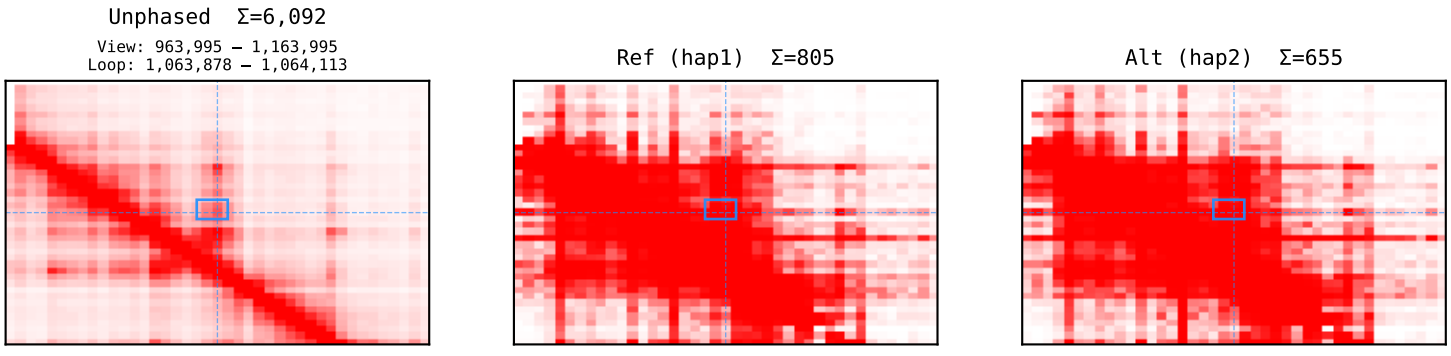
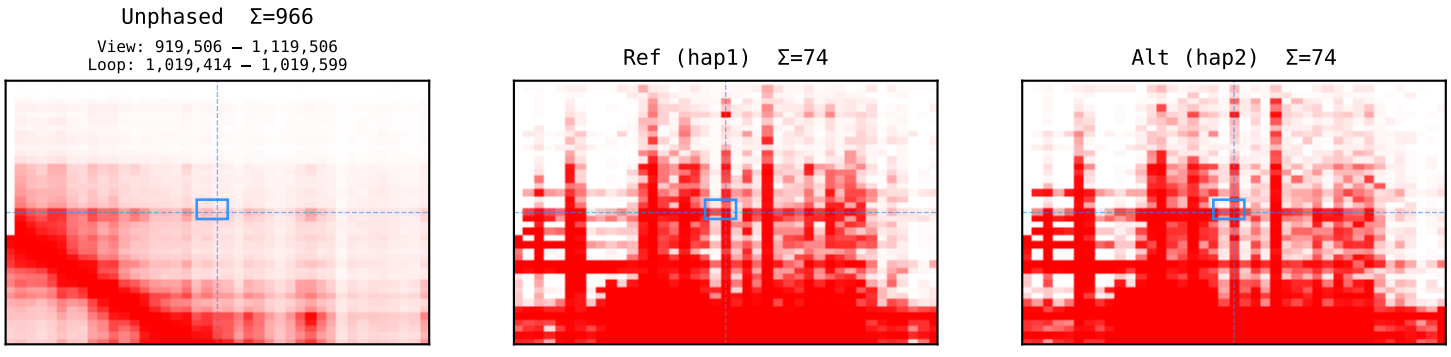
Anchor: chr1:1,019,276–1,019,776 | AlphaGenome pred: -0.0204 | empiric ChIP-seq: 1.17

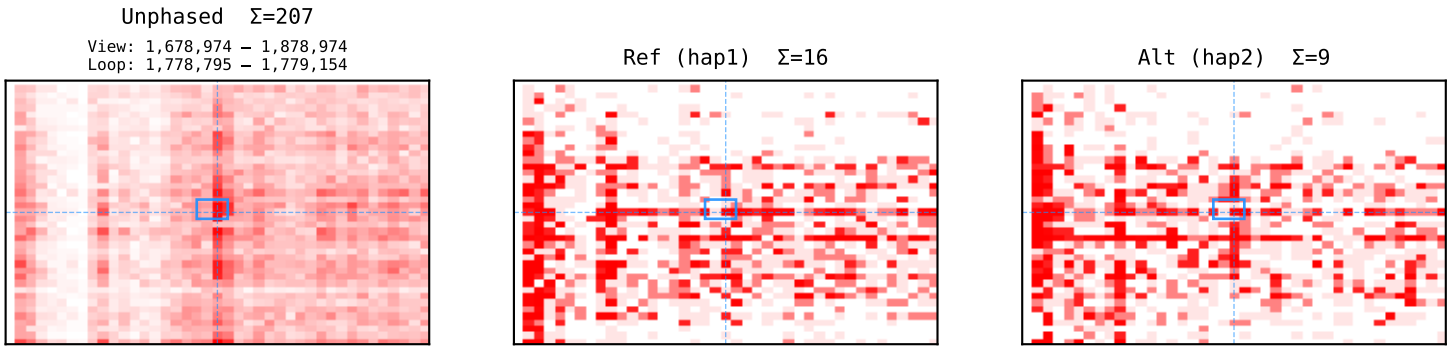
Loop 1
chr1:1,019,414-1,019,599
× chr1:1,063,878-1,064,113
dist: 44 kb
R/A: 1.23× log2FC: +0.30



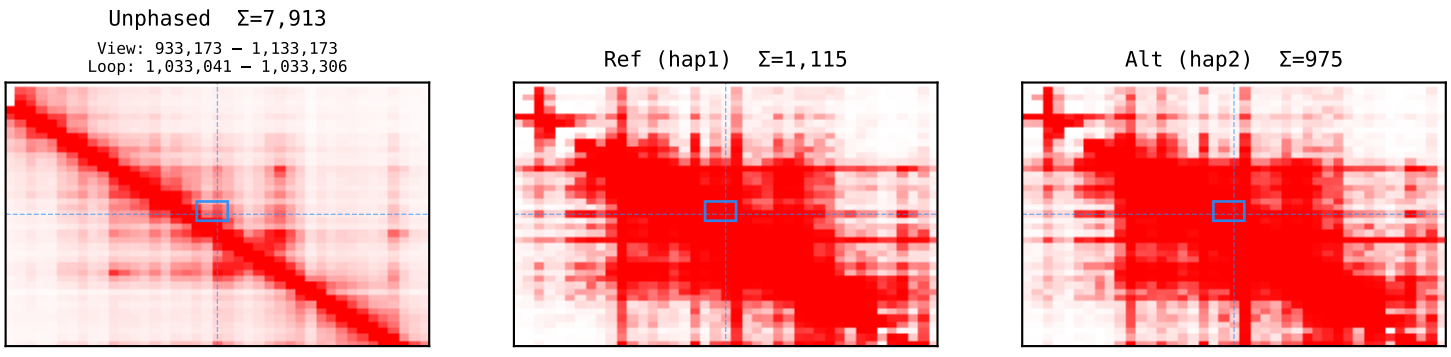
Loop 2
chr1:904,682-904,859
× chr1:1,019,414-1,019,599
dist: 115 kb
R/A: 1.00× log2FC: +0.00



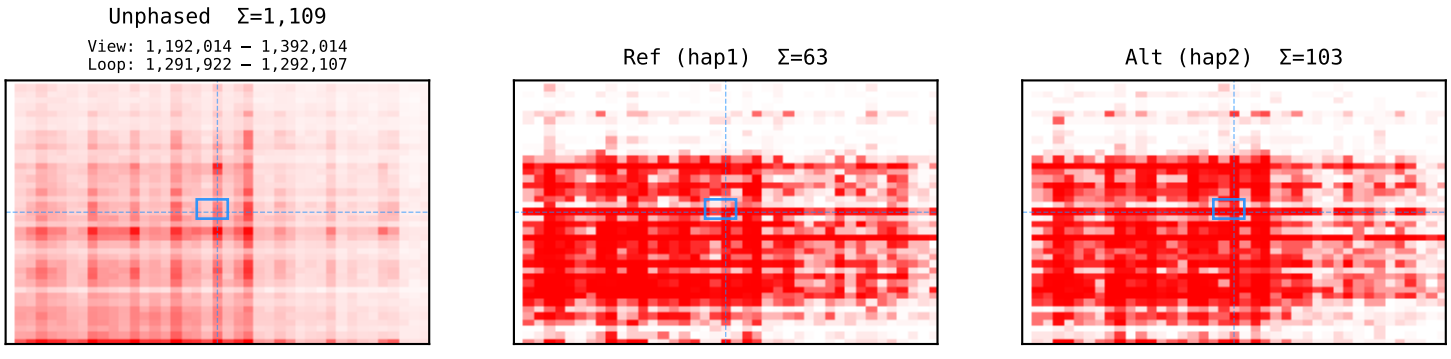
Loop 3
chr1:1,019,414-1,019,599
× chr1:1,778,795-1,779,154
dist: 759 kb
R/A: 1.78× log2FC: +0.83



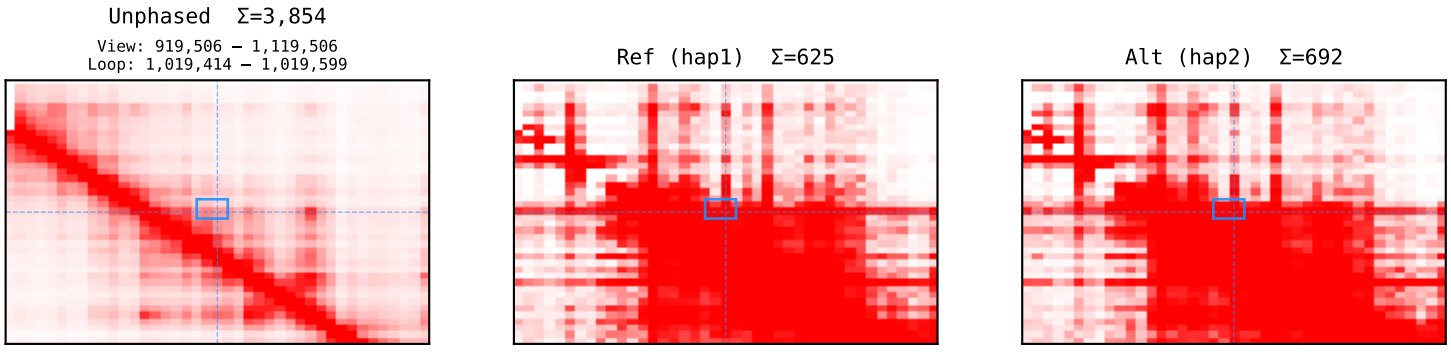
Loop 4
chr1:1,019,414-1,019,599
× chr1:1,033,041-1,033,306
dist: 14 kb
R/A: 1.14× log2FC: +0.19



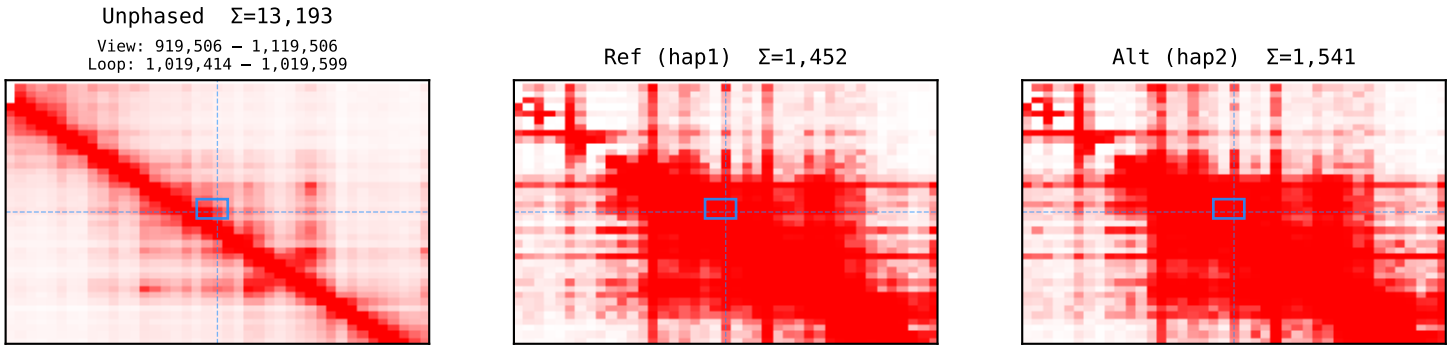
Loop 5
chr1:1,019,414-1,019,599
× chr1:1,291,922-1,292,107
dist: 273 kb
R/A: 0.61× log2FC: -0.71



Loop 6
chr1:984,170-984,457
× chr1:1,019,414-1,019,599
dist: 35 kb
R/A: 0.90× log2FC: -0.15



Loop 7
chr1:1,001,900-1,002,029
× chr1:1,019,414-1,019,599
dist: 18 kb
R/A: 0.94× log2FC: -0.09

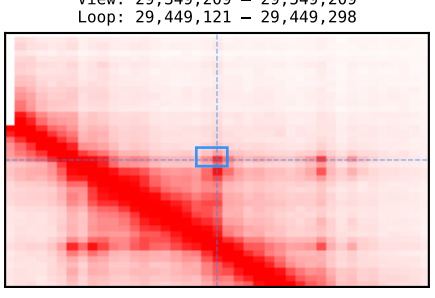


Anchor: chr10:29,449,156–29,449,250 | AlphaGenome pred: 0.00333 | empiric ChIP-seq: 4

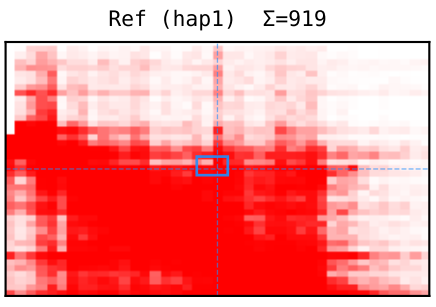
Unphased Σ=4,196

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

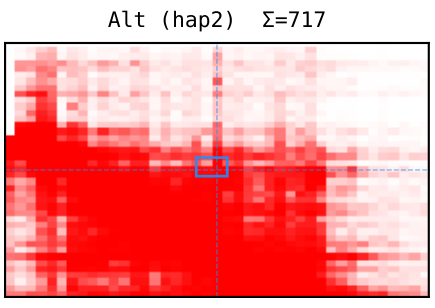
Loop 1
chr10:29,377,500-29,377,597
× chr10:29,449,121-29,449,298
dist: 72 kb
R/A: 1.28× log2FC: +0.36



Ref (hap1) Σ=919



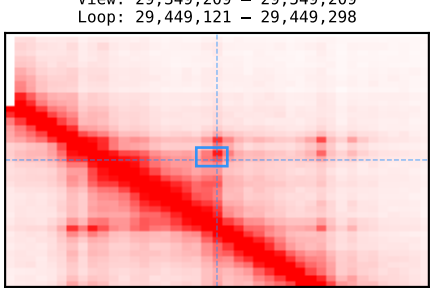
Alt (hap2) Σ=717



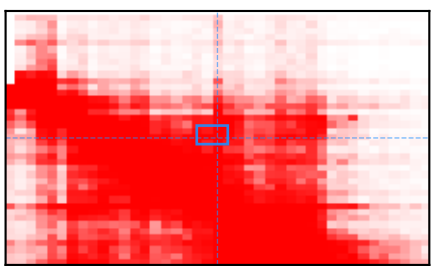
Unphased Σ=6,830

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

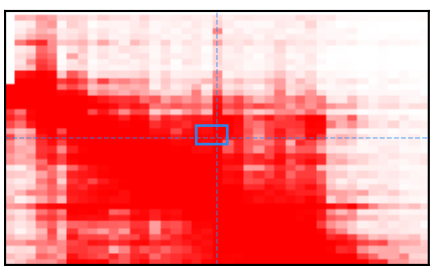
Loop 2
chr10:29,392,524-29,392,607
× chr10:29,449,121-29,449,298
dist: 57 kb
R/A: 1.36× log2FC: +0.44



Ref (hap1) Σ=1,437



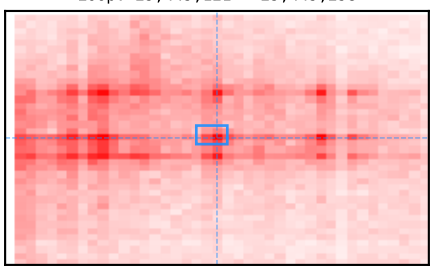
Alt (hap2) Σ=1,058



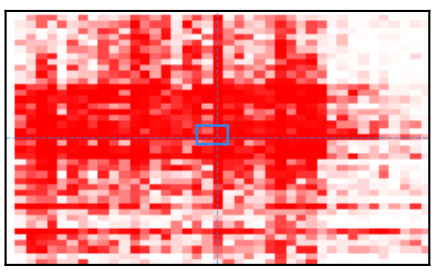
Unphased Σ=748

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

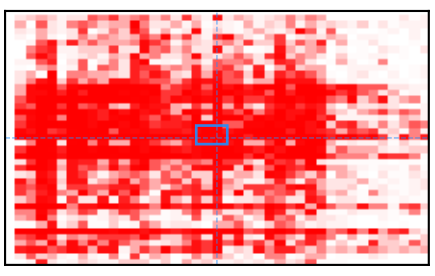
Loop 3
chr10:28,663,987-28,664,094
× chr10:29,449,121-29,449,298
dist: 785 kb
R/A: 0.89× log2FC: -0.17



Ref (hap1) Σ=163



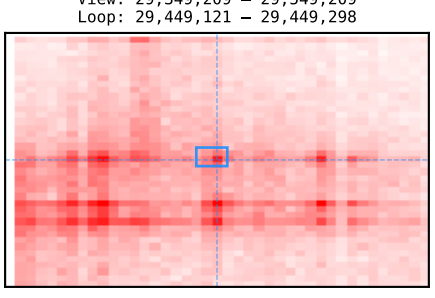
Alt (hap2) Σ=184



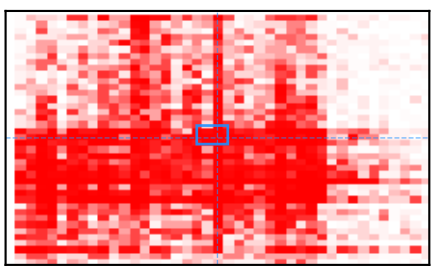
Unphased Σ=623

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

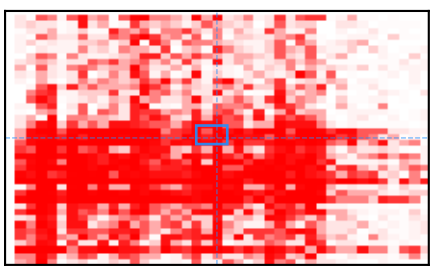
Loop 4
chr10:28,628,211-28,628,374
× chr10:29,449,121-29,449,298
dist: 821 kb
R/A: 1.28× log2FC: +0.35



Ref (hap1) Σ=148



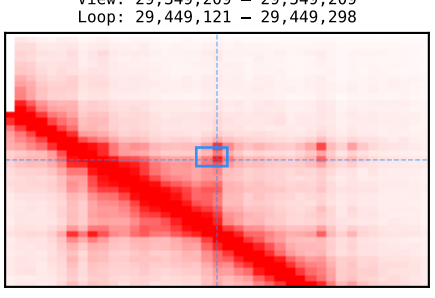
Alt (hap2) Σ=116



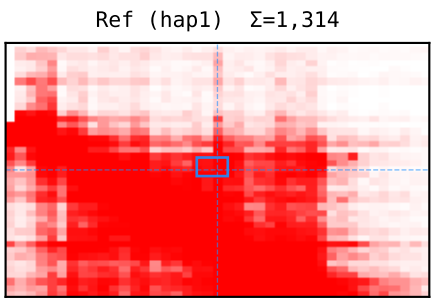
Unphased Σ=5,787

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

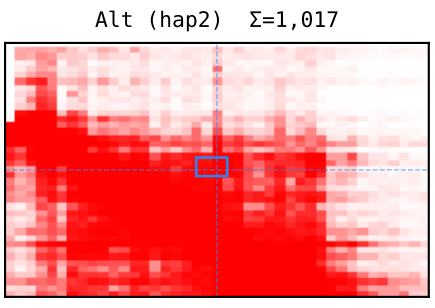
Loop 5
chr10:29,386,488-29,386,665
× chr10:29,449,121-29,449,298
dist: 63 kb
R/A: 1.29× log2FC: +0.37



Ref (hap1) Σ=1,314



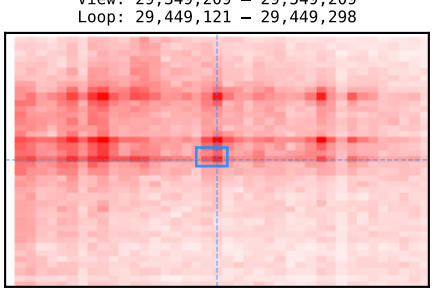
Alt (hap2) Σ=1,017



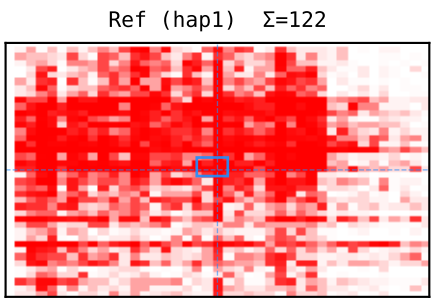
Unphased Σ=671

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

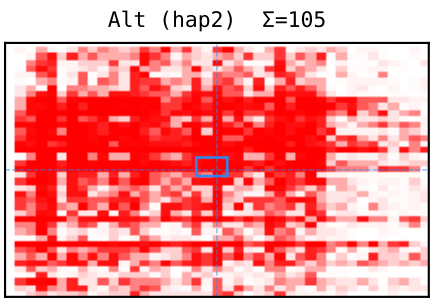
Loop 6
chr10:28,677,980-28,678,190
× chr10:29,449,121-29,449,298
dist: 771 kb
R/A: 1.16× log2FC: +0.22



Ref (hap1) Σ=122



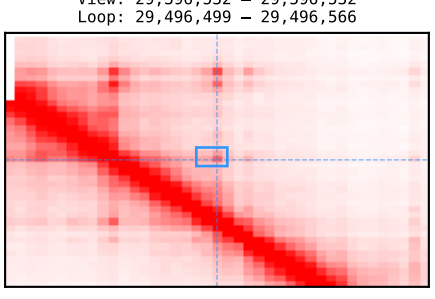
Alt (hap2) Σ=105



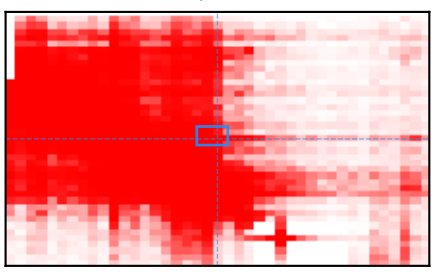
Unphased Σ=3,626

View: 29,396,532 – 29,596,532
Loop: 29,496,499 – 29,496,566

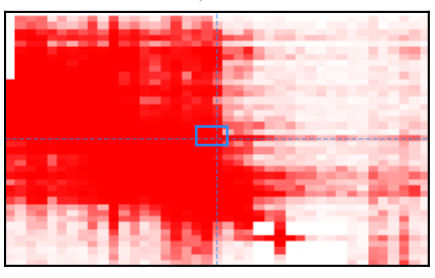
Loop 7
chr10:29,449,121-29,449,298
× chr10:29,496,499-29,496,566
dist: 47 kb
R/A: 1.00× log2FC: +0.00



Ref (hap1) Σ=797



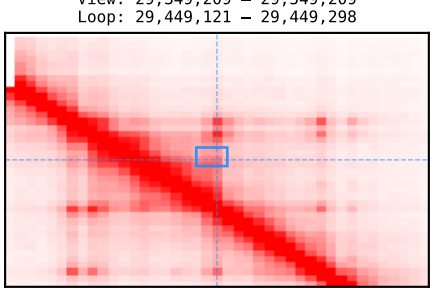
Alt (hap2) Σ=795



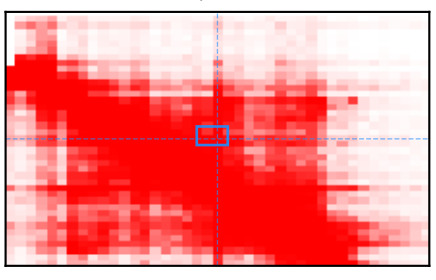
Unphased Σ=4,344

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

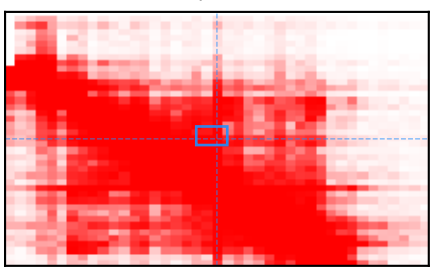
Loop 8
chr10:29,409,835-29,409,874
× chr10:29,449,121-29,449,298
dist: 39 kb
R/A: 1.28× log2FC: +0.36



Ref (hap1) Σ=907



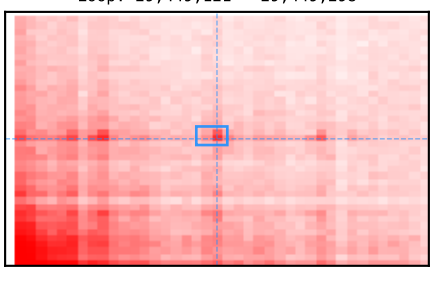
Alt (hap2) Σ=708



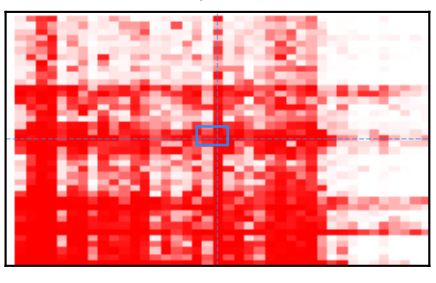
Unphased Σ=926

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

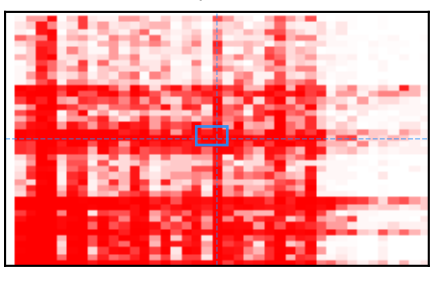
Loop 9
chr10:29,240,042-29,240,107
× chr10:29,449,121-29,449,298
dist: 209 kb
R/A: 0.93× log2FC: -0.11



Ref (hap1) Σ=167



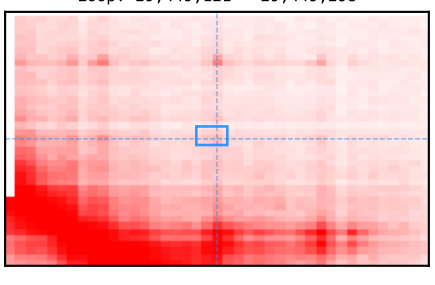
Alt (hap2) Σ=180



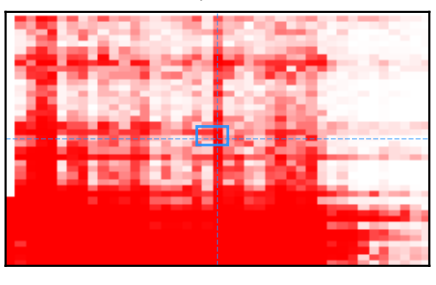
Unphased Σ=646

View: 29,349,209 – 29,549,209
Loop: 29,449,121 – 29,449,298

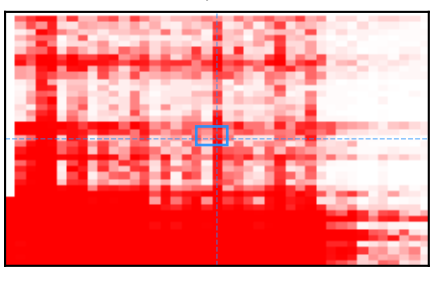
Loop 10
chr10:29,304,620-29,304,769
× chr10:29,449,121-29,449,298
dist: 145 kb
R/A: 1.62× log2FC: +0.69



Ref (hap1) Σ=123

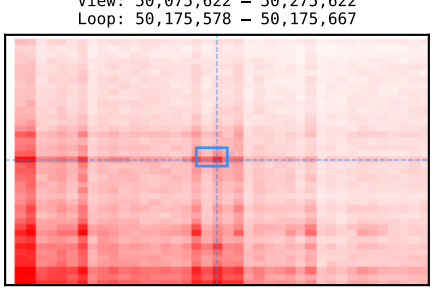


Alt (hap2) Σ=76

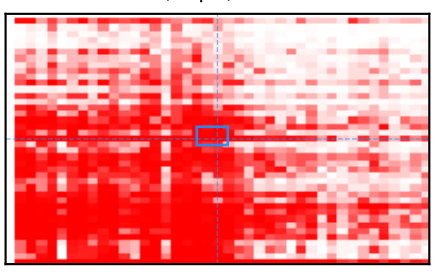


Unphased Σ=1,325
View: 50,075,622 – 50,275,622
Loop: 50,175,578 – 50,175,667

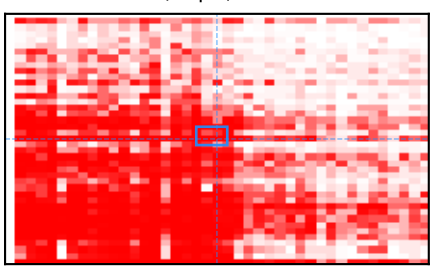
Loop 1
chr12:49,967,455-49,967,494
× chr12:50,175,578-50,175,667
dist: 208 kb
R/A: 1.28× log2FC: +0.36



Ref (hap1) Σ=159

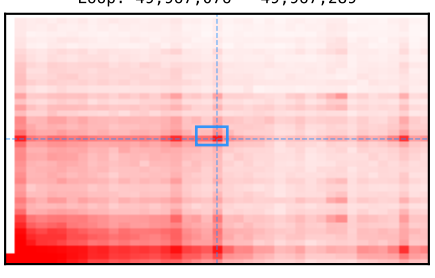


Alt (hap2) Σ=124

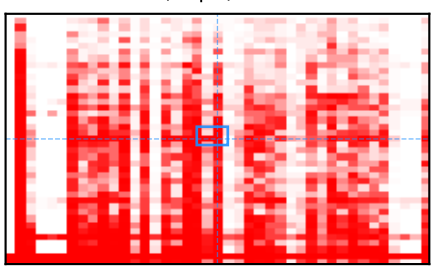


Unphased Σ=2,383
View: 49,867,182 – 50,067,182
Loop: 49,967,076 – 49,967,289

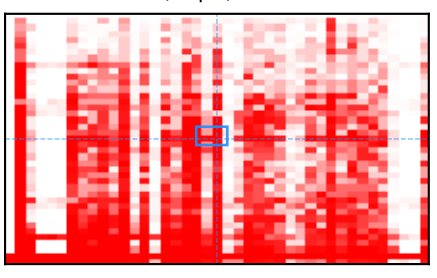
Loop 2
chr12:49,776,443-49,776,618
× chr12:49,967,076-49,967,289
dist: 191 kb
R/A: 0.80× log2FC: -0.32



Ref (hap1) Σ=130

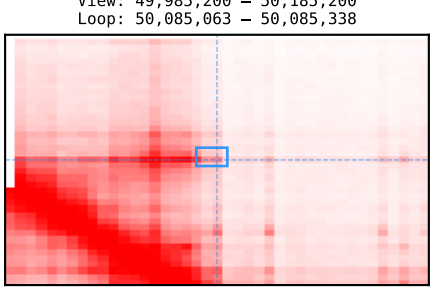


Alt (hap2) Σ=162

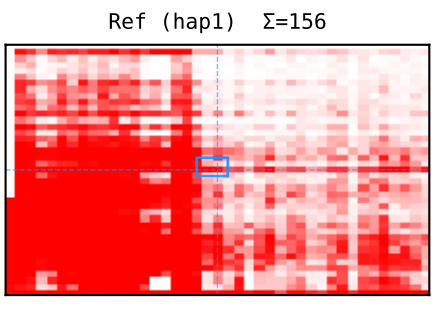


Unphased Σ=2,440
View: 49,985,200 – 50,185,200
Loop: 50,085,063 – 50,085,338

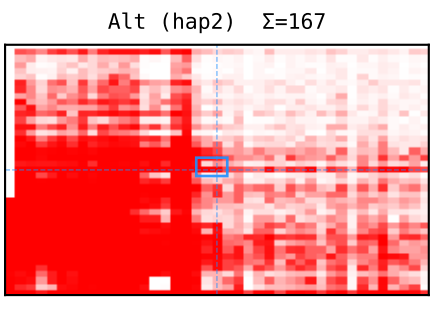
Loop 3
chr12:49,967,076-49,967,289
× chr12:50,085,063-50,085,338
dist: 118 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=156

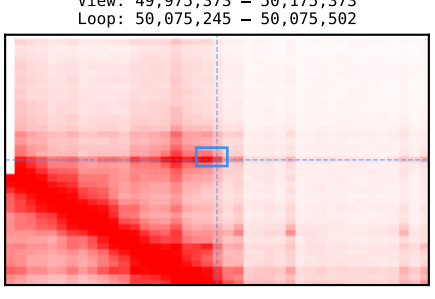


Alt (hap2) Σ=167

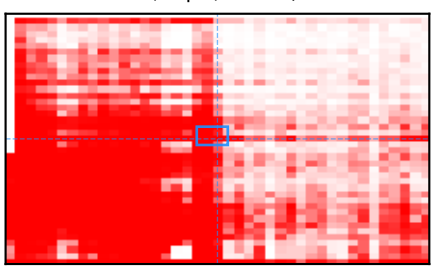


Unphased Σ=8,846
View: 49,975,373 – 50,175,373
Loop: 50,075,245 – 50,075,502

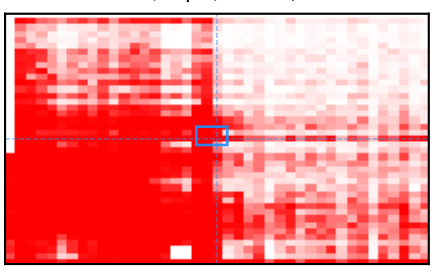
Loop 4
chr12:49,967,076-49,967,289
× chr12:50,075,245-50,075,502
dist: 108 kb
R/A: 0.91× log2FC: -0.14



Ref (hap1) Σ=1,185

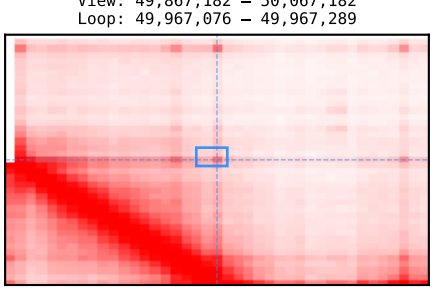


Alt (hap2) Σ=1,306

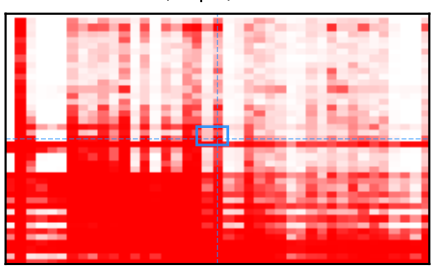


Unphased Σ=2,647
View: 49,867,182 – 50,067,182
Loop: 49,967,076 – 49,967,289

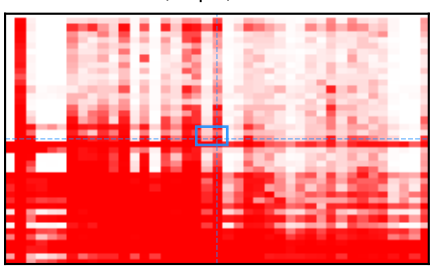
Loop 5
chr12:49,867,236-49,867,339
× chr12:49,967,076-49,967,289
dist: 100 kb
R/A: 0.94× log2FC: -0.09



Ref (hap1) Σ=134

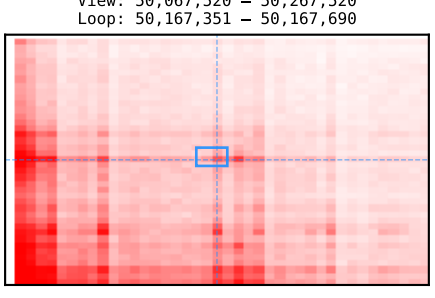


Alt (hap2) Σ=143

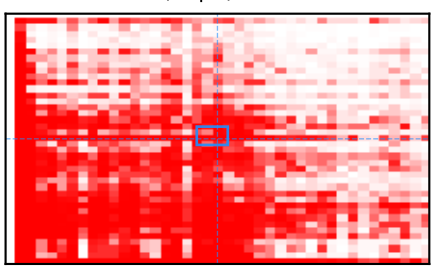


Unphased Σ=1,040
View: 50,067,520 – 50,267,520
Loop: 50,167,351 – 50,167,690

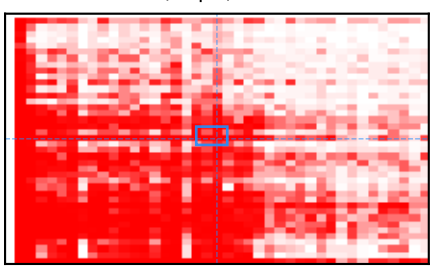
Loop 6
chr12:49,967,600-49,967,610
× chr12:50,167,351-50,167,690
dist: 200 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=109

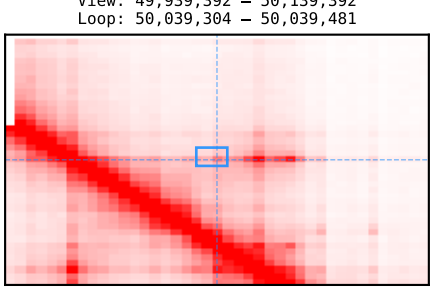


Alt (hap2) Σ=117

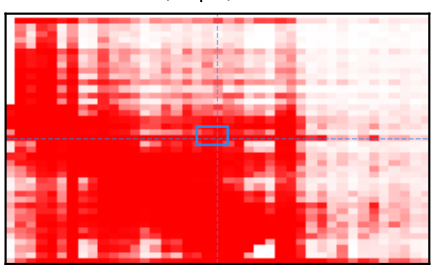


Unphased Σ=4,285
View: 49,939,392 – 50,139,392
Loop: 50,039,304 – 50,039,481

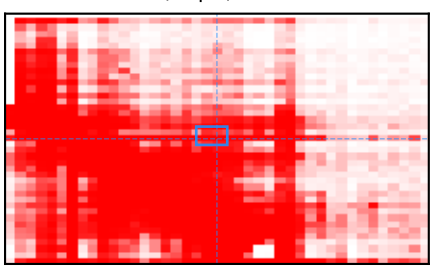
Loop 7
chr12:49,967,076-49,967,289
× chr12:50,039,304-50,039,481
dist: 72 kb
R/A: 0.85× log2FC: -0.23



Ref (hap1) Σ=472

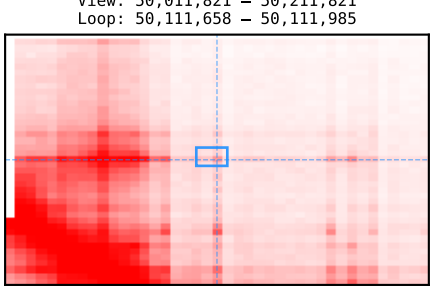


Alt (hap2) Σ=554

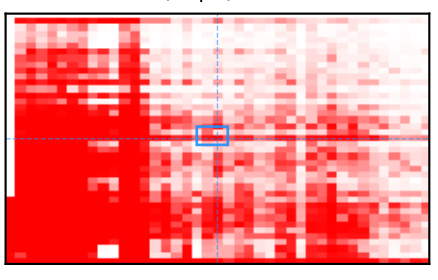


Unphased Σ=1,200
View: 50,011,821 – 50,211,821
Loop: 50,111,658 – 50,111,985

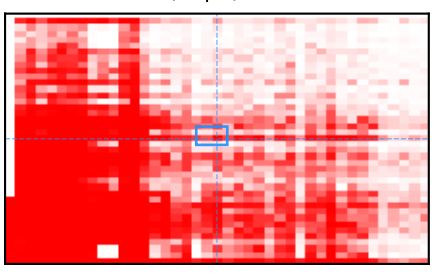
Loop 8
chr12:49,967,076-49,967,289
× chr12:50,111,658-50,111,985
dist: 145 kb
R/A: 1.29× log2FC: +0.37



Ref (hap1) Σ=115

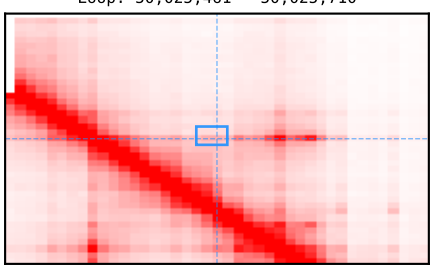


Alt (hap2) Σ=89

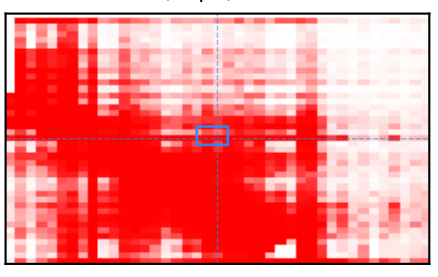


Unphased Σ=3,416
View: 49,925,585 – 50,125,585
Loop: 50,025,461 – 50,025,710

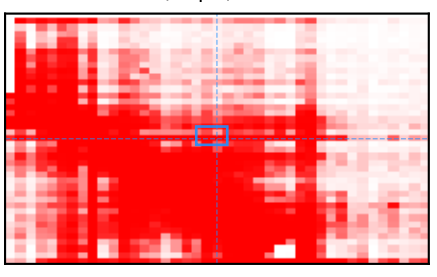
Loop 9
chr12:49,967,600-49,967,610
× chr12:50,025,461-50,025,710
dist: 58 kb
R/A: 0.92× log2FC: -0.12



Ref (hap1) Σ=387

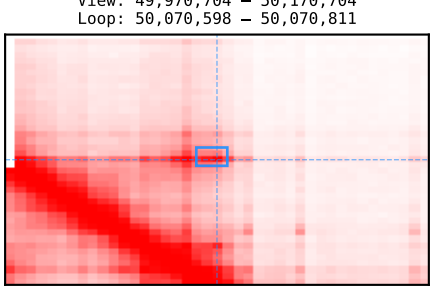


Alt (hap2) Σ=421

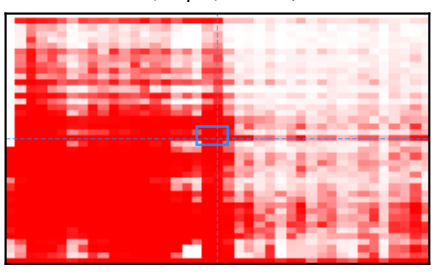


Unphased Σ=10,390
View: 49,970,704 – 50,170,704
Loop: 50,070,598 – 50,070,811

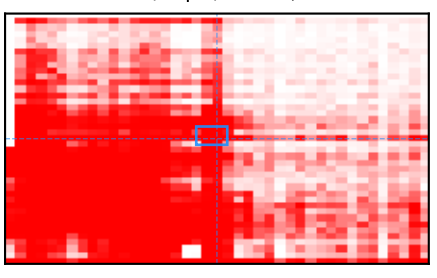
Loop 10
chr12:49,967,076-49,967,289
× chr12:50,070,598-50,070,811
dist: 104 kb
R/A: 0.94× log2FC: -0.10



Ref (hap1) Σ=1,454

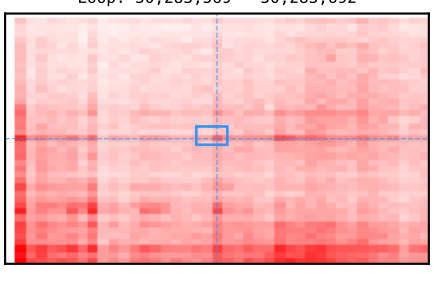


Alt (hap2) Σ=1,555

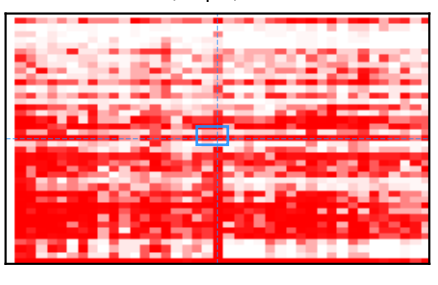


Unphased Σ=473
View: 50,183,630 – 50,383,630
Loop: 50,283,569 – 50,283,692

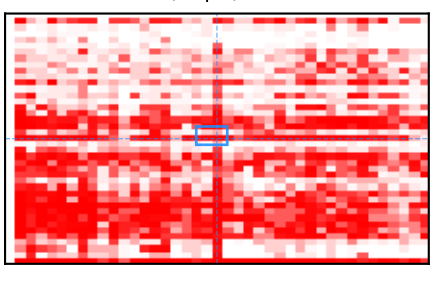
Loop 11
chr12:49,967,076-49,967,289
× chr12:50,283,569-50,283,692
dist: 316 kb
R/A: 1.20× log2FC: +0.26



Ref (hap1) Σ=48

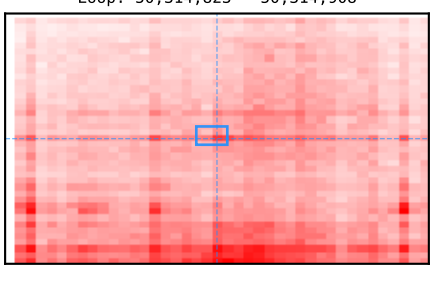


Alt (hap2) Σ=40

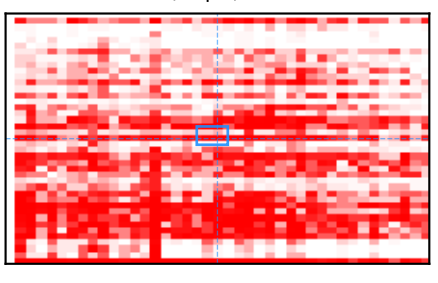


Unphased Σ=604
View: 50,214,865 – 50,414,865
Loop: 50,314,823 – 50,314,908

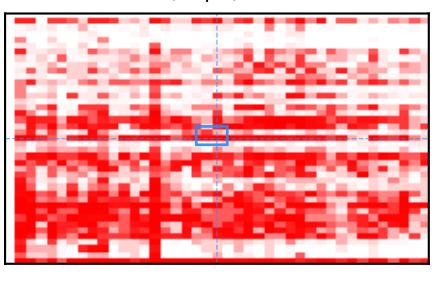
Loop 12
chr12:49,967,076-49,967,289
× chr12:50,314,823-50,314,908
dist: 348 kb
R/A: 1.10× log2FC: +0.14



Ref (hap1) Σ=44

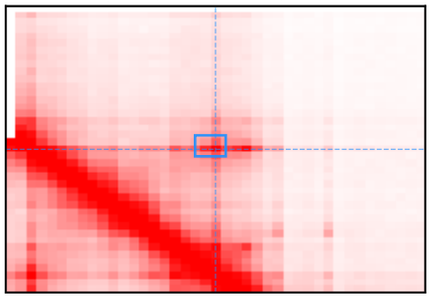


Alt (hap2) Σ=40

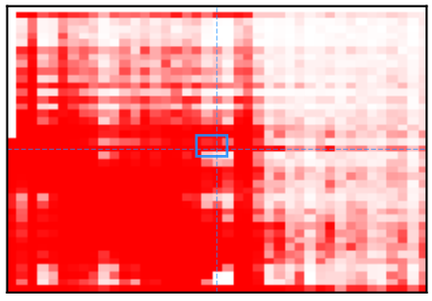


Loop 1
chr12:49,967,076-49,967,289
× chr12:50,056,117-50,056,274
dist: 89 kb
R/A: 0.85× log2FC: -0.23

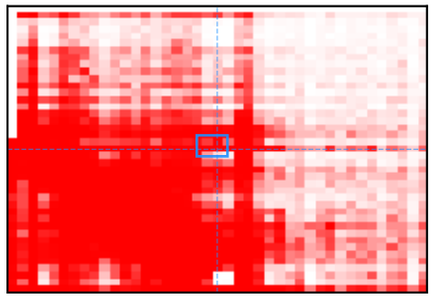
Unphased Σ=14,088
View: 49,956,195 – 50,156,195
Loop: 50,056,117 – 50,056,274



Ref (hap1) Σ=825

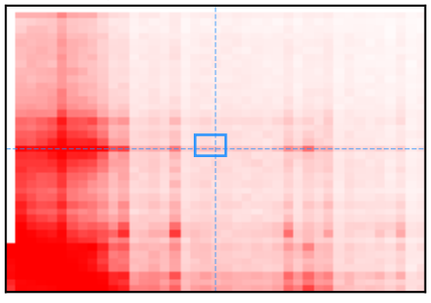


Alt (hap2) Σ=970

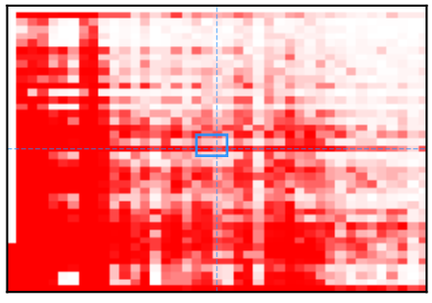


Loop 2
chr12:49,967,076-49,967,289
× chr12:50,132,687-50,132,752
dist: 166 kb
R/A: 0.58× log2FC: -0.78

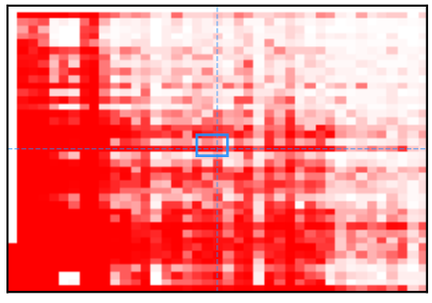
Unphased Σ=788
View: 50,032,719 – 50,232,719
Loop: 50,132,687 – 50,132,752



Ref (hap1) Σ=61

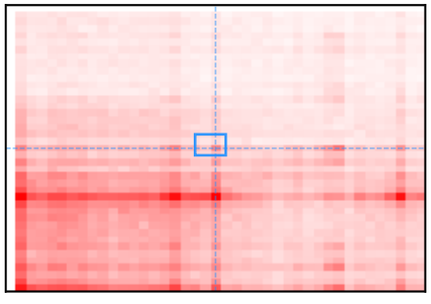


Alt (hap2) Σ=105

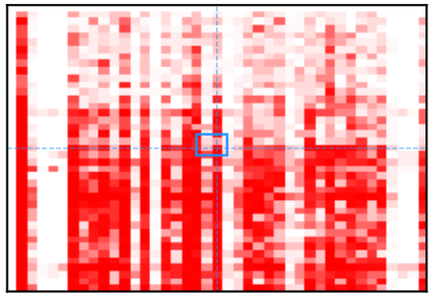


Loop 3
chr12:49,741,258-49,741,573
× chr12:49,967,076-49,967,289
dist: 226 kb
R/A: 1.15× log2FC: +0.20

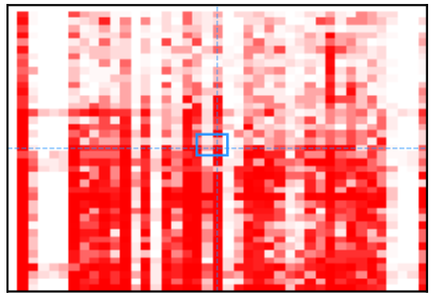
Unphased Σ=666
View: 49,867,182 – 50,067,182
Loop: 49,967,076 – 49,967,289



Ref (hap1) Σ=38

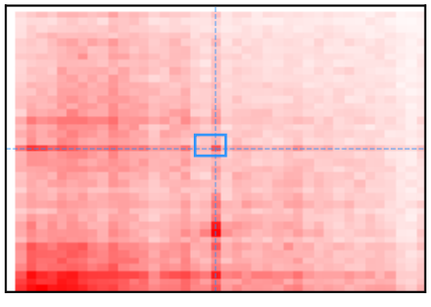


Alt (hap2) Σ=33

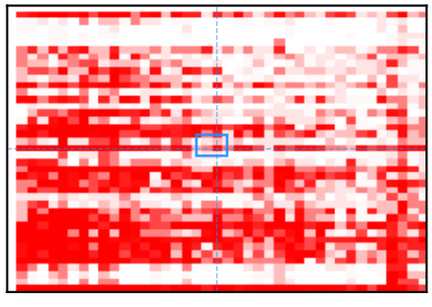


Loop 4
chr12:49,967,076-49,967,289
× chr12:50,400,697-50,400,996
dist: 434 kb
R/A: 0.69× log2FC: -0.53

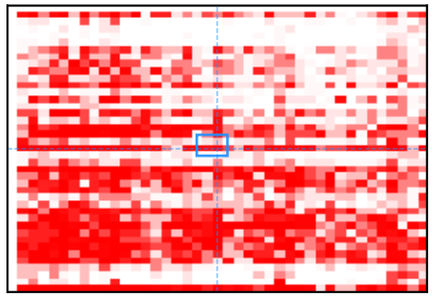
Unphased Σ=440
View: 50,300,846 – 50,500,846
Loop: 50,400,697 – 50,400,996



Ref (hap1) Σ=18



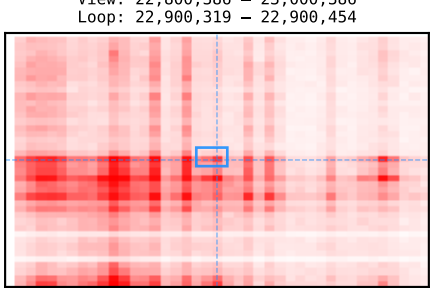
Alt (hap2) Σ=26



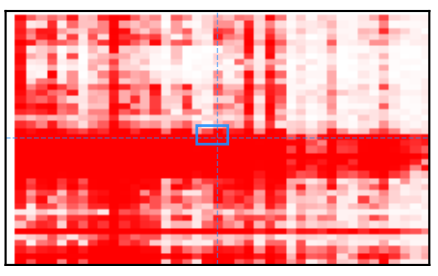
Anchor: chr14:22,556,999–22,557,483 | AlphaGenome pred: -0.0172 | empiric ChIP-seq: 1.4

Unphased Σ=1,627

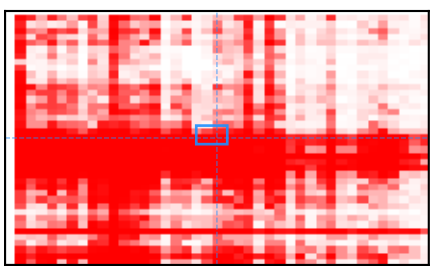
View: 22,800,386 – 23,000,386
Loop: 22,900,319 – 22,900,454



Ref (hap1) Σ=397



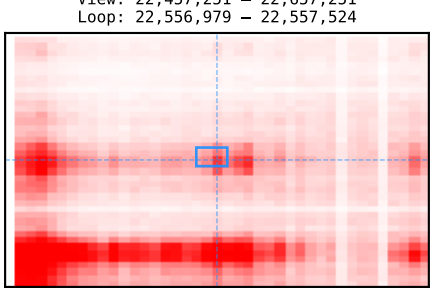
Alt (hap2) Σ=336



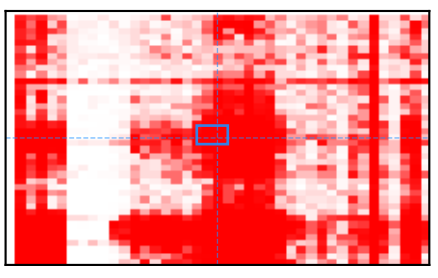
Loop 1
chr14:22,556,979-22,557,524
× chr14:22,900,319-22,900,454
dist: 343 kb
R/A: 1.18× log2FC: +0.24

Unphased Σ=2,118

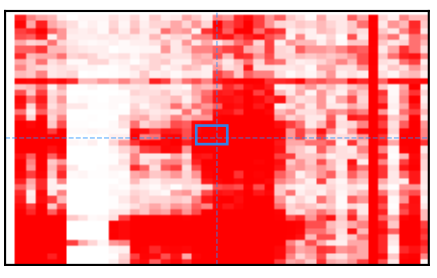
View: 22,457,251 – 22,657,251
Loop: 22,556,979 – 22,557,524



Ref (hap1) Σ=424



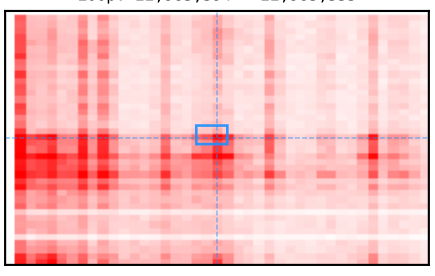
Alt (hap2) Σ=362



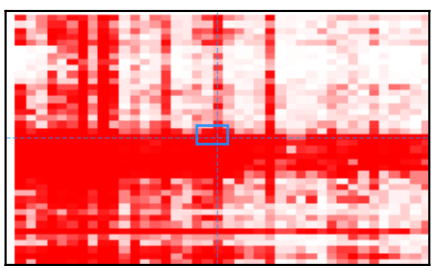
Loop 2
chr14:22,314,322-22,314,477
× chr14:22,556,979-22,557,524
dist: 243 kb
R/A: 1.17× log2FC: +0.23

Unphased Σ=1,566

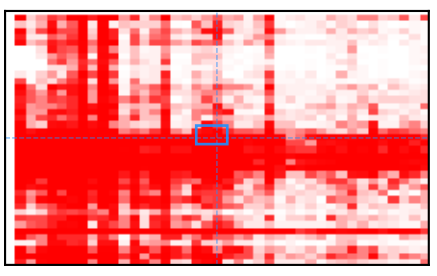
View: 22,883,474 – 23,083,474
Loop: 22,983,394 – 22,983,555



Ref (hap1) Σ=371



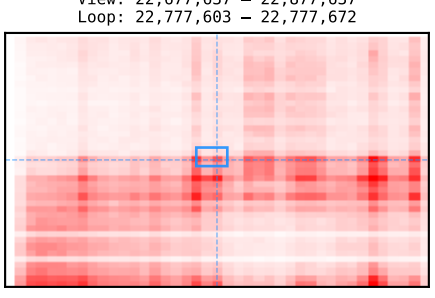
Alt (hap2) Σ=341



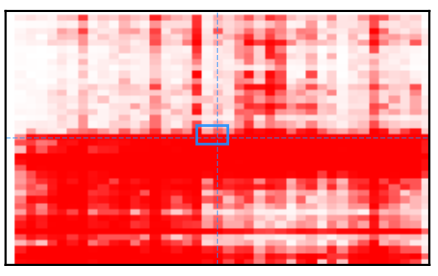
Loop 3
chr14:22,556,979-22,557,524
× chr14:22,983,394-22,983,555
dist: 426 kb
R/A: 1.09× log2FC: +0.12

Unphased Σ=1,841

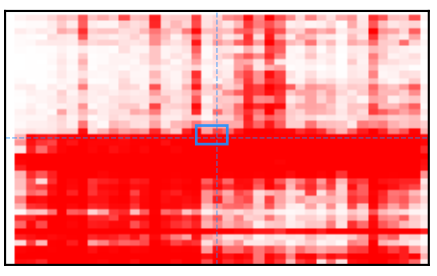
View: 22,677,637 – 22,877,637
Loop: 22,777,603 – 22,777,672



Ref (hap1) Σ=438



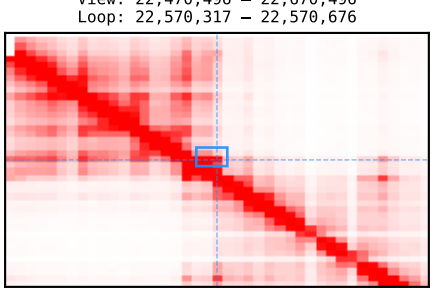
Alt (hap2) Σ=361



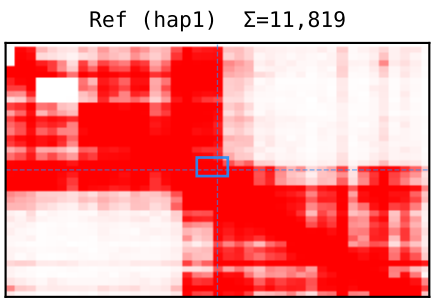
Loop 4
chr14:22,556,979-22,557,524
× chr14:22,777,603-22,777,672
dist: 220 kb
R/A: 1.21× log2FC: +0.28

Unphased Σ=37,363

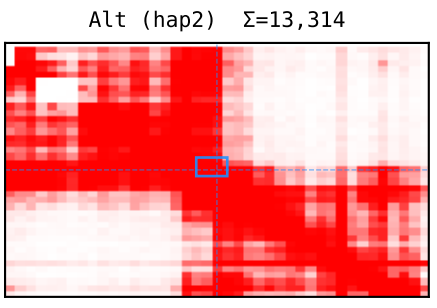
View: 22,470,496 – 22,670,496
Loop: 22,570,317 – 22,570,676



Ref (hap1) Σ=11,819



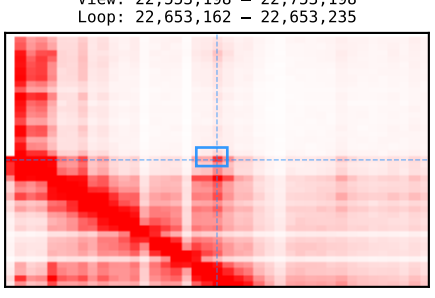
Alt (hap2) Σ=13,314



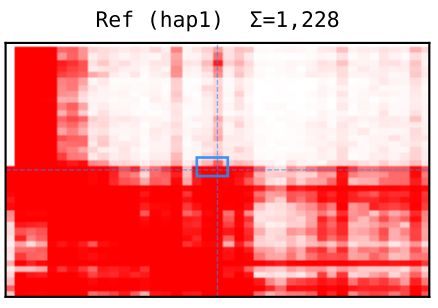
Loop 5
chr14:22,556,979-22,557,524
× chr14:22,570,317-22,570,676
dist: 13 kb
R/A: 0.89× log2FC: -0.17

Unphased Σ=4,543

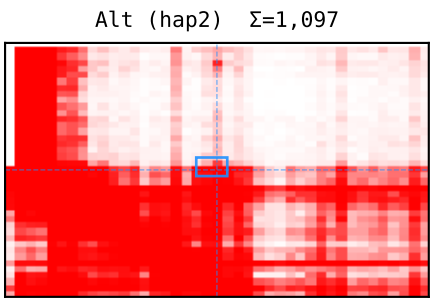
View: 22,553,198 – 22,753,198
Loop: 22,653,162 – 22,653,235



Ref (hap1) Σ=1,228



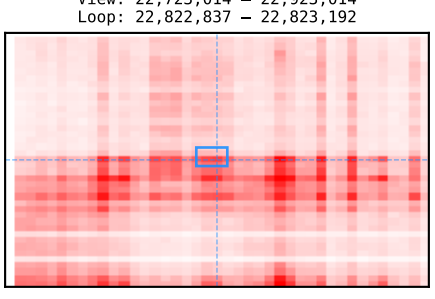
Alt (hap2) Σ=1,097



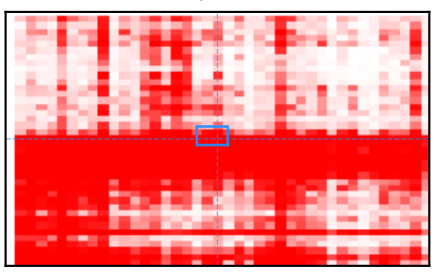
Loop 6
chr14:22,556,979-22,557,524
× chr14:22,653,162-22,653,235
dist: 96 kb
R/A: 1.12× log2FC: +0.16

Unphased Σ=2,491

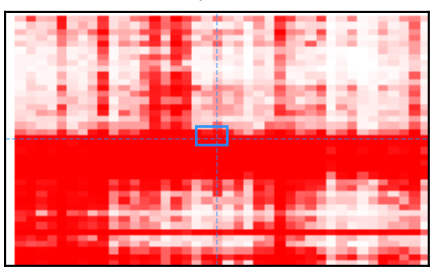
View: 22,723,014 – 22,923,014
Loop: 22,822,837 – 22,823,192



Ref (hap1) Σ=588



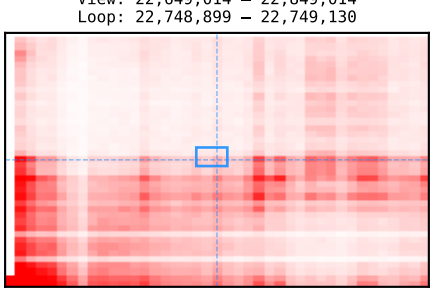
Alt (hap2) Σ=460



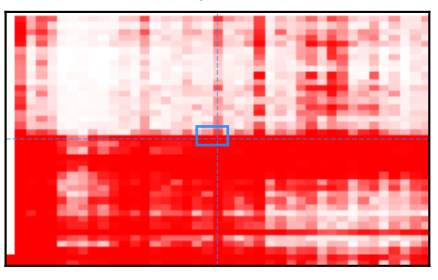
Loop 7
chr14:22,556,979-22,557,524
× chr14:22,822,837-22,823,192
dist: 266 kb
R/A: 1.28× log2FC: +0.35

Unphased Σ=994

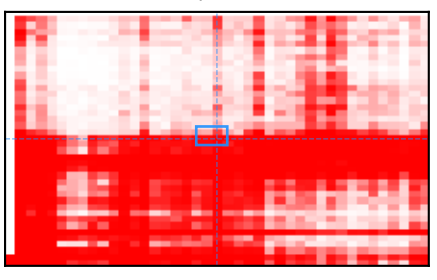
View: 22,649,014 – 22,849,014
Loop: 22,748,899 – 22,749,130



Ref (hap1) Σ=298



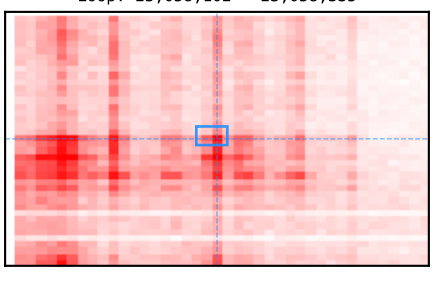
Alt (hap2) Σ=242



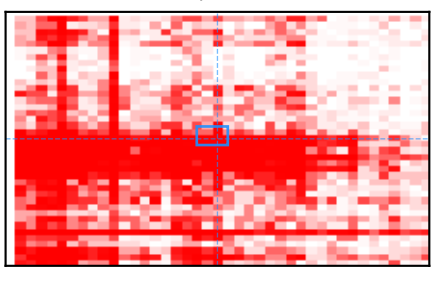
Loop 8
chr14:22,556,979-22,557,524
× chr14:22,748,899-22,749,130
dist: 192 kb
R/A: 1.23× log2FC: +0.30

Unphased Σ=1,035

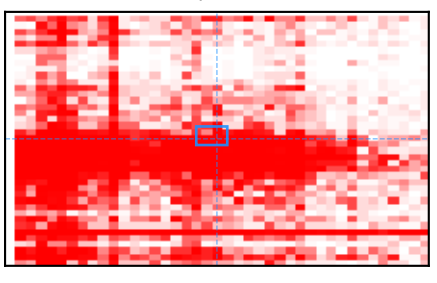
View: 22,958,218 – 23,158,218
Loop: 23,058,102 – 23,058,335



Ref (hap1) Σ=223



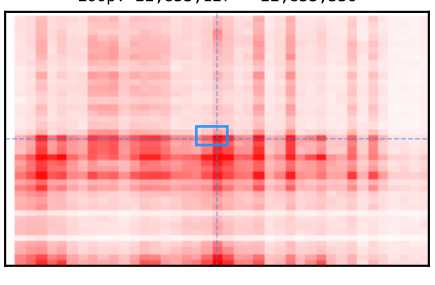
Alt (hap2) Σ=166



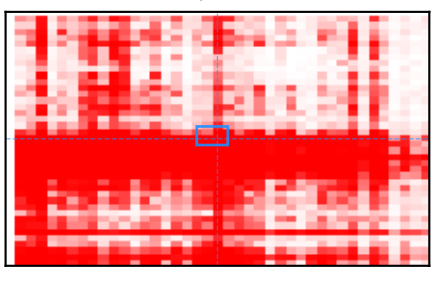
Loop 9
chr14:22,556,979-22,557,524
× chr14:23,058,102-23,058,335
dist: 501 kb
R/A: 1.34× log2FC: +0.43

Unphased Σ=3,565

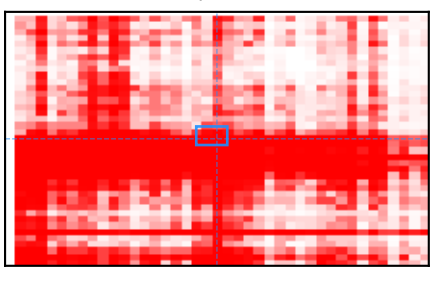
View: 22,753,231 – 22,953,231
Loop: 22,853,127 – 22,853,336



Ref (hap1) Σ=871



Alt (hap2) Σ=830

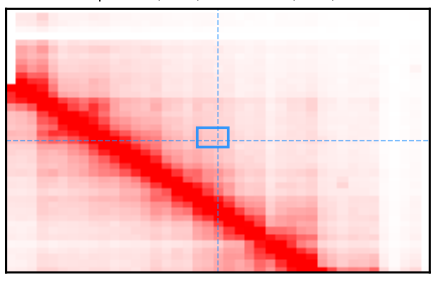


Loop 10
chr14:22,556,979-22,557,524
× chr14:22,853,127-22,853,336
dist: 296 kb
R/A: 1.05× log2FC: +0.07

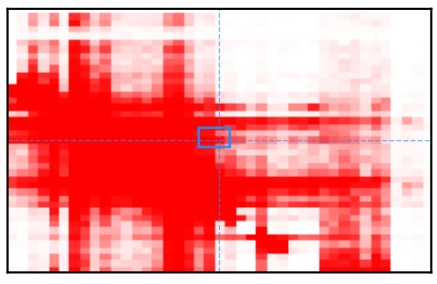
Anchor: chr14:106,564,201–106,564,275 | AlphaGenome pred: 0.007 | empiric ChIP-seq: -4.17

Loop 1
chr14:106,564,172-106,564,279
× chr14:106,618,522-106,618,681
dist: 54 kb
R/A: 1.04× log2FC: +0.06

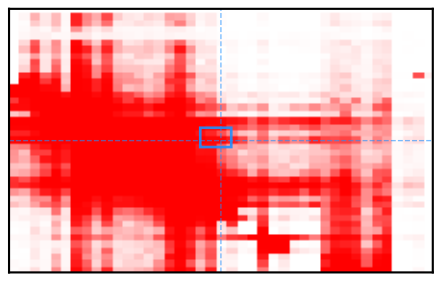
Unphased Σ=2,784
View: 106,518,601 – 106,718,601
Loop: 106,618,522 – 106,618,681



Ref (hap1) Σ=433

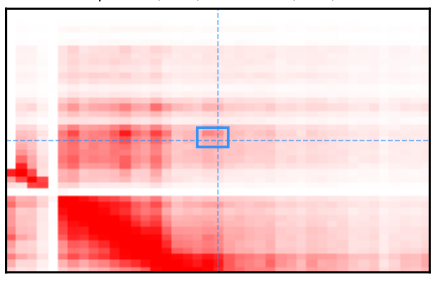


Alt (hap2) Σ=415

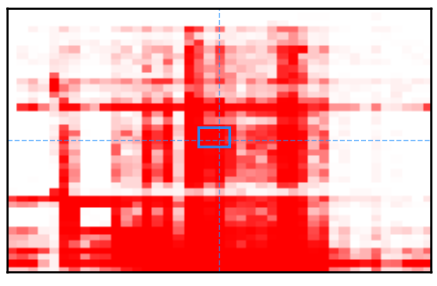


Loop 2
chr14:106,443,090-106,443,109
× chr14:106,564,172-106,564,279
dist: 121 kb
R/A: 0.88× log2FC: -0.19

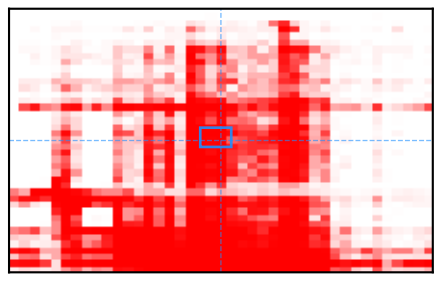
Unphased Σ=2,539
View: 106,464,225 – 106,664,225
Loop: 106,564,172 – 106,564,279



Ref (hap1) Σ=259

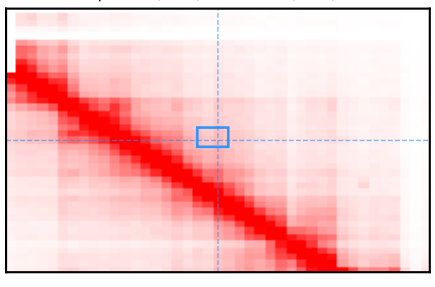


Alt (hap2) Σ=296

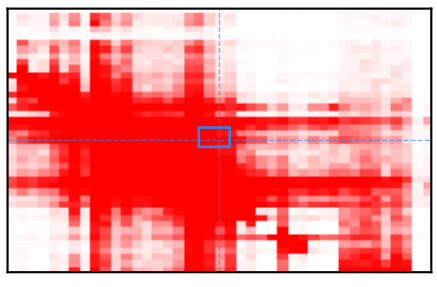


Loop 3
chr14:106,564,172-106,564,279
× chr14:106,605,933-106,606,086
dist: 42 kb
R/A: 0.97× log2FC: -0.04

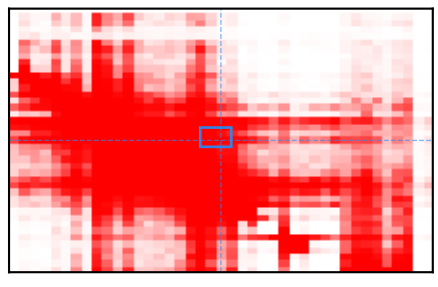
Unphased Σ=3,072
View: 106,506,009 – 106,706,009
Loop: 106,605,933 – 106,606,086



Ref (hap1) Σ=617

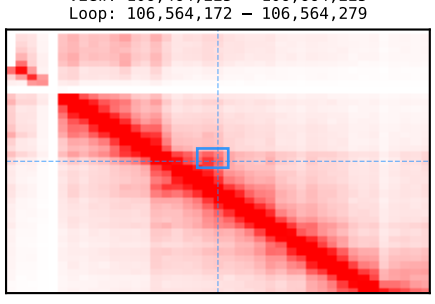


Alt (hap2) Σ=634

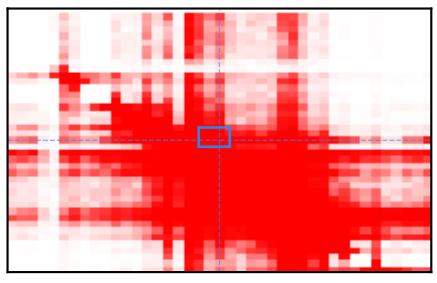


Loop 4
chr14:106,539,594-106,540,157
× chr14:106,564,172-106,564,279
dist: 24 kb
R/A: 1.17× log2FC: +0.23

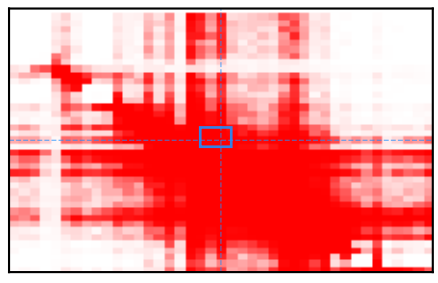
Unphased Σ=9,453
View: 106,464,225 – 106,664,225
Loop: 106,564,172 – 106,564,279



Ref (hap1) Σ=1,992

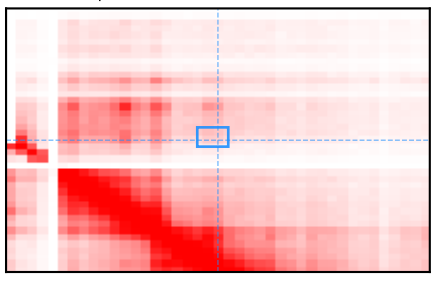


Alt (hap2) Σ=1,697

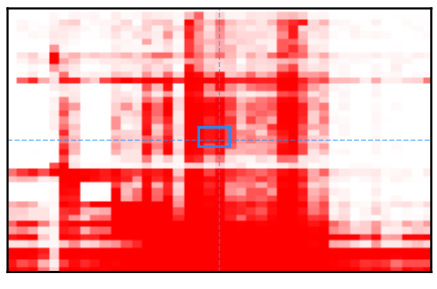


Loop 5
chr14:106,460,105-106,460,200
× chr14:106,564,172-106,564,279
dist: 104 kb
R/A: 0.85× log2FC: -0.23

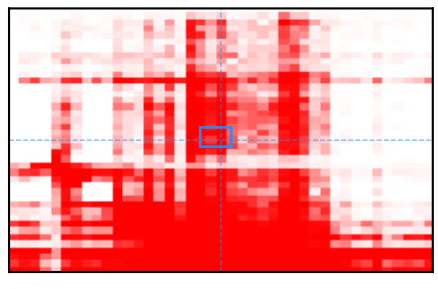
Unphased Σ=1,644
View: 106,464,225 – 106,664,225
Loop: 106,564,172 – 106,564,279



Ref (hap1) Σ=200

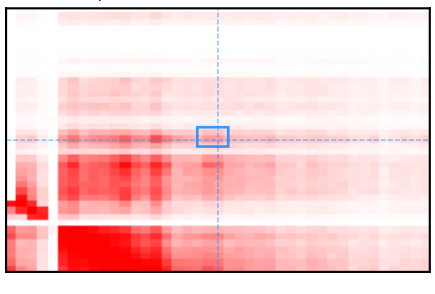


Alt (hap2) Σ=234

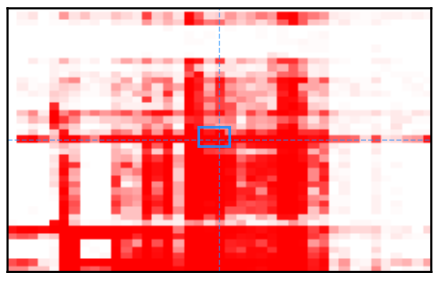


Loop 6
chr14:106,417,706-106,417,883
× chr14:106,564,172-106,564,279
dist: 146 kb
R/A: 1.08× log2FC: +0.12

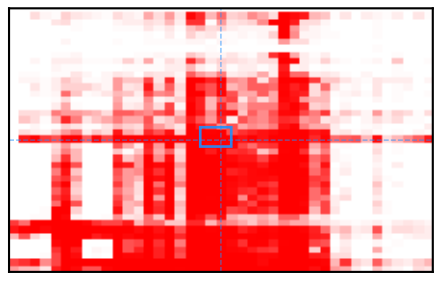
Unphased Σ=1,457
View: 106,464,225 – 106,664,225
Loop: 106,564,172 – 106,564,279



Ref (hap1) Σ=218

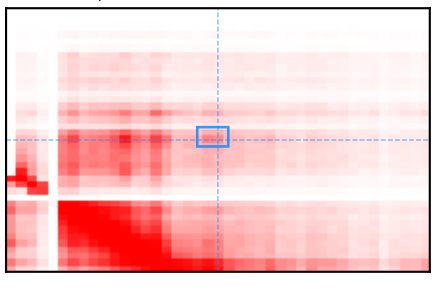


Alt (hap2) Σ=201

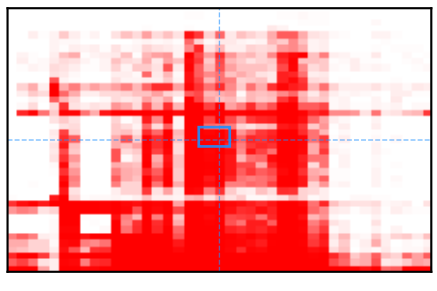


Loop 7
chr14:106,436,070-106,436,280
× chr14:106,564,172-106,564,279
dist: 128 kb
R/A: 0.82× log2FC: -0.28

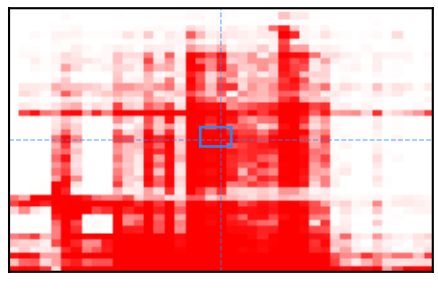
Unphased Σ=2,449
View: 106,464,225 – 106,664,225
Loop: 106,564,172 – 106,564,279



Ref (hap1) Σ=230

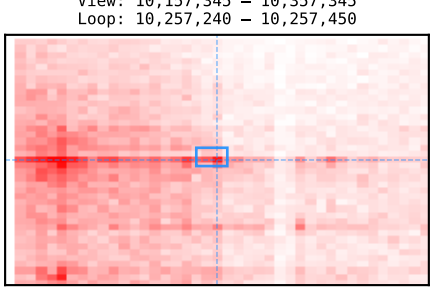


Alt (hap2) Σ=279

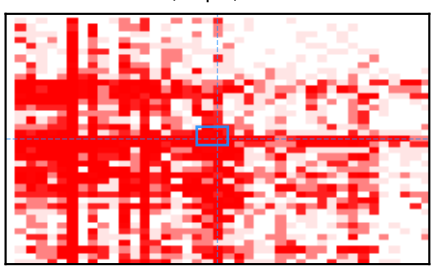


Unphased Σ=274
View: 10,157,345 – 10,357,345
Loop: 10,257,240 – 10,257,450

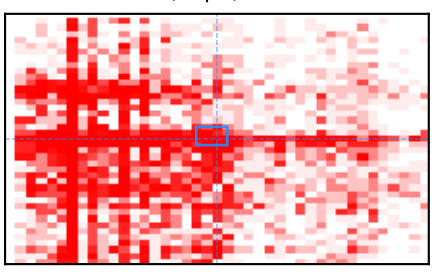
Loop 1
chr16:8,567,746-8,567,983
× chr16:10,257,240-10,257,450
dist: 1689 kb
R/A: 0.97× log2FC: -0.05



Ref (hap1) Σ=83

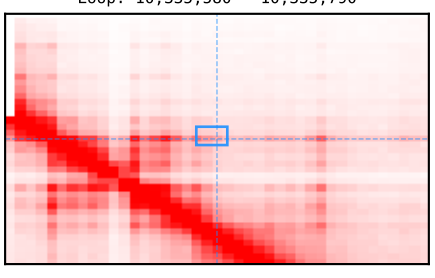


Alt (hap2) Σ=86

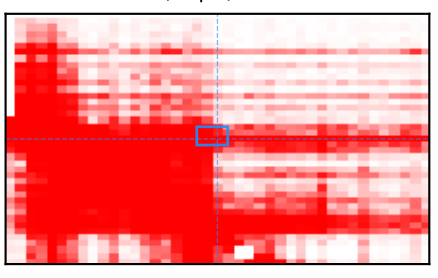


Unphased Σ=2,839
View: 10,235,685 – 10,435,685
Loop: 10,335,580 – 10,335,790

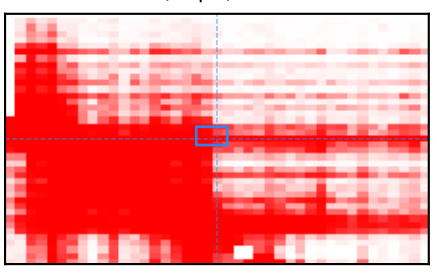
Loop 2
chr16:10,256,833-10,257,128
× chr16:10,335,580-10,335,790
dist: 79 kb
R/A: 1.12× log2FC: +0.17



Ref (hap1) Σ=617

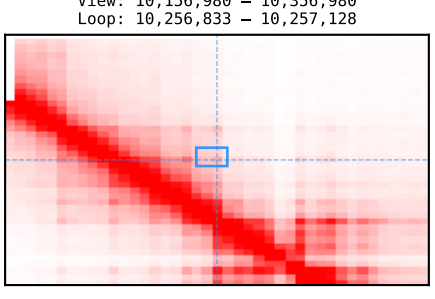


Alt (hap2) Σ=549

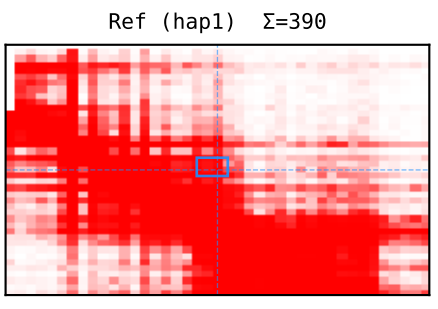


Unphased Σ=2,183
View: 10,156,980 – 10,356,980
Loop: 10,256,833 – 10,257,128

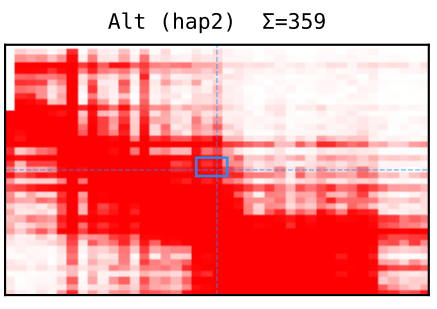
Loop 3
chr16:10,206,348-10,206,521
× chr16:10,256,833-10,257,128
dist: 51 kb
R/A: 1.09× log2FC: +0.12



Ref (hap1) Σ=390

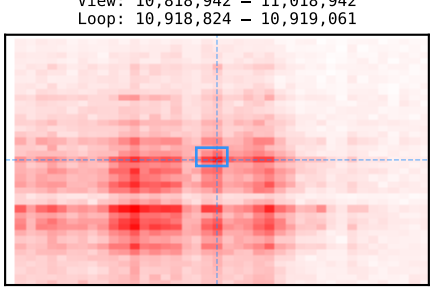


Alt (hap2) Σ=359

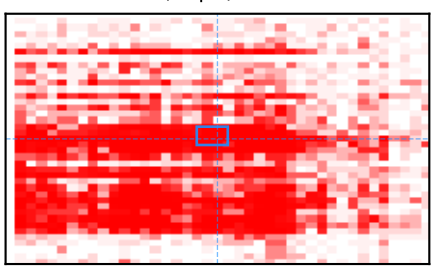


Unphased Σ=847
View: 10,818,942 – 11,018,942
Loop: 10,918,824 – 10,919,061

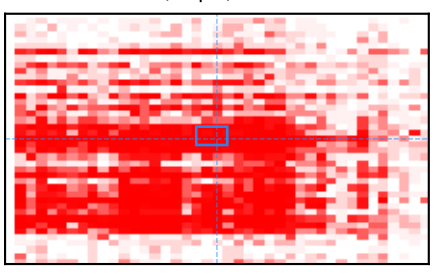
Loop 4
chr16:10,256,833-10,257,128
× chr16:10,918,824-10,919,061
dist: 662 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=128

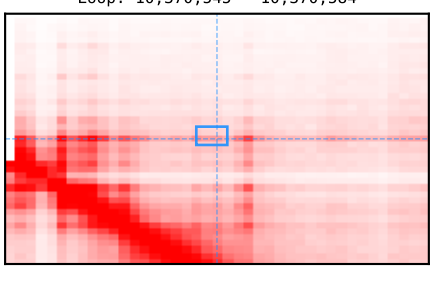


Alt (hap2) Σ=137

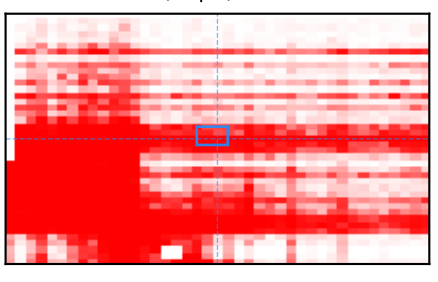


Unphased Σ=2,190
View: 10,270,564 – 10,470,564
Loop: 10,370,545 – 10,370,584

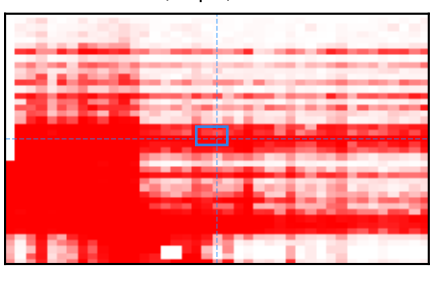
Loop 5
chr16:10,256,833-10,257,128
× chr16:10,370,545-10,370,584
dist: 114 kb
R/A: 1.20× log2FC: +0.27



Ref (hap1) Σ=394

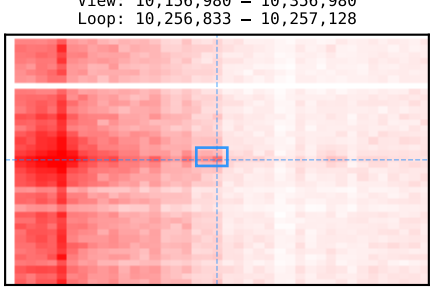


Alt (hap2) Σ=327

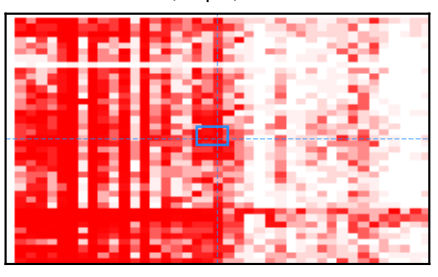


Unphased Σ=439
View: 10,156,980 – 10,356,980
Loop: 10,256,833 – 10,257,128

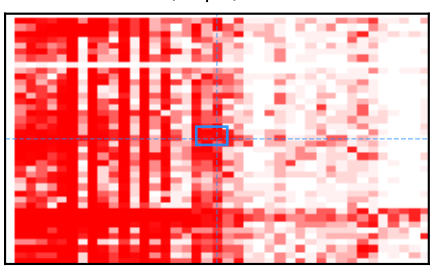
Loop 6
chr16:9,648,211-9,648,392
× chr16:10,256,833-10,257,128
dist: 609 kb
R/A: 1.24× log2FC: +0.31



Ref (hap1) Σ=68

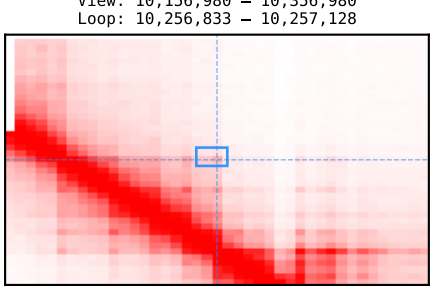


Alt (hap2) Σ=55

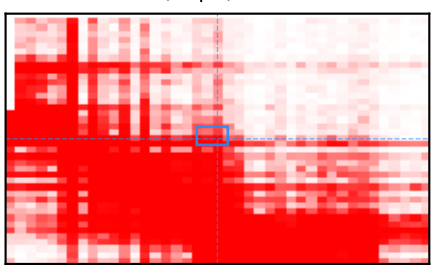


Unphased Σ=1,493
View: 10,156,980 – 10,356,980
Loop: 10,256,833 – 10,257,128

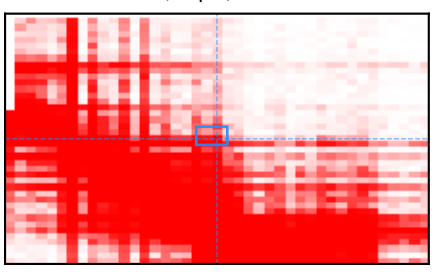
Loop 7
chr16:10,181,590-10,181,797
× chr16:10,256,833-10,257,128
dist: 75 kb
R/A: 1.42× log2FC: +0.51



Ref (hap1) Σ=281

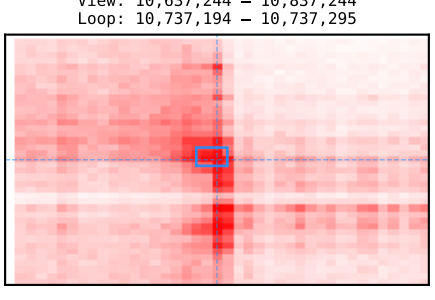


Alt (hap2) Σ=198

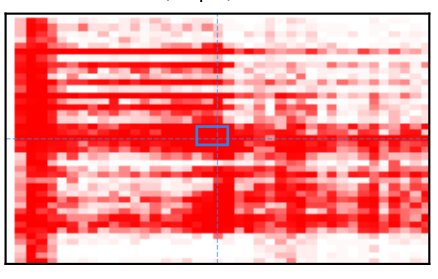


Unphased Σ=3,257
View: 10,637,244 – 10,837,244
Loop: 10,737,194 – 10,737,295

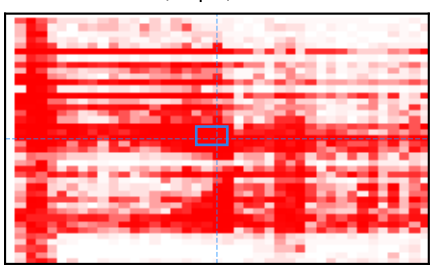
Loop 8
chr16:10,256,833-10,257,128
× chr16:10,737,194-10,737,295
dist: 480 kb
R/A: 0.86× log2FC: -0.21



Ref (hap1) Σ=512

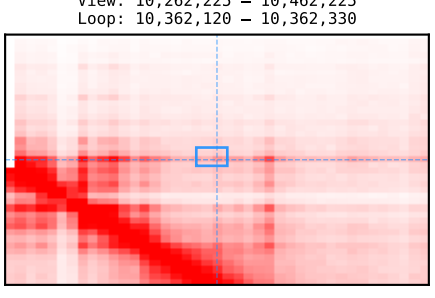


Alt (hap2) Σ=593

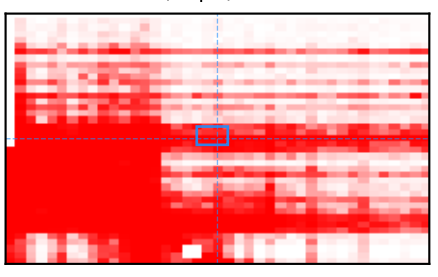


Unphased Σ=2,198
View: 10,262,225 – 10,462,225
Loop: 10,362,120 – 10,362,330

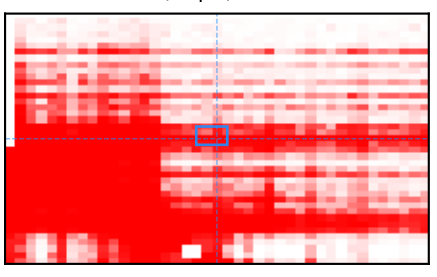
Loop 9
chr16:10,256,833-10,257,128
× chr16:10,362,120-10,362,330
dist: 105 kb
R/A: 1.04× log2FC: +0.05



Ref (hap1) Σ=373

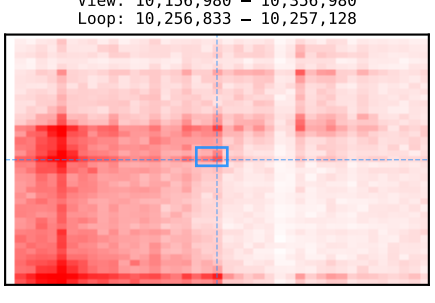


Alt (hap2) Σ=360

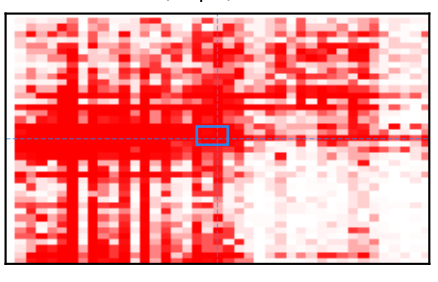


Unphased Σ=525
View: 10,156,980 – 10,356,980
Loop: 10,256,833 – 10,257,128

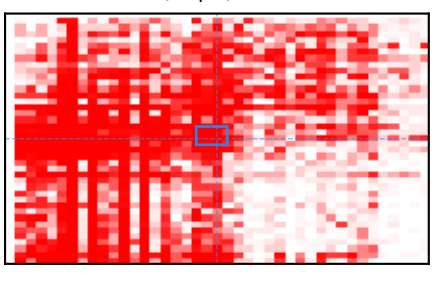
Loop 10
chr16:9,160,250-9,160,355
× chr16:10,256,833-10,257,128
dist: 1097 kb
R/A: 1.70× log2FC: +0.76



Ref (hap1) Σ=173

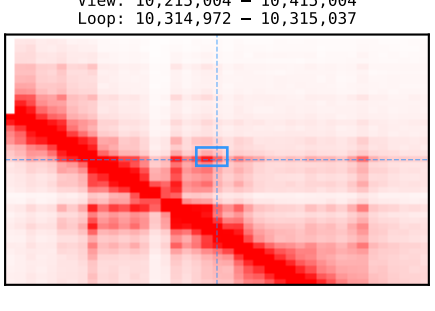


Alt (hap2) Σ=102

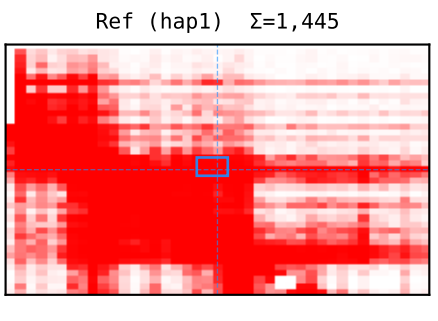


Unphased Σ=6,744
View: 10,215,004 – 10,415,004
Loop: 10,314,972 – 10,315,037

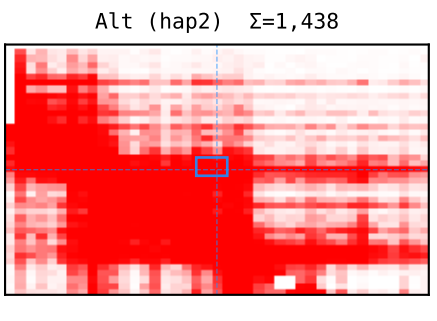
Loop 11
chr16:10,256,833-10,257,128
× chr16:10,314,972-10,315,037
dist: 58 kb
R/A: 1.00× log2FC: +0.01



Ref (hap1) Σ=1,445

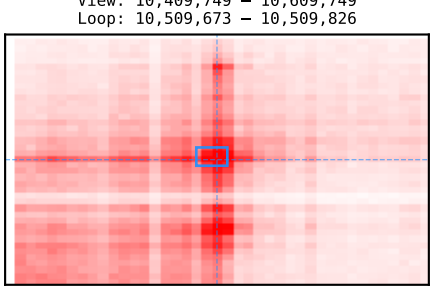


Alt (hap2) Σ=1,438

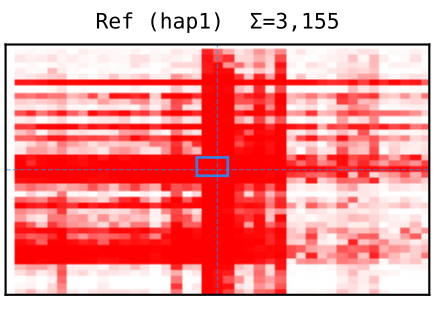


Unphased Σ=13,876
View: 10,409,749 – 10,609,749
Loop: 10,509,673 – 10,509,826

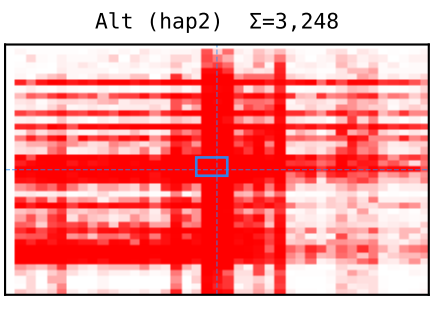
Loop 12
chr16:10,256,833-10,257,128
× chr16:10,509,673-10,509,826
dist: 253 kb
R/A: 0.97× log2FC: -0.04



Ref (hap1) Σ=3,155



Alt (hap2) Σ=3,248



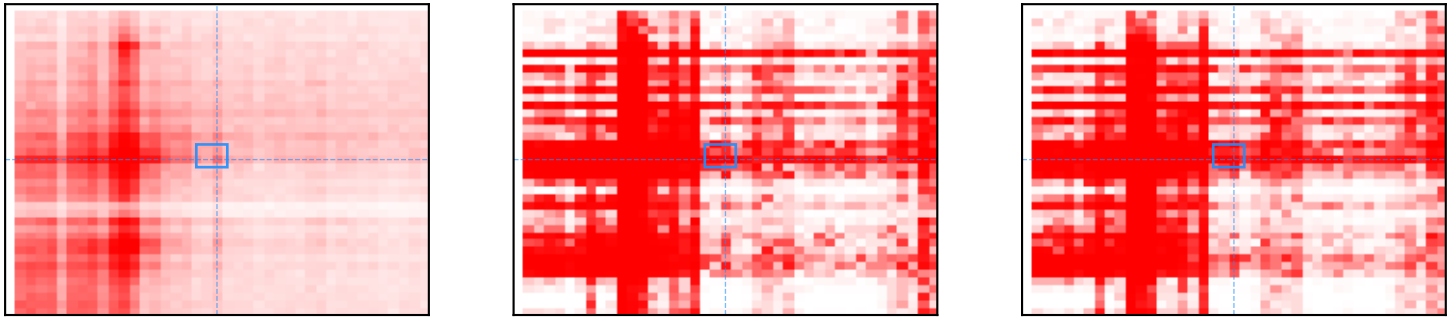
Unphased Σ=992

View: 10,453,338 – 10,653,338
Loop: 10,553,217 – 10,553,460

Ref (hap1) Σ=141

Alt (hap2) Σ=134

Loop 1
chr16:10,256,833-10,257,128
× chr16:10,553,217-10,553,460
dist: 296 kb
R/A: 1.05× log2FC: +0.07



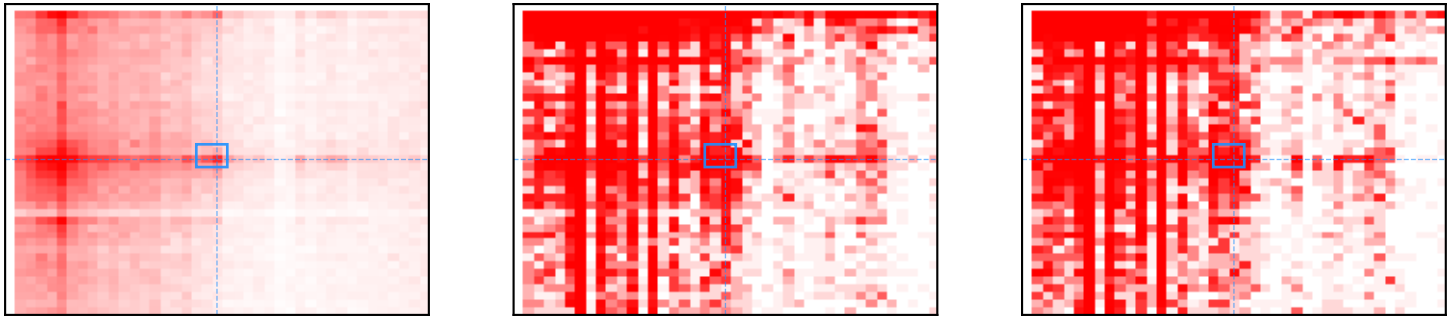
Unphased Σ=717

View: 10,156,980 – 10,356,980
Loop: 10,256,833 – 10,257,128

Ref (hap1) Σ=103

Alt (hap2) Σ=112

Loop 2
chr16:9,258,549-9,258,774
× chr16:10,256,833-10,257,128
dist: 998 kb
R/A: 0.92× log2FC: -0.12



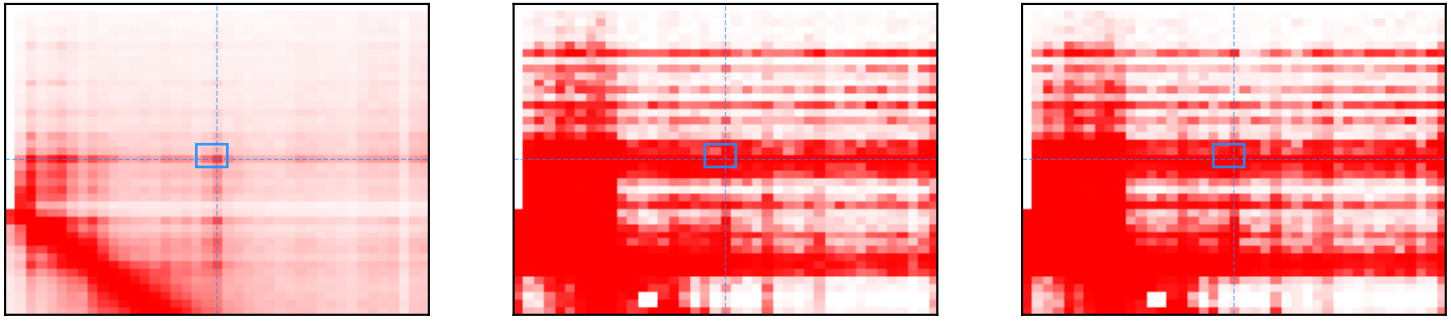
Unphased Σ=3,682

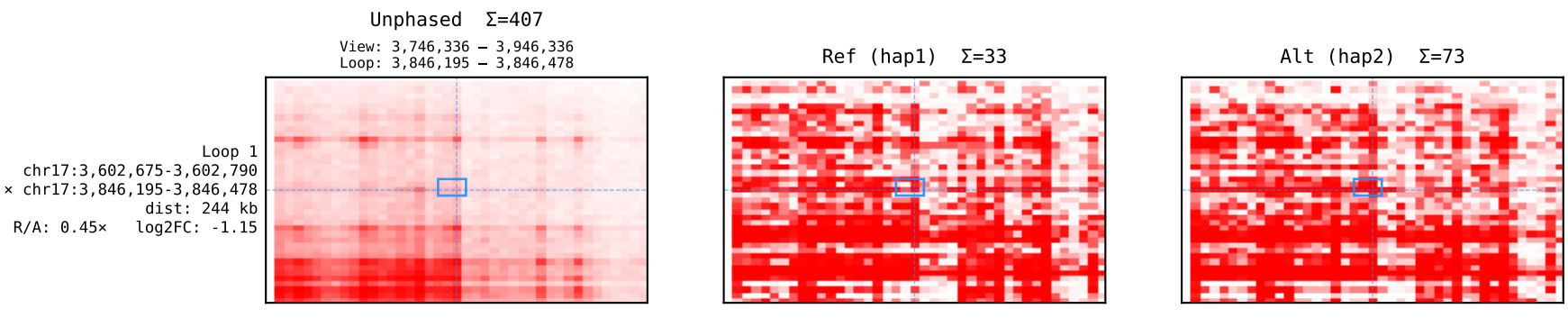
View: 10,286,029 – 10,486,029
Loop: 10,385,953 – 10,386,106

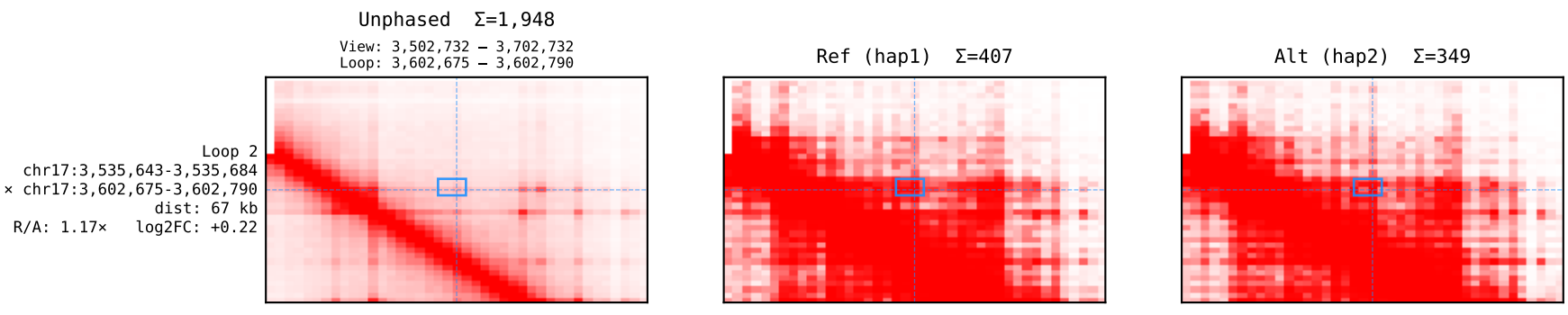
Ref (hap1) Σ=606

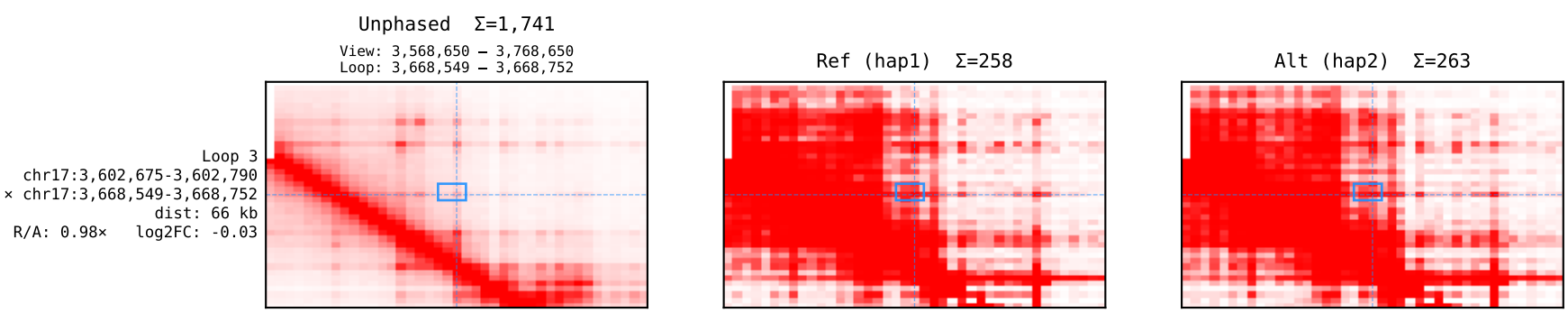
Alt (hap2) Σ=622

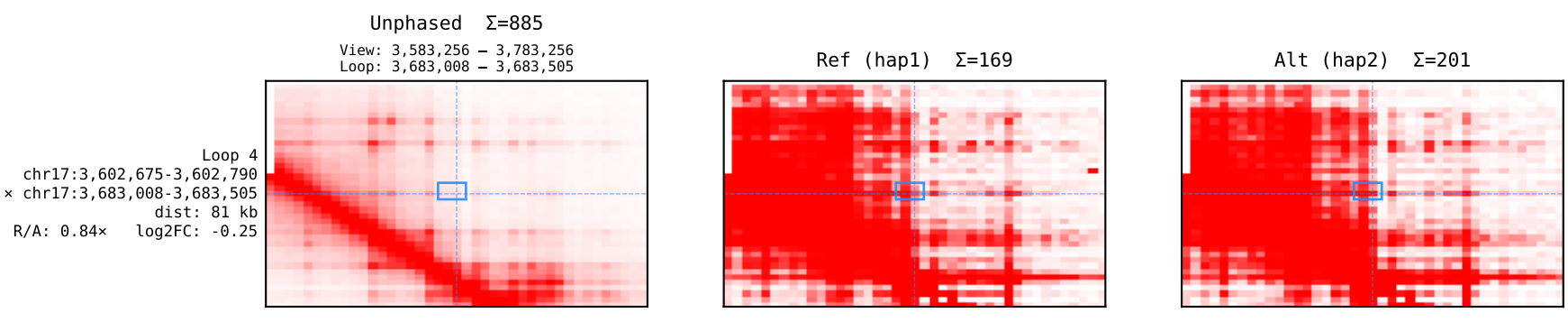
Loop 3
chr16:10,256,833-10,257,128
× chr16:10,385,953-10,386,106
dist: 129 kb
R/A: 0.97× log2FC: -0.04

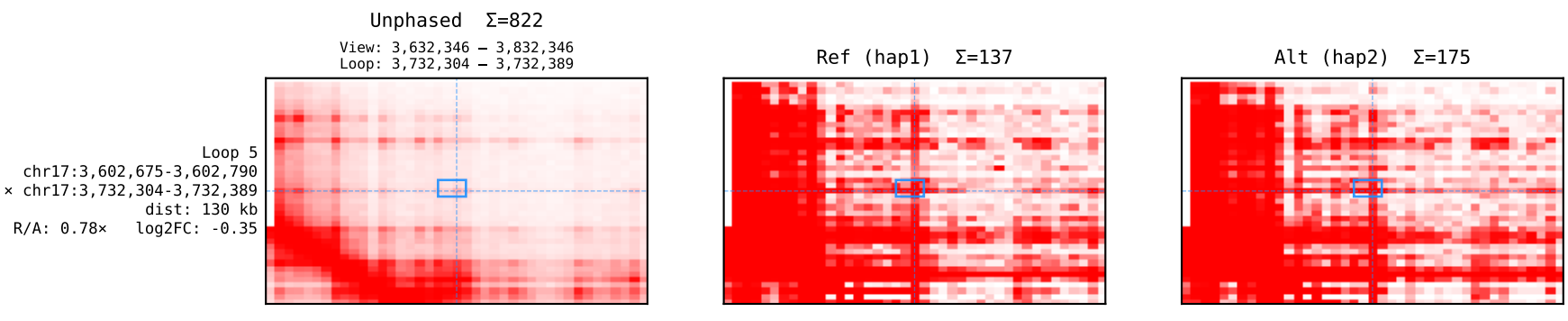


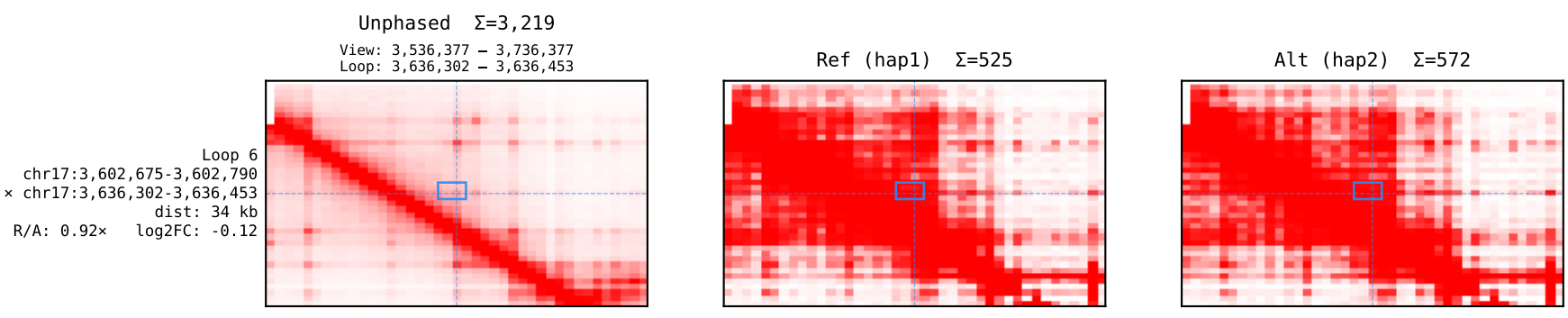

Unphased Σ=407
View: 3,746,336 – 3,946,336
Loop: 3,846,195 – 3,846,478
Ref (hap1) Σ=33
Alt (hap2) Σ=73
Loop 1
chr17:3,602,675-3,602,790
× chr17:3,846,195-3,846,478
dist: 244 kb
R/A: 0.45× log2FC: -1.15

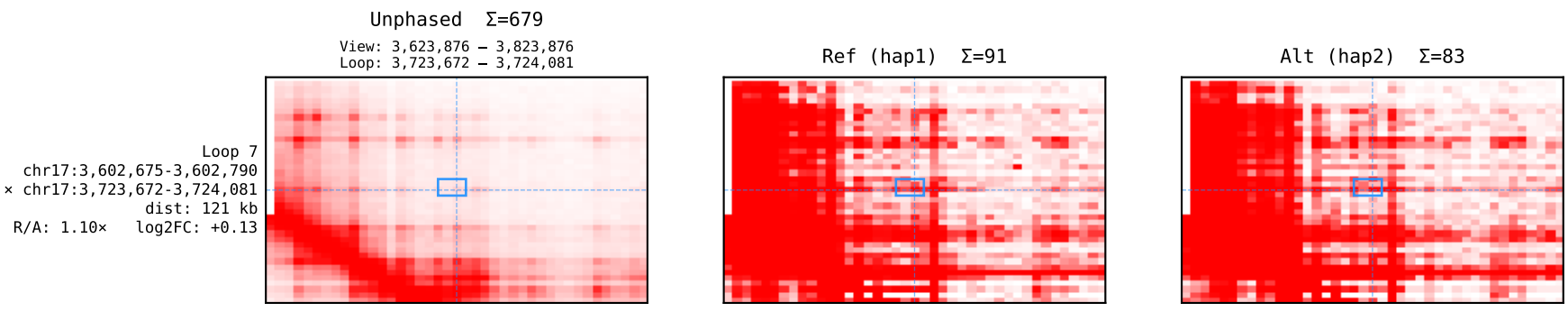

Unphased Σ=1,948
View: 3,502,732 – 3,702,732
Loop: 3,602,675 – 3,602,790
Ref (hap1) Σ=407
Alt (hap2) Σ=349
Loop 2
chr17:3,535,643-3,535,684
× chr17:3,602,675-3,602,790
dist: 67 kb
R/A: 1.17× log2FC: +0.22

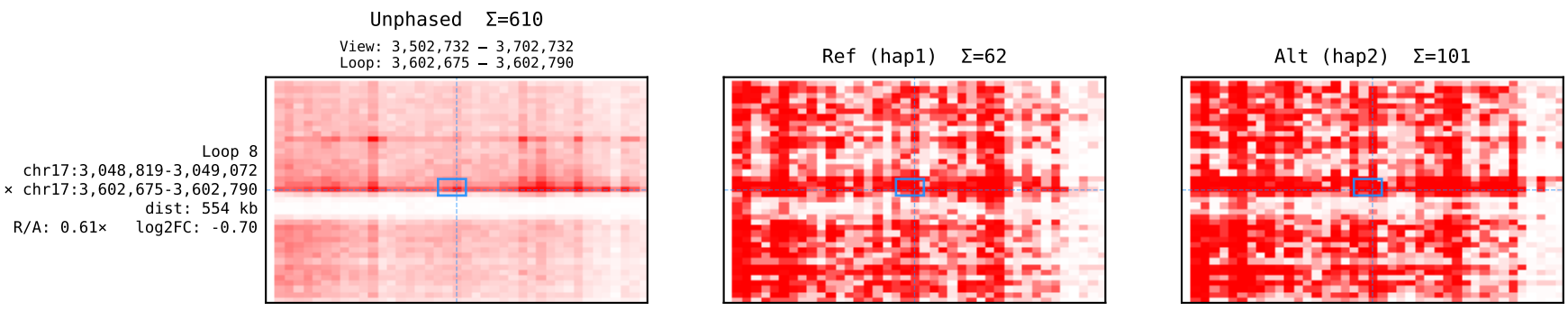

Unphased Σ=1,741
View: 3,568,650 – 3,768,650
Loop: 3,668,549 – 3,668,752
Ref (hap1) Σ=258
Alt (hap2) Σ=263
Loop 3
chr17:3,602,675-3,602,790
× chr17:3,668,549-3,668,752
dist: 66 kb
R/A: 0.98× log2FC: -0.03

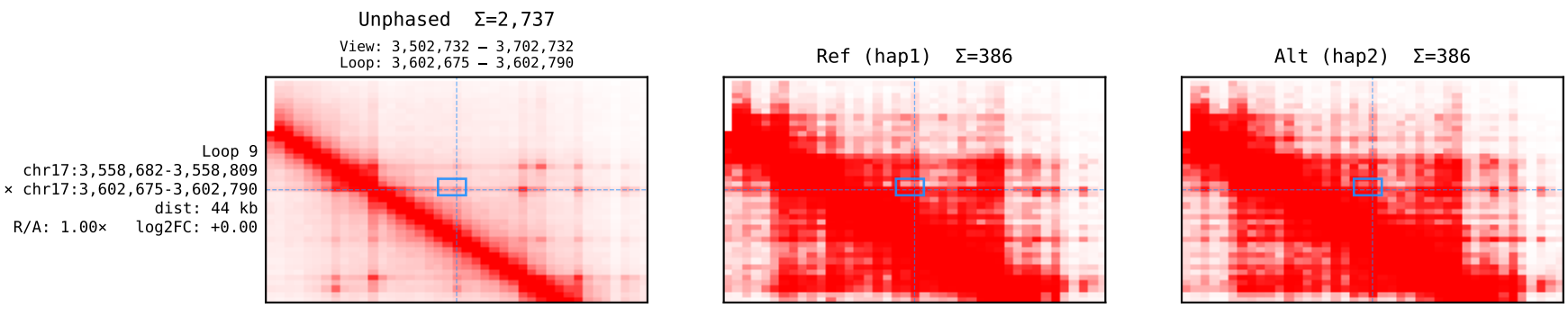

Unphased Σ=885
View: 3,583,256 – 3,783,256
Loop: 3,683,008 – 3,683,505
Ref (hap1) Σ=169
Alt (hap2) Σ=201
Loop 4
chr17:3,602,675-3,602,790
× chr17:3,683,008-3,683,505
dist: 81 kb
R/A: 0.84× log2FC: -0.25

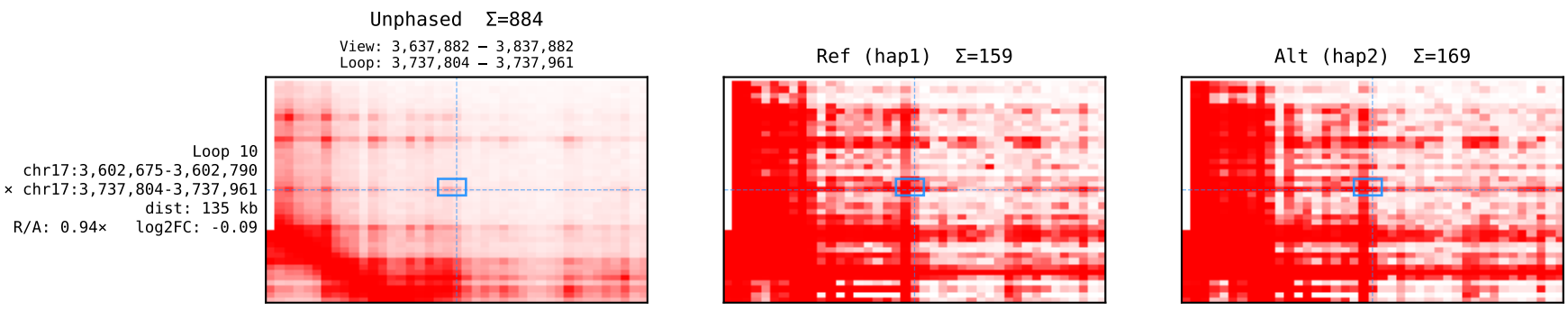

Unphased Σ=822
View: 3,632,346 – 3,832,346
Loop: 3,732,304 – 3,732,389
Ref (hap1) Σ=137
Alt (hap2) Σ=175
Loop 5
chr17:3,602,675-3,602,790
× chr17:3,732,304-3,732,389
dist: 130 kb
R/A: 0.78× log2FC: -0.35

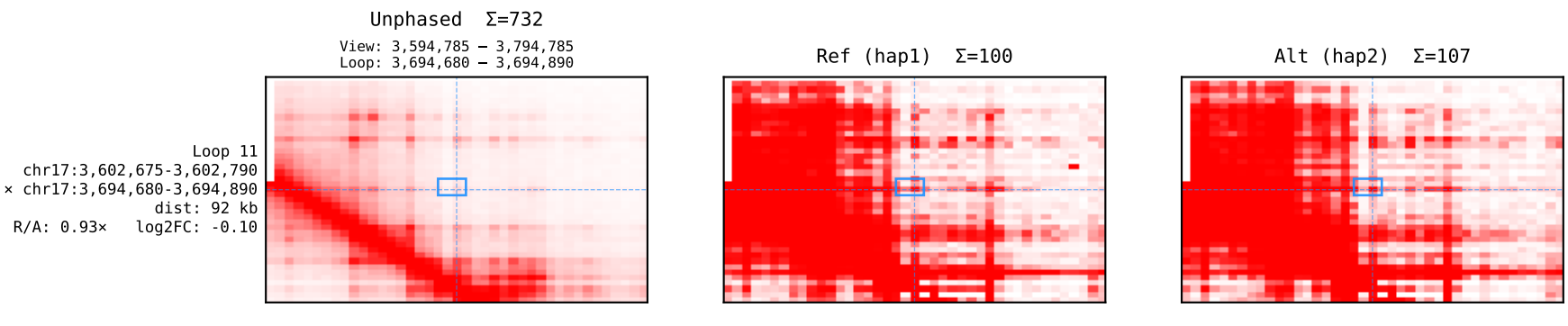

Unphased Σ=3,219
View: 3,536,377 – 3,736,377
Loop: 3,636,302 – 3,636,453
Ref (hap1) Σ=525
Alt (hap2) Σ=572
Loop 6
chr17:3,602,675-3,602,790
× chr17:3,636,302-3,636,453
dist: 34 kb
R/A: 0.92× log2FC: -0.12

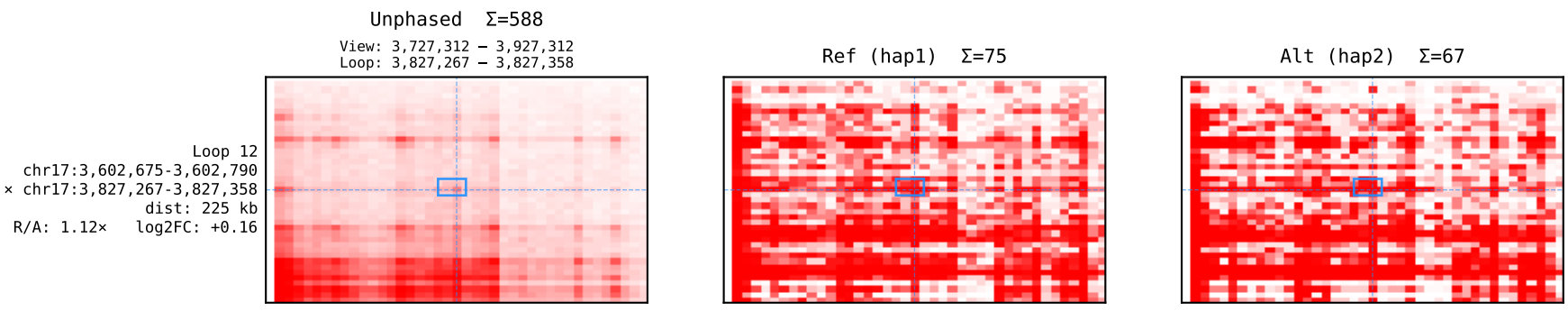

Unphased Σ=679
View: 3,623,876 – 3,823,876
Loop: 3,723,672 – 3,724,081
Ref (hap1) Σ=91
Alt (hap2) Σ=83
Loop 7
chr17:3,602,675-3,602,790
× chr17:3,723,672-3,724,081
dist: 121 kb
R/A: 1.10× log2FC: +0.13


Unphased Σ=610
View: 3,502,732 – 3,702,732
Loop: 3,602,675 – 3,602,790
Ref (hap1) Σ=62
Alt (hap2) Σ=101
Loop 8
chr17:3,048,819-3,049,072
× chr17:3,602,675-3,602,790
dist: 554 kb
R/A: 0.61× log2FC: -0.70

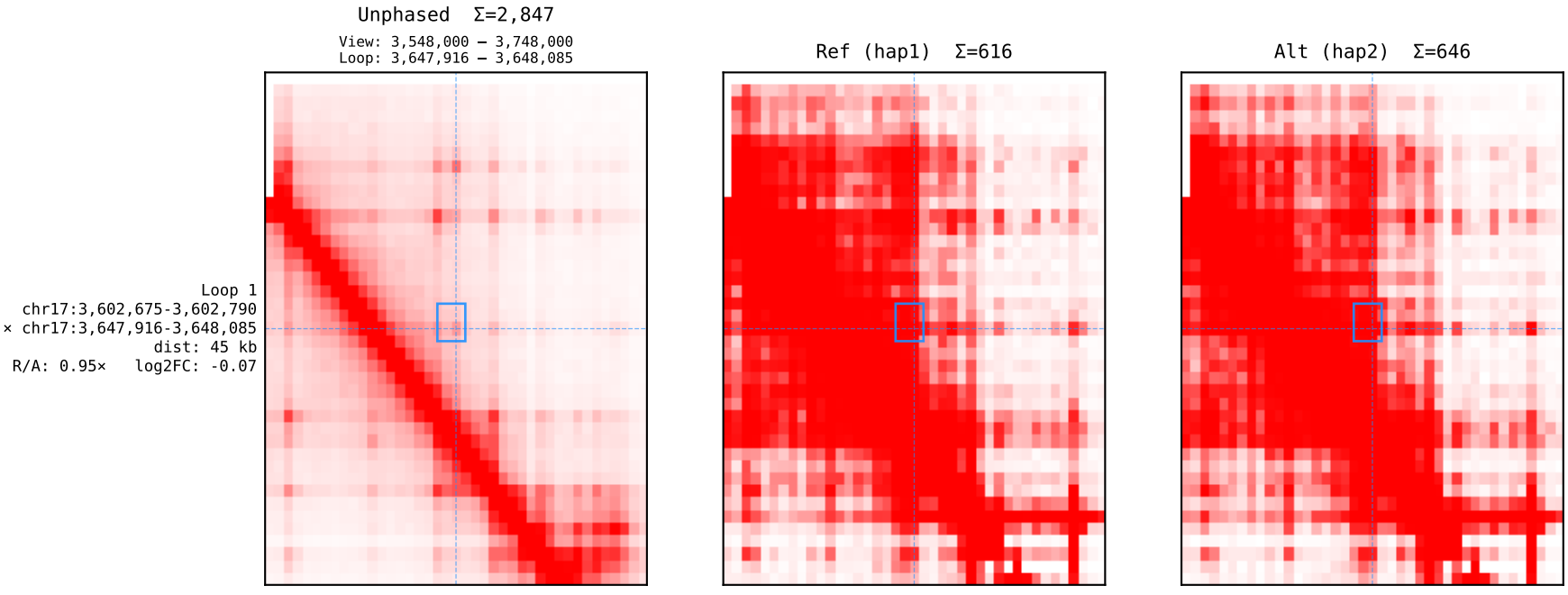

Unphased Σ=2,737
View: 3,502,732 – 3,702,732
Loop: 3,602,675 – 3,602,790
Ref (hap1) Σ=386
Alt (hap2) Σ=386
Loop 9
chr17:3,558,682-3,558,809
× chr17:3,602,675-3,602,790
dist: 44 kb
R/A: 1.00× log2FC: +0.00


Unphased Σ=884
View: 3,637,882 – 3,837,882
Loop: 3,737,804 – 3,737,961
Ref (hap1) Σ=159
Alt (hap2) Σ=169
Loop 10
chr17:3,602,675-3,602,790
× chr17:3,737,804-3,737,961
dist: 135 kb
R/A: 0.94× log2FC: -0.09


Unphased Σ=732
View: 3,594,785 – 3,794,785
Loop: 3,694,680 – 3,694,890
Ref (hap1) Σ=100
Alt (hap2) Σ=107
Loop 11
chr17:3,602,675-3,602,790
× chr17:3,694,680-3,694,890
dist: 92 kb
R/A: 0.93× log2FC: -0.10


Unphased Σ=588
View: 3,727,312 – 3,927,312
Loop: 3,827,267 – 3,827,358
Ref (hap1) Σ=75
Alt (hap2) Σ=67
Loop 12
chr17:3,602,675-3,602,790
× chr17:3,827,267-3,827,358
dist: 225 kb
R/A: 1.12× log2FC: +0.16

Anchor: chr17:3,602,688–3,602,760 | AlphaGenome pred: -0.0574 | empiric ChIP-seq: 1.32 | page 2 of 2

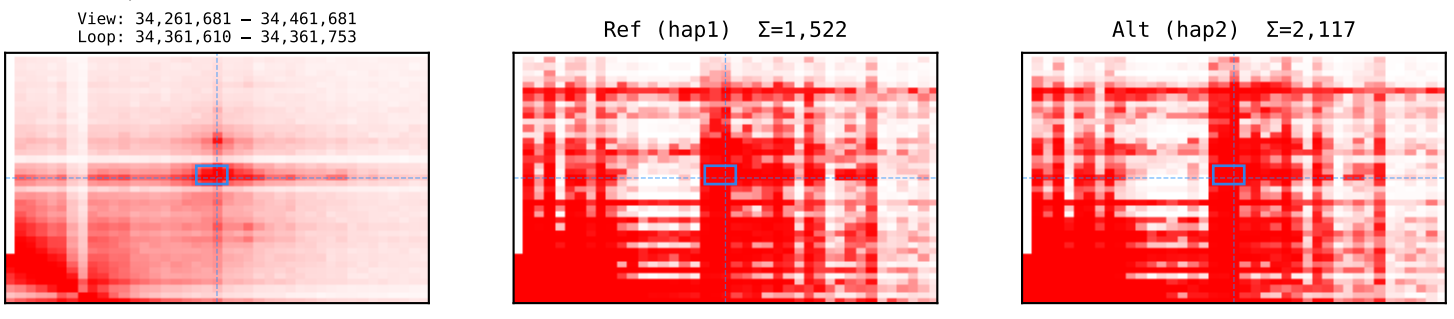


Anchor: chr17:34,361,633–34,361,717 | AlphaGenome pred: -0.0583 | empiric ChIP-seq: -4.91

Unphased Σ=15,275 | Ref (hap1) Σ=1,522 | Alt (hap2) Σ=2,117

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

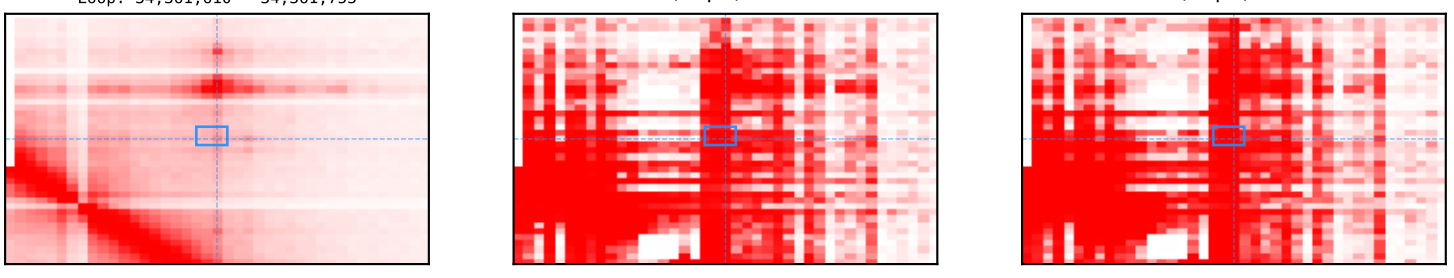
Loop 1
chr17:34,200,028-34,200,233
× chr17:34,361,610-34,361,753
dist: 162 kb
R/A: 0.72× log2FC: -0.48



Unphased Σ=2,187 | Ref (hap1) Σ=285 | Alt (hap2) Σ=289

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

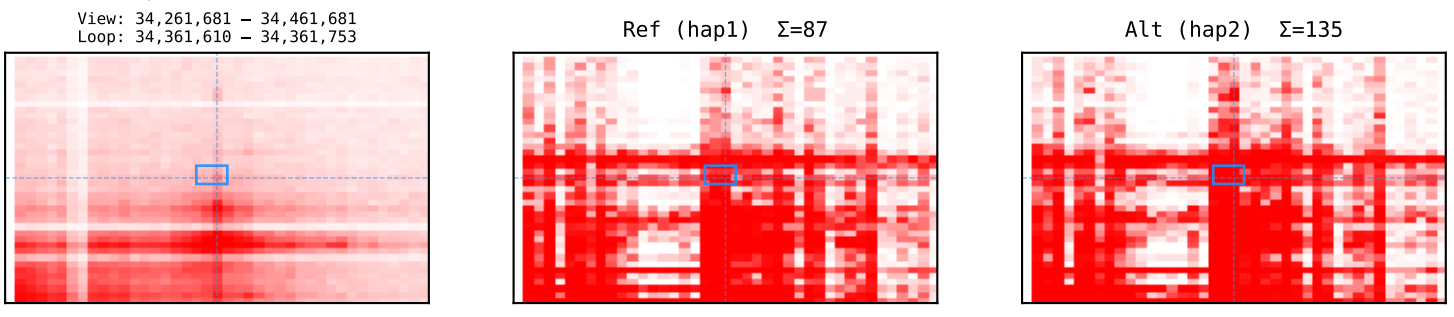
Loop 2
chr17:34,242,717-34,242,798
× chr17:34,361,610-34,361,753
dist: 119 kb
R/A: 0.99× log2FC: -0.02



Unphased Σ=811 | Ref (hap1) Σ=87 | Alt (hap2) Σ=135

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

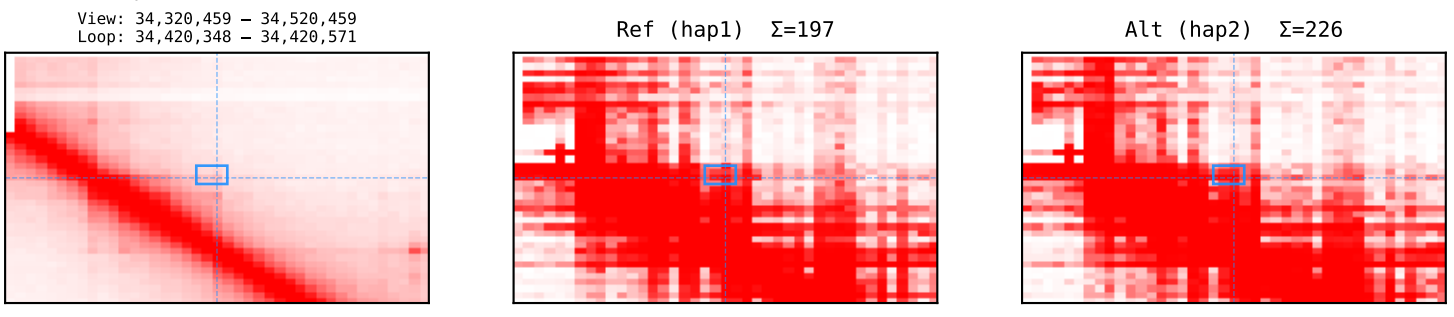
Loop 3
chr17:34,146,840-34,147,050
× chr17:34,361,610-34,361,753
dist: 215 kb
R/A: 0.64× log2FC: -0.63



Unphased Σ=1,633 | Ref (hap1) Σ=197 | Alt (hap2) Σ=226

View: 34,320,459 – 34,520,459
Loop: 34,420,348 – 34,420,571

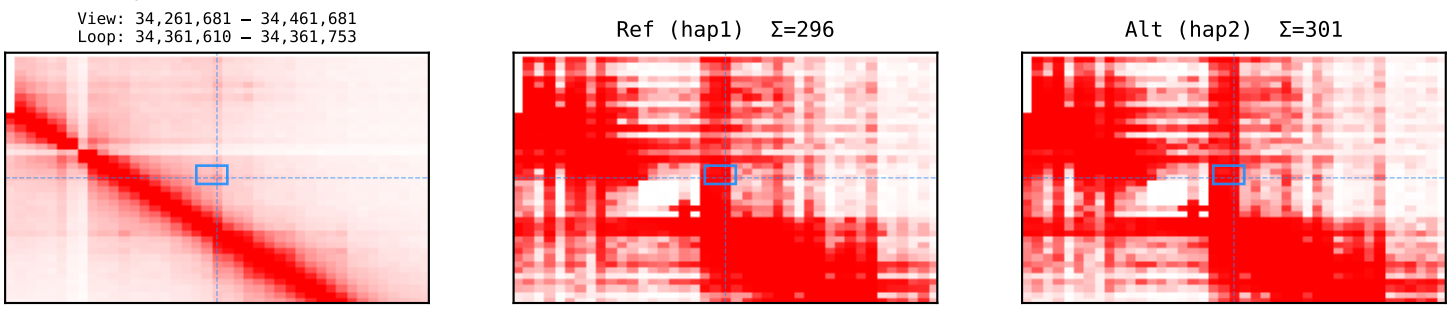
Loop 4
chr17:34,361,610-34,361,753
× chr17:34,420,348-34,420,571
dist: 59 kb
R/A: 0.87× log2FC: -0.20



Unphased Σ=2,874 | Ref (hap1) Σ=296 | Alt (hap2) Σ=301

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

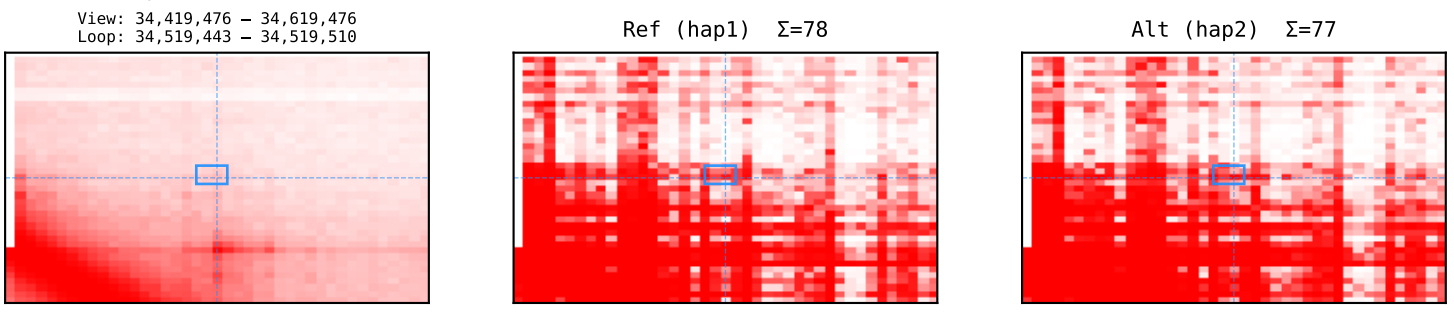
Loop 5
chr17:34,318,710-34,318,720
× chr17:34,361,610-34,361,753
dist: 43 kb
R/A: 0.98× log2FC: -0.02



Unphased Σ=729 | Ref (hap1) Σ=78 | Alt (hap2) Σ=77

View: 34,419,476 – 34,619,476
Loop: 34,519,443 – 34,519,510

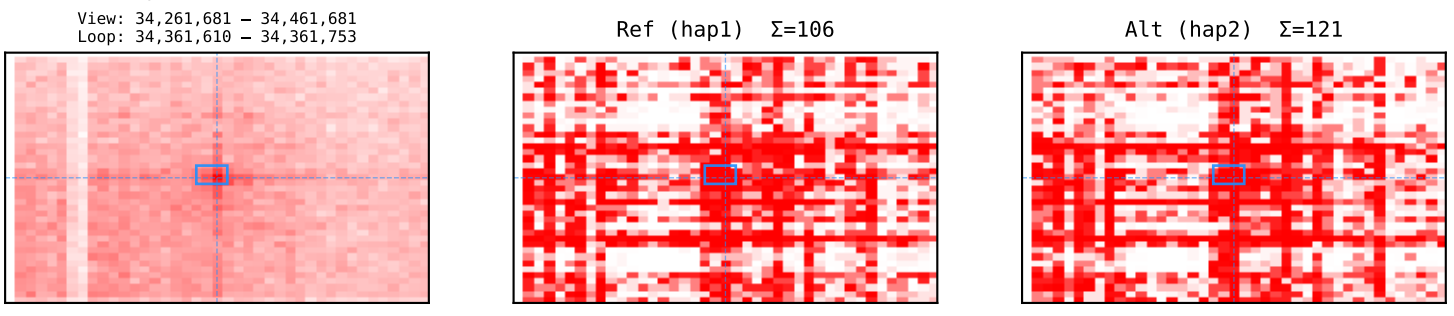
Loop 6
chr17:34,361,610-34,361,753
× chr17:34,519,443-34,519,510
dist: 158 kb
R/A: 1.01× log2FC: +0.02



Unphased Σ=642 | Ref (hap1) Σ=106 | Alt (hap2) Σ=121

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

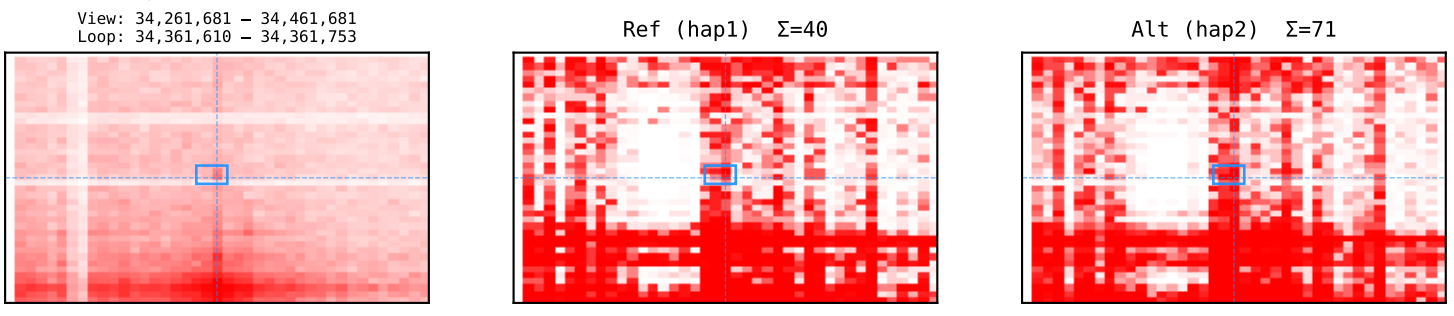
Loop 7
chr17:33,779,193-33,779,368
× chr17:34,361,610-34,361,753
dist: 582 kb
R/A: 0.88× log2FC: -0.19



Unphased Σ=472 | Ref (hap1) Σ=40 | Alt (hap2) Σ=71

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

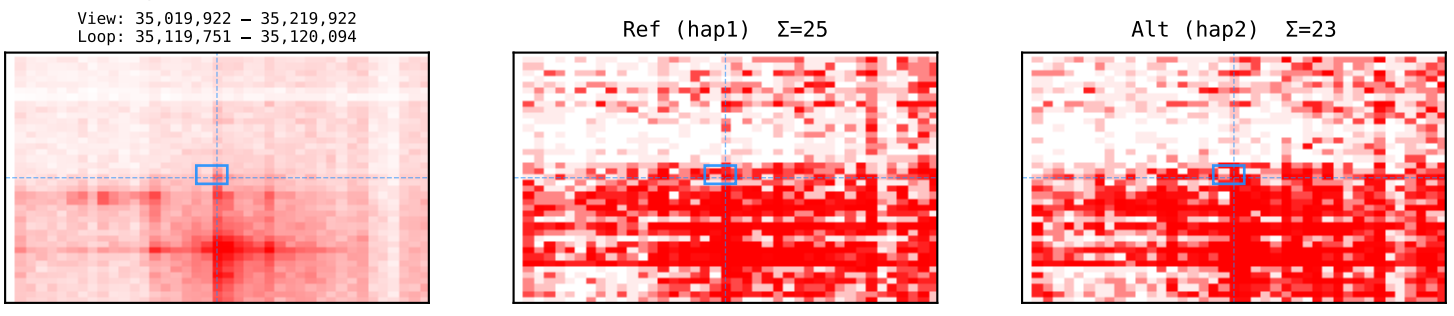
Loop 8
chr17:34,081,452-34,081,517
× chr17:34,361,610-34,361,753
dist: 280 kb
R/A: 0.56× log2FC: -0.83



Unphased Σ=261 | Ref (hap1) Σ=25 | Alt (hap2) Σ=23

View: 35,019,922 – 35,219,922
Loop: 35,119,751 – 35,120,094

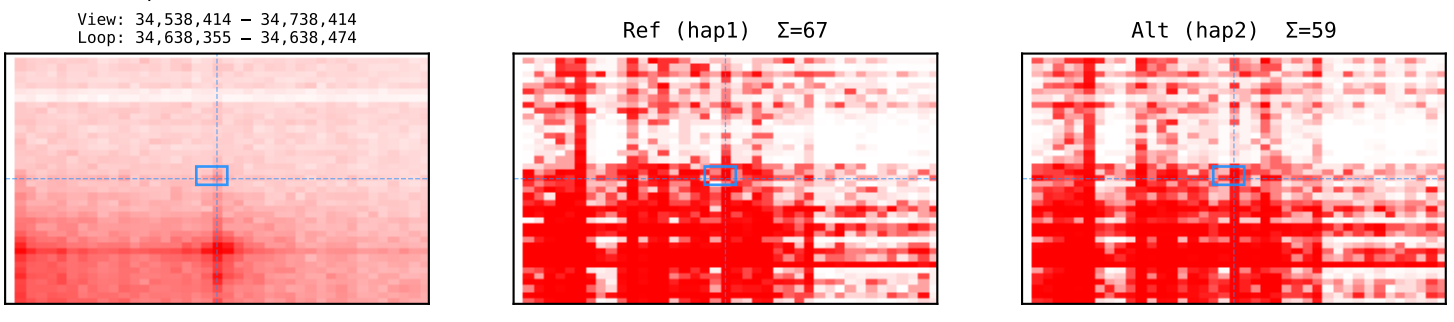
Loop 9
chr17:34,361,610-34,361,753
× chr17:35,119,751-35,120,094
dist: 758 kb
R/A: 1.09× log2FC: +0.12



Unphased Σ=515 | Ref (hap1) Σ=67 | Alt (hap2) Σ=59

View: 34,538,414 – 34,738,414
Loop: 34,638,355 – 34,638,474

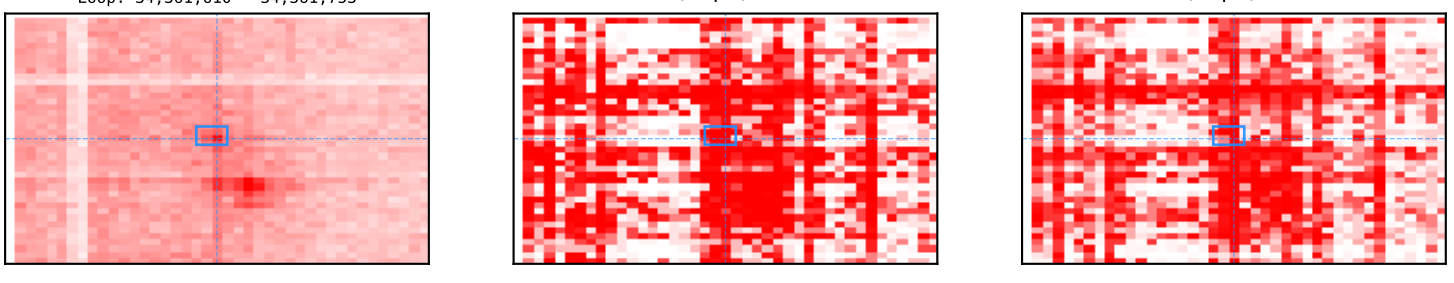
Loop 10
chr17:34,361,610-34,361,753
× chr17:34,638,355-34,638,474
dist: 277 kb
R/A: 1.14× log2FC: +0.18



Unphased Σ=601 | Ref (hap1) Σ=60 | Alt (hap2) Σ=78

View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

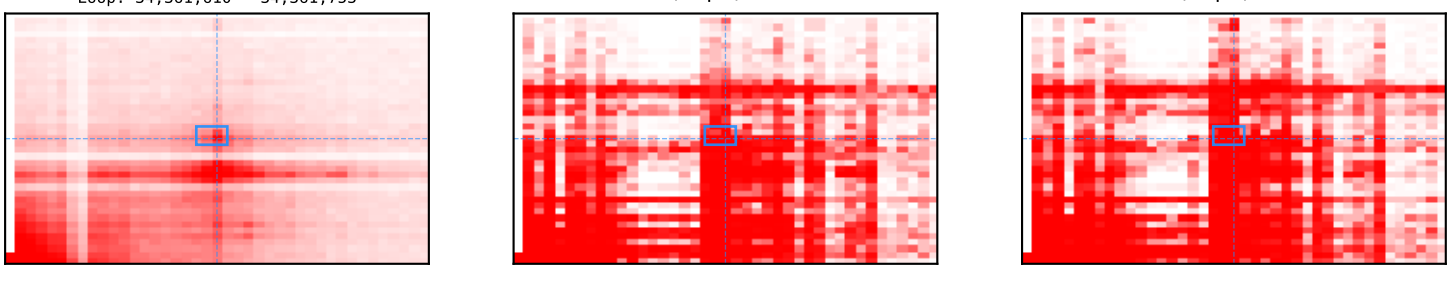
Loop 11
chr17:33,927,988-33,928,095
× chr17:34,361,610-34,361,753
dist: 434 kb
R/A: 0.77× log2FC: -0.38



Unphased Σ=2,995 | Ref (hap1) Σ=291 | Alt (hap2) Σ=437

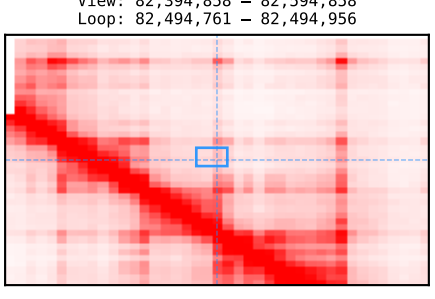
View: 34,261,681 – 34,461,681
Loop: 34,361,610 – 34,361,753

Loop 12
chr17:34,171,237-34,171,572
× chr17:34,361,610-34,361,753
dist: 190 kb
R/A: 0.67× log2FC: -0.59

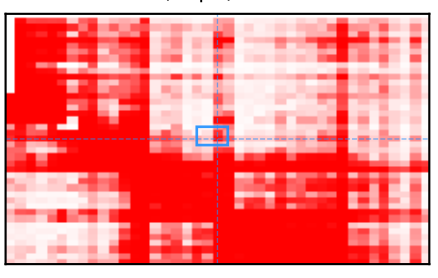


Unphased Σ=2,034
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

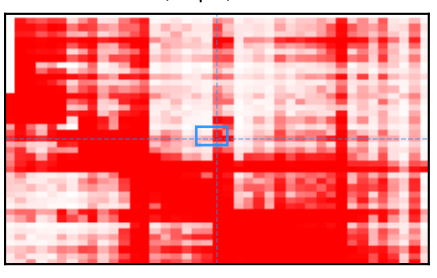
Loop 1
chr17:82,434,413-82,434,462
× chr17:82,494,761-82,494,956
dist: 60 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=220

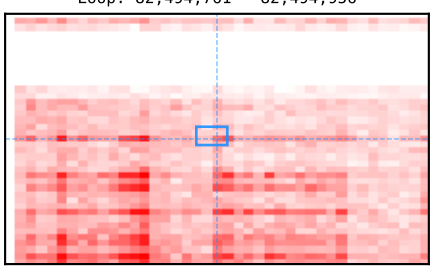


Alt (hap2) Σ=236

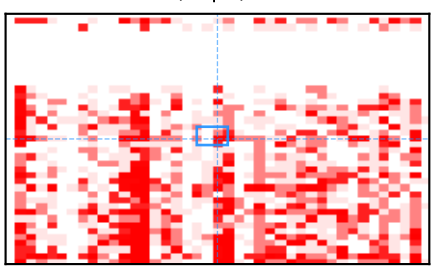


Unphased Σ=91
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

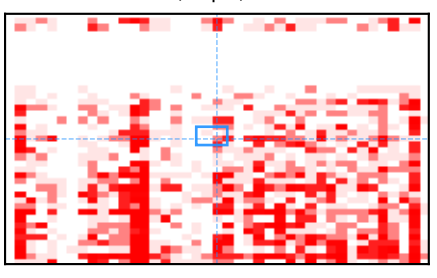
Loop 2
chr17:81,833,108-81,833,485
× chr17:82,494,761-82,494,956
dist: 662 kb
R/A: 2.00× log2FC: +1.00



Ref (hap1) Σ=6

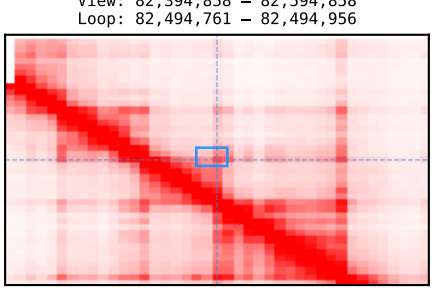


Alt (hap2) Σ=3

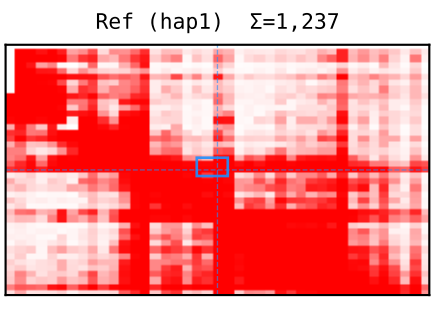


Unphased Σ=5,611
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

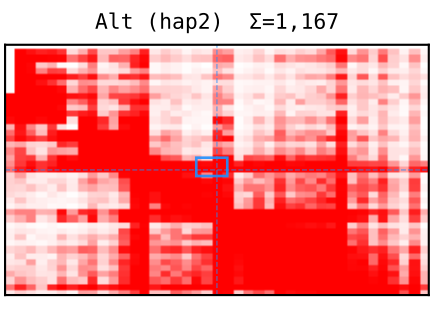
Loop 3
chr17:82,458,465-82,458,714
× chr17:82,494,761-82,494,956
dist: 36 kb
R/A: 1.06× log2FC: +0.08



Ref (hap1) Σ=1,237

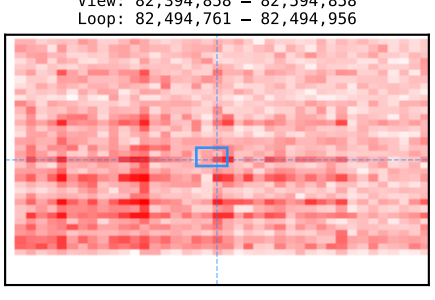


Alt (hap2) Σ=1,167

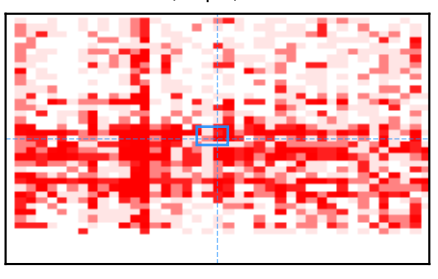


Unphased Σ=75
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

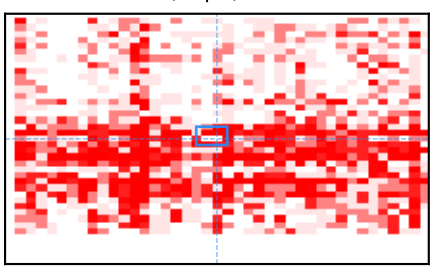
Loop 4
chr17:81,666,535-81,666,760
× chr17:82,494,761-82,494,956
dist: 828 kb
R/A: 1.00× log2FC: +0.00



Ref (hap1) Σ=11

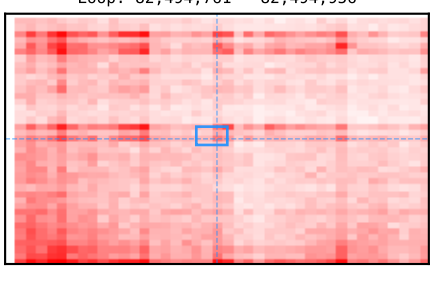


Alt (hap2) Σ=11

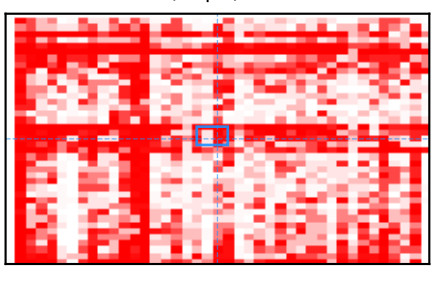


Unphased Σ=233
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

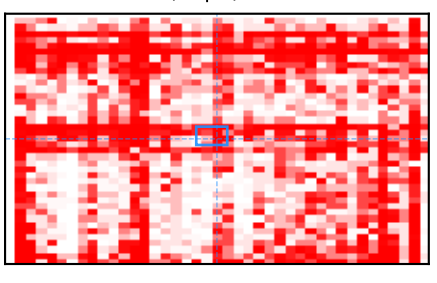
Loop 5
chr17:82,106,337-82,106,480
× chr17:82,494,761-82,494,956
dist: 388 kb
R/A: 0.74× log2FC: -0.43



Ref (hap1) Σ=23

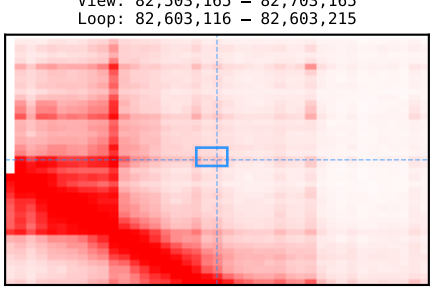


Alt (hap2) Σ=31

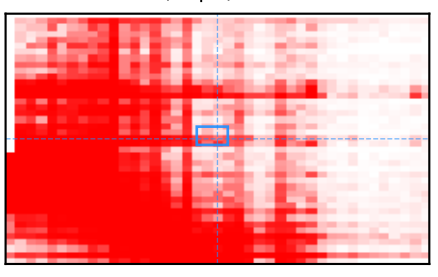


Unphased Σ=1,070
View: 82,503,165 – 82,703,165
Loop: 82,603,116 – 82,603,215

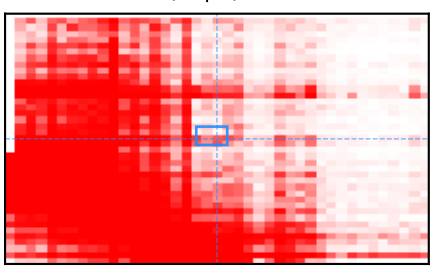
Loop 6
chr17:82,494,761-82,494,956
× chr17:82,603,116-82,603,215
dist: 108 kb
R/A: 1.25× log2FC: +0.32



Ref (hap1) Σ=105

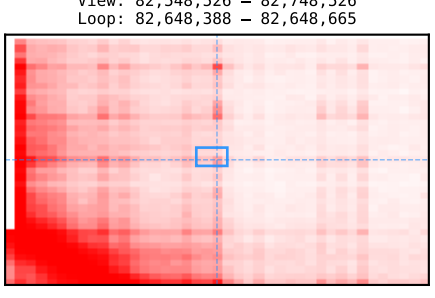


Alt (hap2) Σ=84

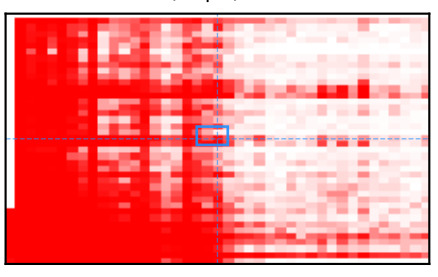


Unphased Σ=747
View: 82,548,526 – 82,748,526
Loop: 82,648,388 – 82,648,665

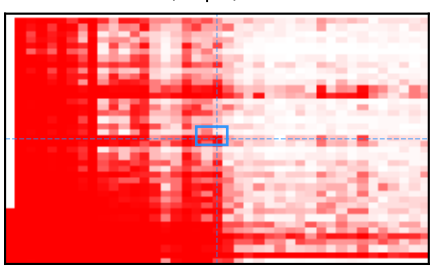
Loop 7
chr17:82,494,761-82,494,956
× chr17:82,648,388-82,648,665
dist: 154 kb
R/A: 0.78× log2FC: -0.36



Ref (hap1) Σ=63

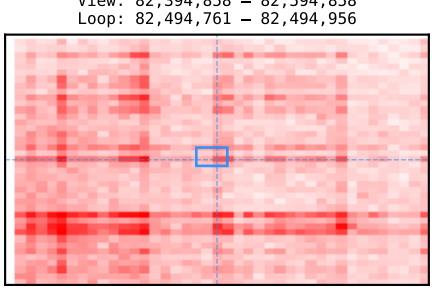


Alt (hap2) Σ=81

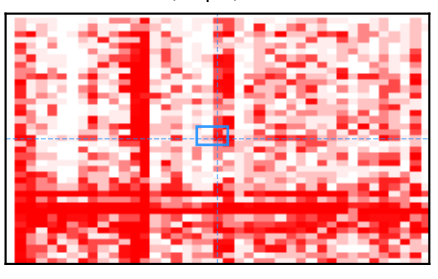


Unphased Σ=187
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

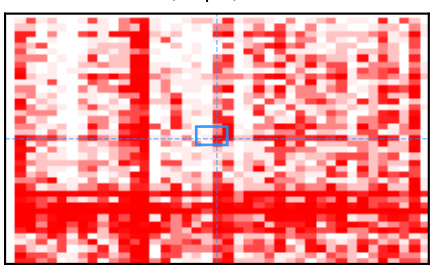
Loop 8
chr17:81,977,441-81,977,672
× chr17:82,494,761-82,494,956
dist: 517 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=14

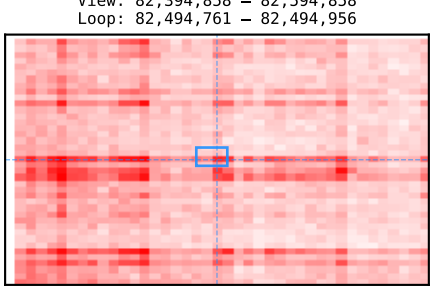


Alt (hap2) Σ=15

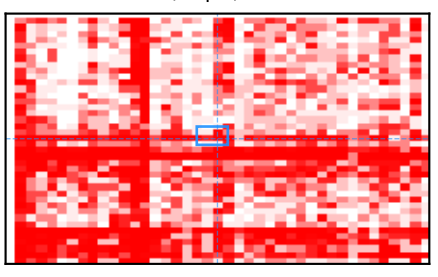


Unphased Σ=262
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

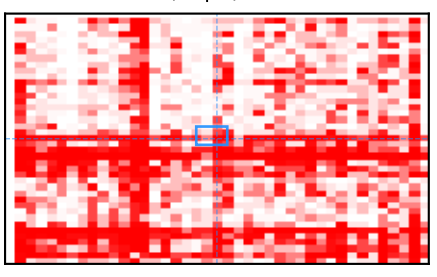
Loop 9
chr17:82,022,908-82,023,391
× chr17:82,494,761-82,494,956
dist: 472 kb
R/A: 0.85× log2FC: -0.23



Ref (hap1) Σ=29

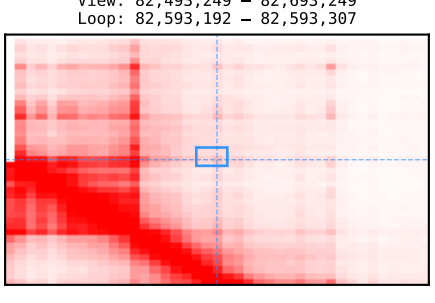


Alt (hap2) Σ=34

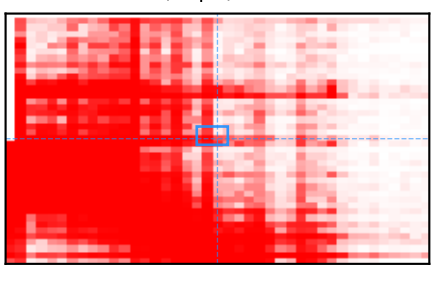


Unphased Σ=1,313
View: 82,493,249 – 82,693,249
Loop: 82,593,192 – 82,593,307

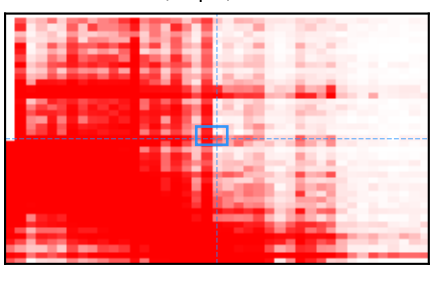
Loop 10
chr17:82,494,761-82,494,956
× chr17:82,593,192-82,593,307
dist: 98 kb
R/A: 0.91× log2FC: -0.14



Ref (hap1) Σ=183

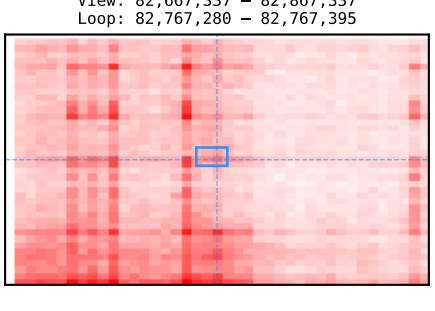


Alt (hap2) Σ=201

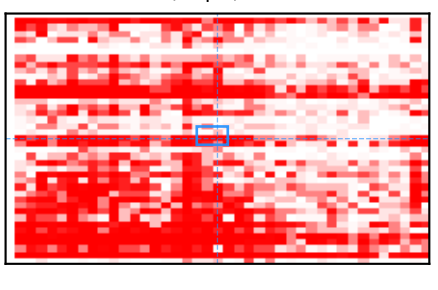


Unphased Σ=432
View: 82,667,337 – 82,867,337
Loop: 82,767,280 – 82,767,395

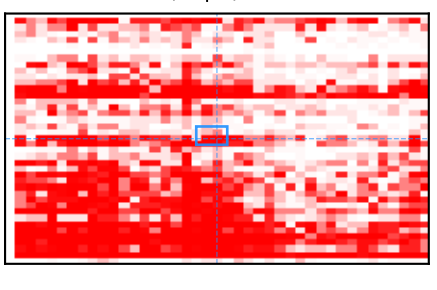
Loop 11
chr17:82,494,761-82,494,956
× chr17:82,767,280-82,767,395
dist: 272 kb
R/A: 0.81× log2FC: -0.31



Ref (hap1) Σ=29

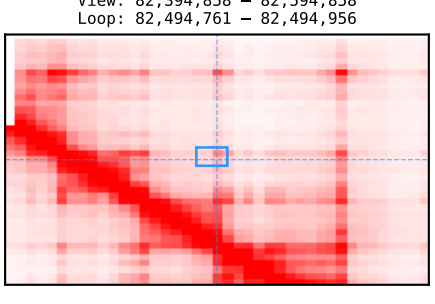


Alt (hap2) Σ=36

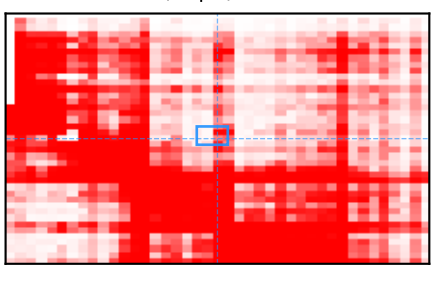


Unphased Σ=2,516
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956

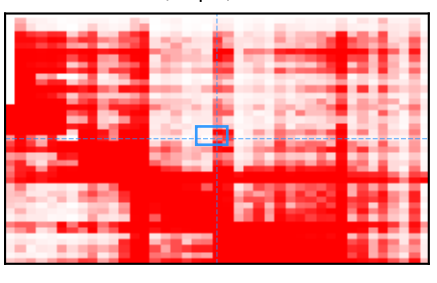
Loop 12
chr17:82,424,690-82,424,900
× chr17:82,494,761-82,494,956
dist: 70 kb
R/A: 0.95× log2FC: -0.08



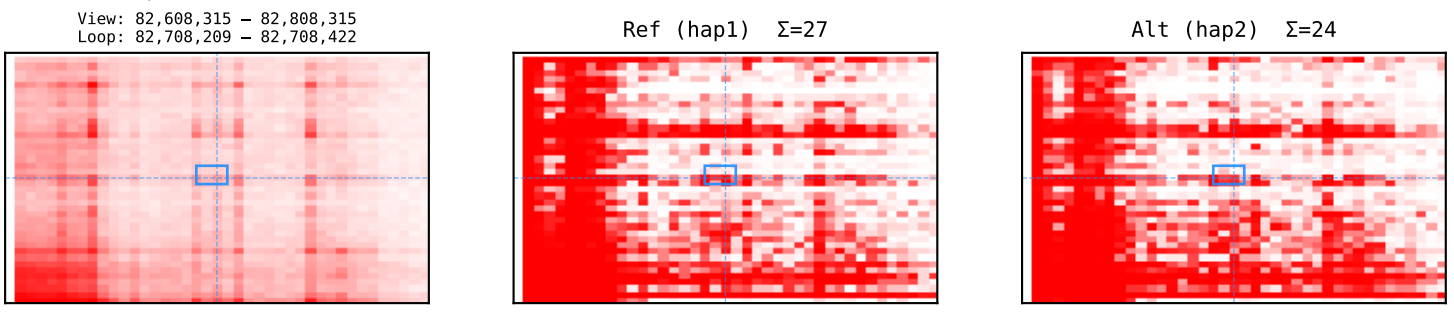
Ref (hap1) Σ=167



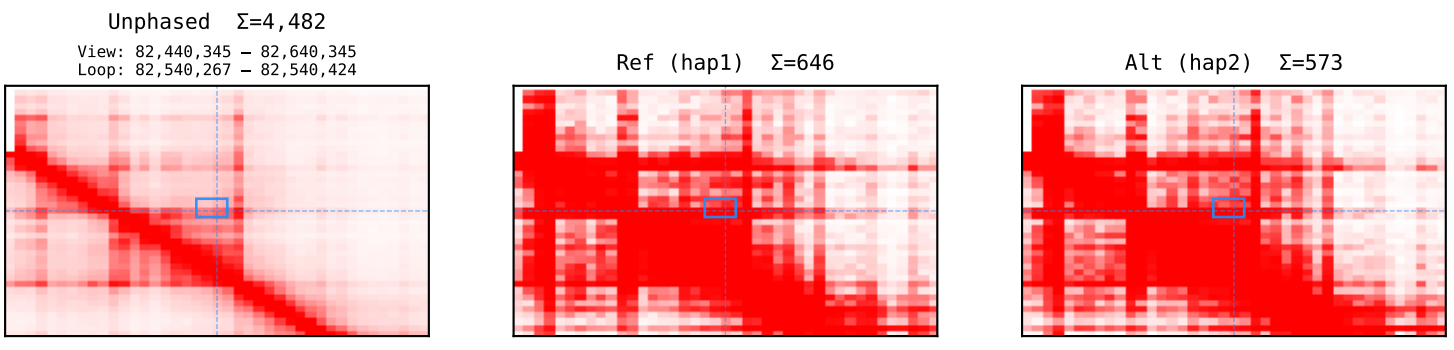
Alt (hap2) Σ=176



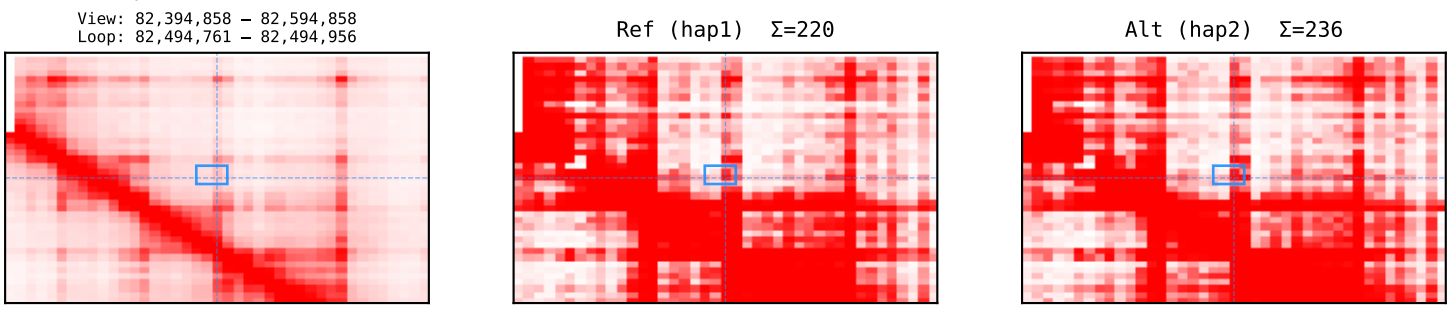
Loop 1
chr17:82,494,761-82,494,956
× chr17:82,708,209-82,708,422
dist: 213 kb
R/A: 1.12× log2FC: +0.17



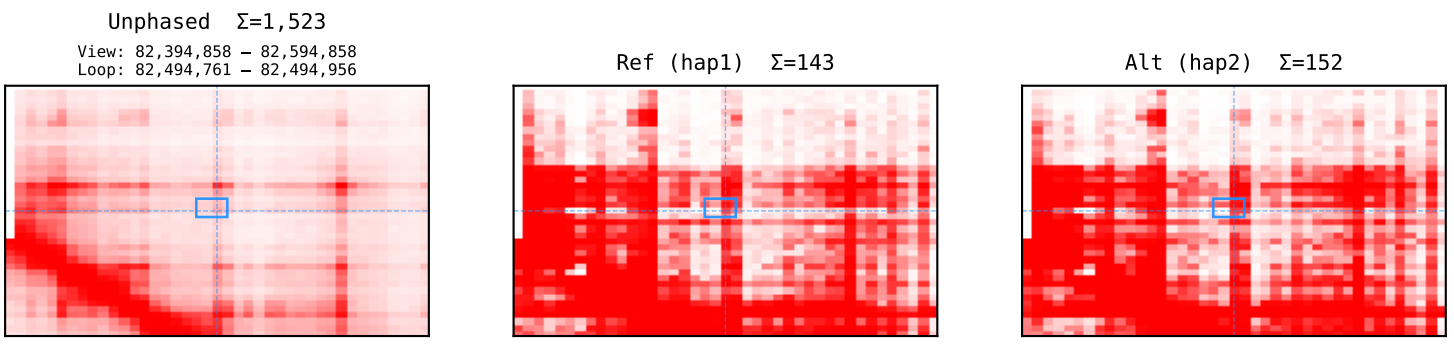
Loop 2
chr17:82,494,761-82,494,956
× chr17:82,540,267-82,540,424
dist: 45 kb
R/A: 1.13× log2FC: +0.17



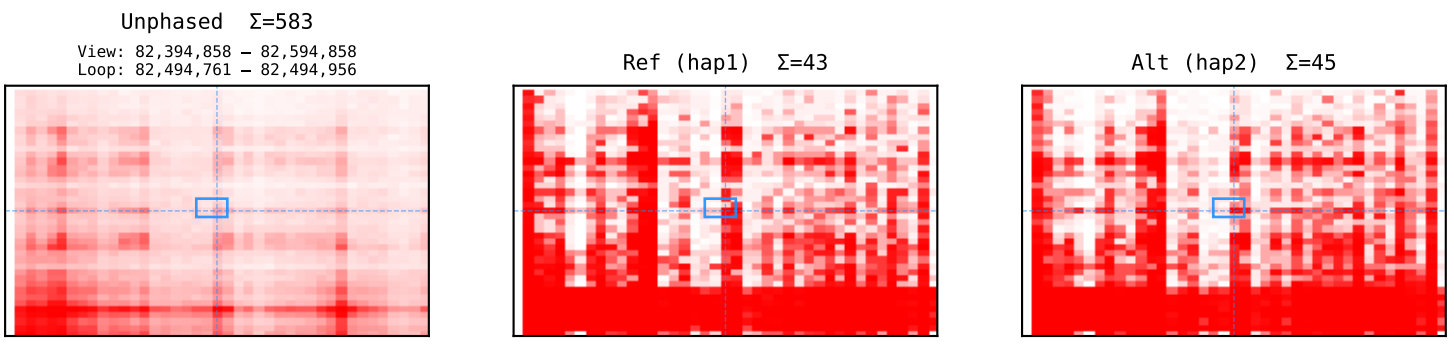
Loop 3
chr17:82,430,010-82,430,029
× chr17:82,494,761-82,494,956
dist: 65 kb
R/A: 0.93× log2FC: -0.10



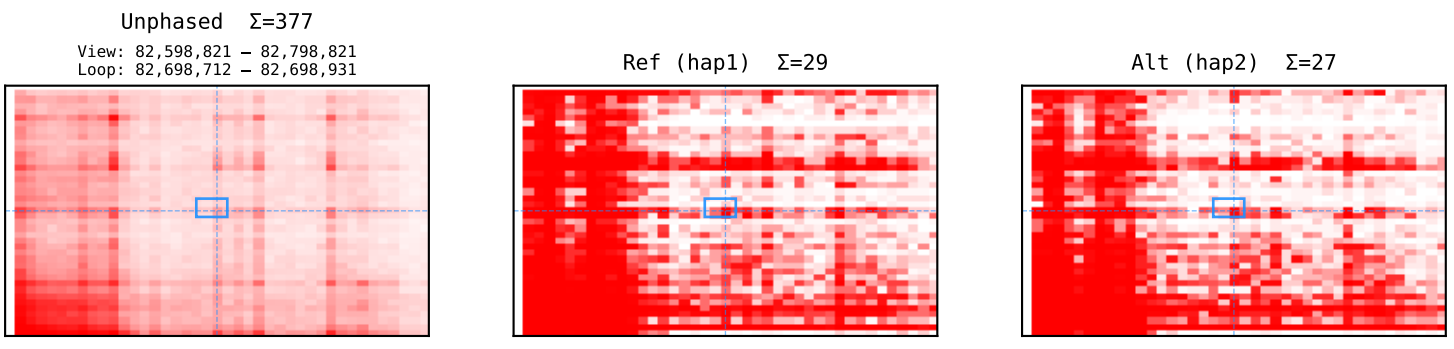
Loop 4
chr17:82,370,556-82,370,689
× chr17:82,494,761-82,494,956
dist: 124 kb
R/A: 0.94× log2FC: -0.09



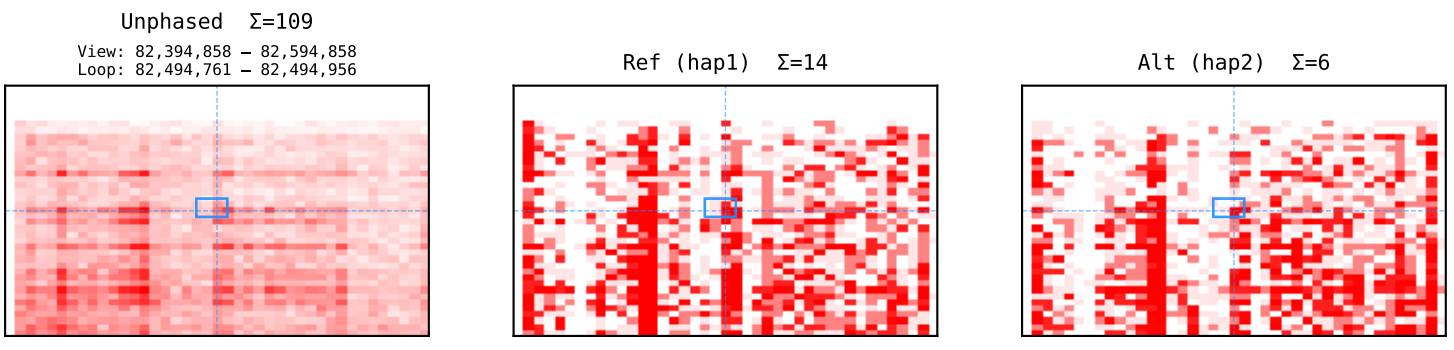
Loop 5
chr17:82,273,289-82,273,870
× chr17:82,494,761-82,494,956
dist: 221 kb
R/A: 0.96× log2FC: -0.07



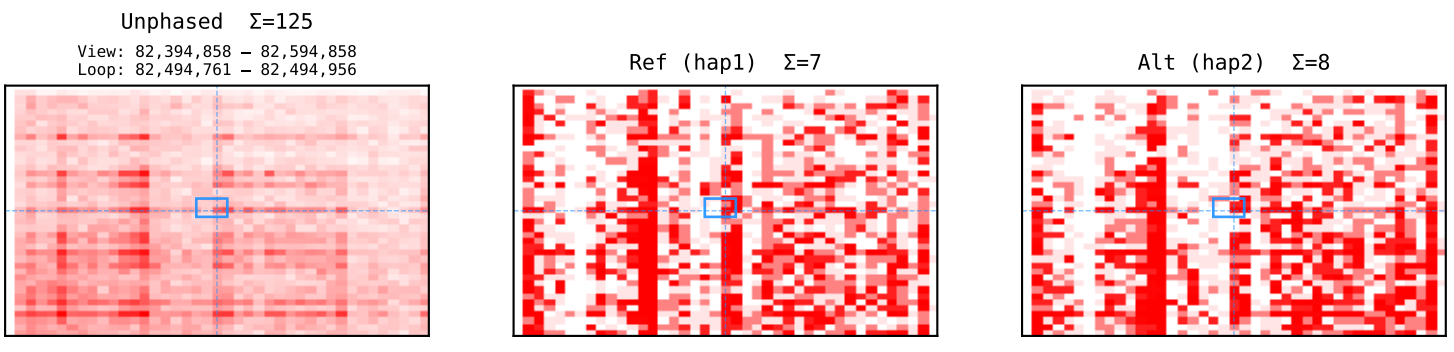
Loop 6
chr17:82,494,761-82,494,956
× chr17:82,698,712-82,698,931
dist: 204 kb
R/A: 1.07× log2FC: +0.10



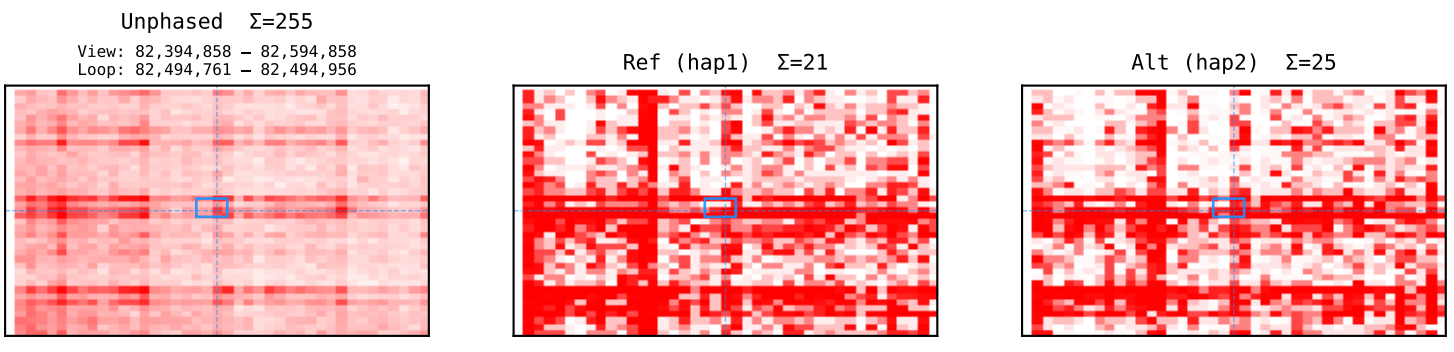
Loop 7
chr17:81,860,608-81,861,067
× chr17:82,494,761-82,494,956
dist: 634 kb
R/A: 2.33× log2FC: +1.22



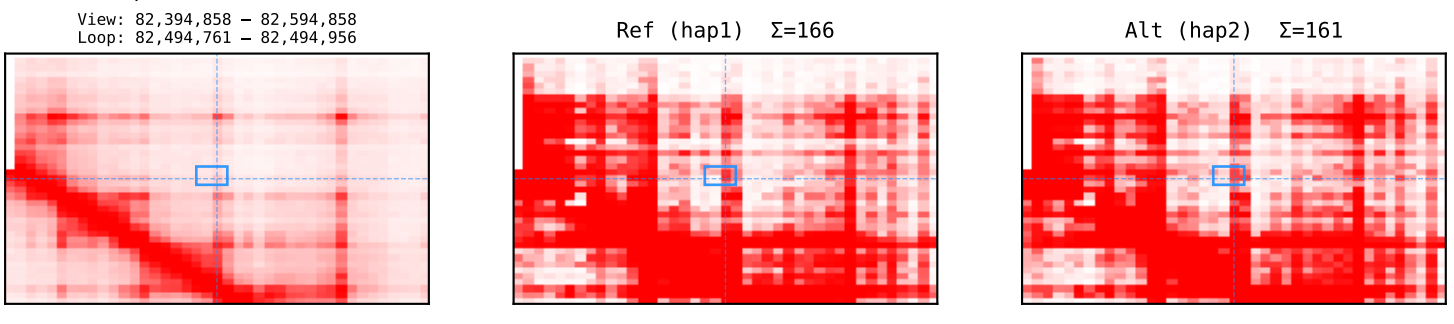
Loop 8
chr17:81,891,508-81,891,827
× chr17:82,494,761-82,494,956
dist: 603 kb
R/A: 0.88× log2FC: -0.19



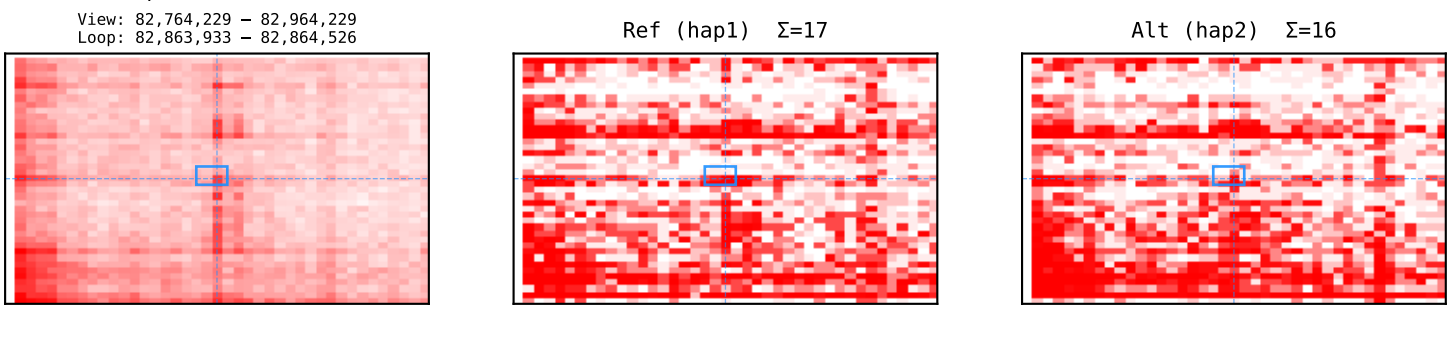
Loop 9
chr17:82,030,640-82,030,659
× chr17:82,494,761-82,494,956
dist: 464 kb
R/A: 0.84× log2FC: -0.25



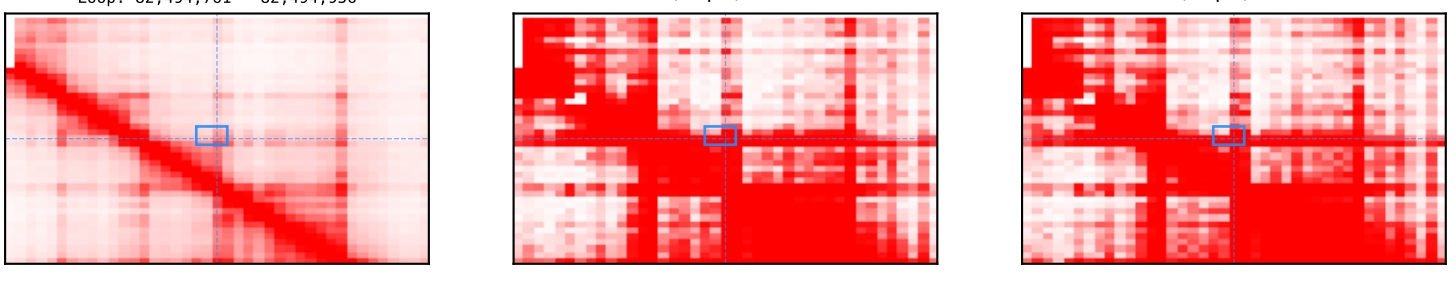
Loop 10
chr17:82,400,944-82,401,087
× chr17:82,494,761-82,494,956
dist: 94 kb
R/A: 1.03× log2FC: +0.04



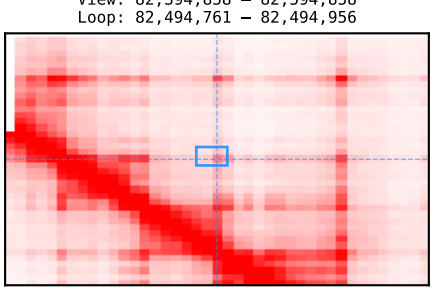
Loop 11
chr17:82,494,761-82,494,956
× chr17:82,863,933-82,864,526
dist: 369 kb
R/A: 1.06× log2FC: +0.09



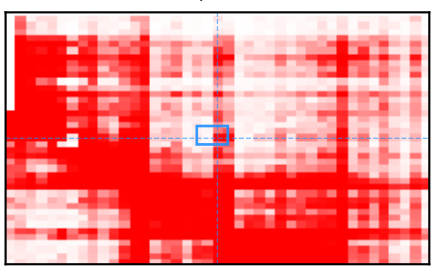
Loop 12
chr17:82,450,600-82,450,949
× chr17:82,494,761-82,494,956
dist: 44 kb
R/A: 1.16× log2FC: +0.22



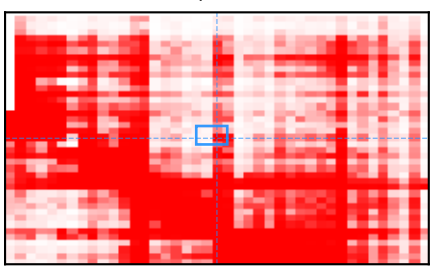
Unphased Σ=2,289
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=170

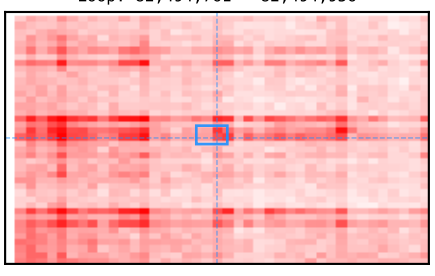


Alt (hap2) Σ=178

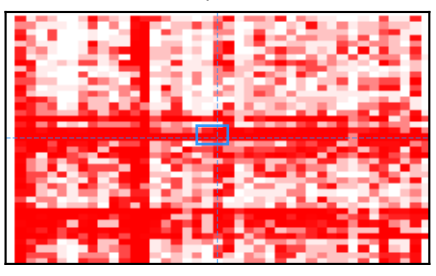


Loop 1
chr17:82,418,242-82,418,819
× chr17:82,494,761-82,494,956
dist: 76 kb
R/A: 0.96× log2FC: -0.07

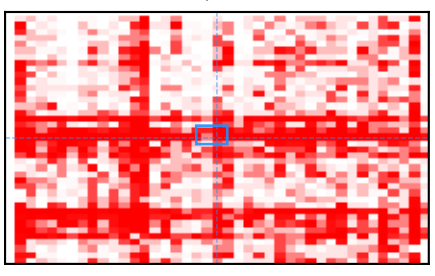
Unphased Σ=319
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=48

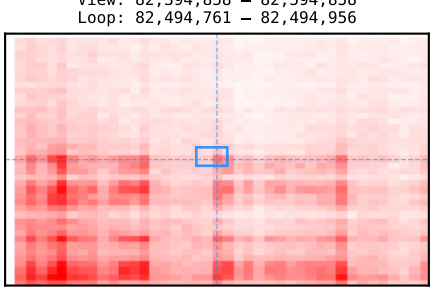


Alt (hap2) Σ=48

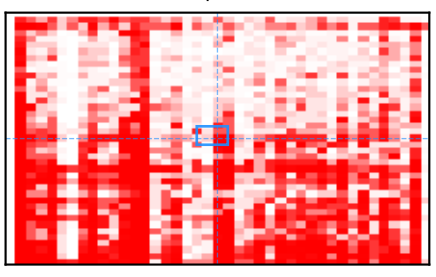


Loop 2
chr17:82,037,345-82,037,948
× chr17:82,494,761-82,494,956
dist: 457 kb
R/A: 1.00× log2FC: +0.00

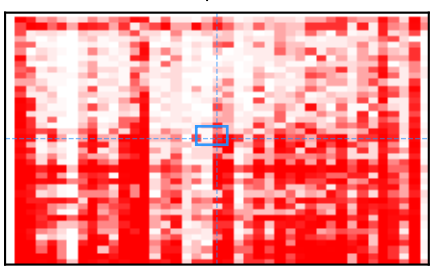
Unphased Σ=465
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=35

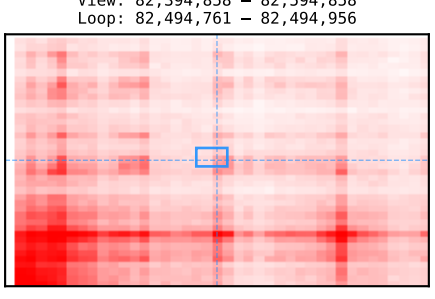


Alt (hap2) Σ=18

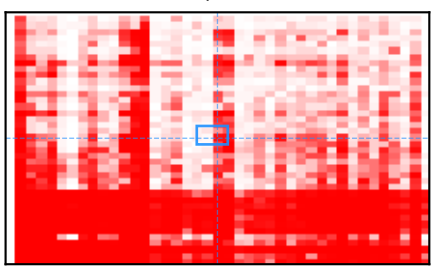


Loop 3
chr17:82,205,996-82,206,101
× chr17:82,494,761-82,494,956
dist: 289 kb
R/A: 1.94× log2FC: +0.96

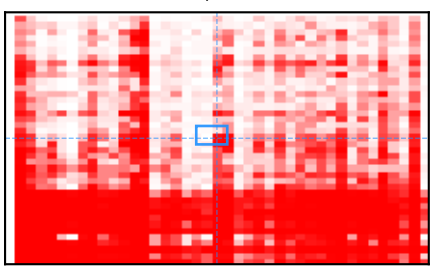
Unphased Σ=666
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=44

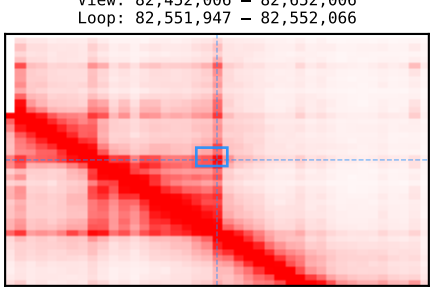


Alt (hap2) Σ=36

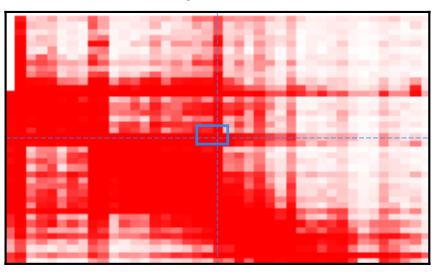


Loop 4
chr17:82,292,780-82,292,983
× chr17:82,494,761-82,494,956
dist: 202 kb
R/A: 1.22× log2FC: +0.29

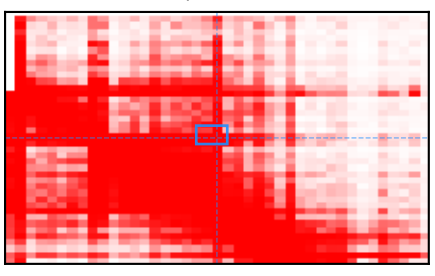
Unphased Σ=8,635
View: 82,452,006 – 82,652,006
Loop: 82,551,947 – 82,552,066



Ref (hap1) Σ=1,101

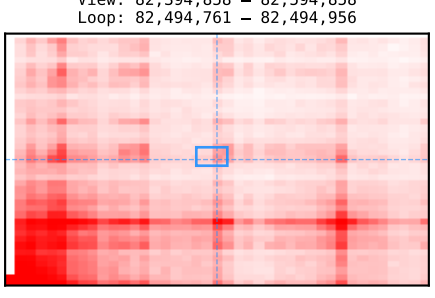


Alt (hap2) Σ=1,081

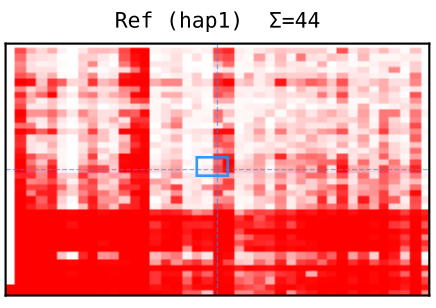


Loop 5
chr17:82,494,761-82,494,956
× chr17:82,551,947-82,552,066
dist: 57 kb
R/A: 1.02× log2FC: +0.03

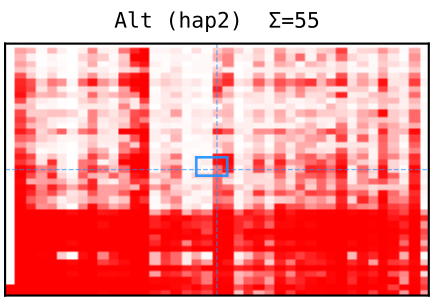
Unphased Σ=973
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=44

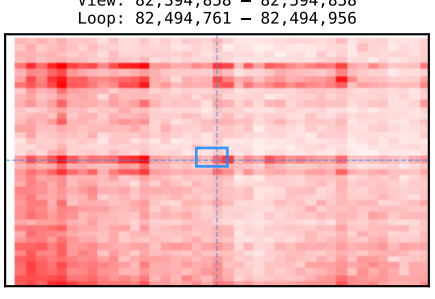


Alt (hap2) Σ=55

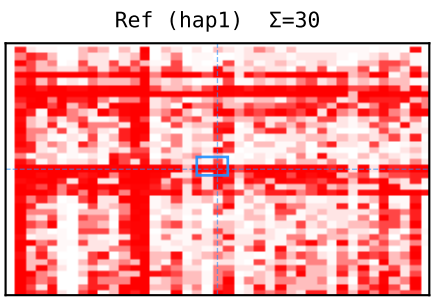


Loop 6
chr17:82,300,935-82,301,070
× chr17:82,494,761-82,494,956
dist: 194 kb
R/A: 0.80× log2FC: -0.32

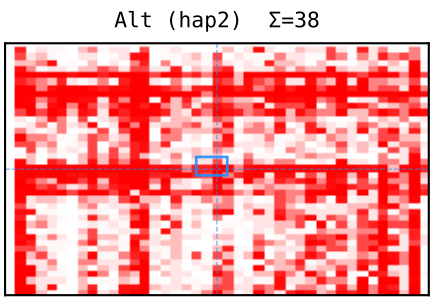
Unphased Σ=192
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=30

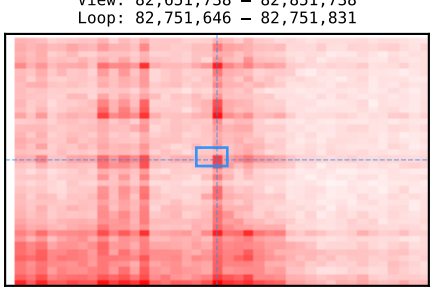


Alt (hap2) Σ=38

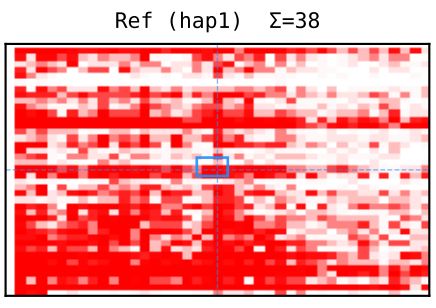


Loop 7
chr17:82,098,227-82,098,480
× chr17:82,494,761-82,494,956
dist: 397 kb
R/A: 0.79× log2FC: -0.34

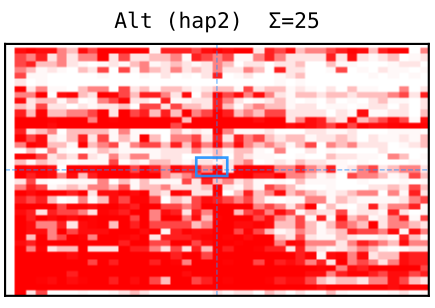
Unphased Σ=431
View: 82,651,738 – 82,851,738
Loop: 82,751,646 – 82,751,831



Ref (hap1) Σ=38

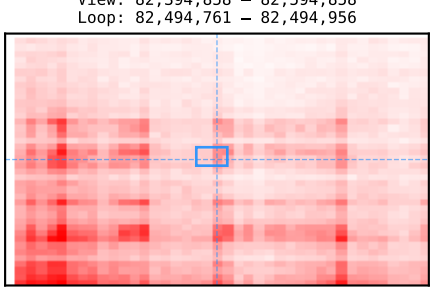


Alt (hap2) Σ=25

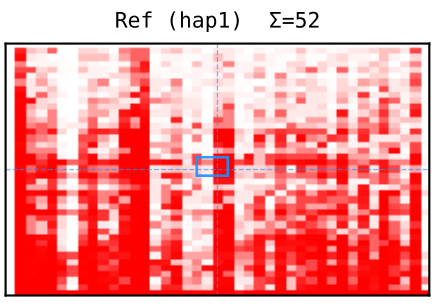


Loop 8
chr17:82,494,761-82,494,956
× chr17:82,751,646-82,751,831
dist: 257 kb
R/A: 1.52× log2FC: +0.60

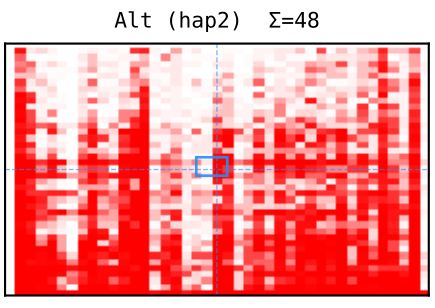
Unphased Σ=652
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=52

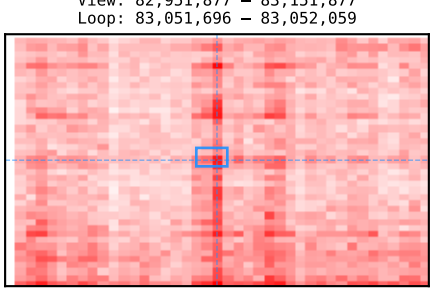


Alt (hap2) Σ=48

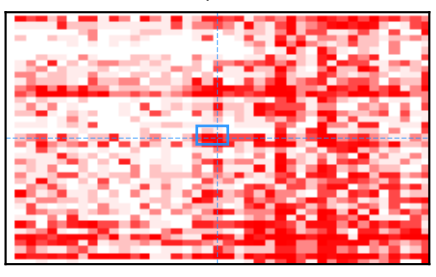


Loop 9
chr17:82,236,546-82,236,597
× chr17:82,494,761-82,494,956
dist: 258 kb
R/A: 1.08× log2FC: +0.12

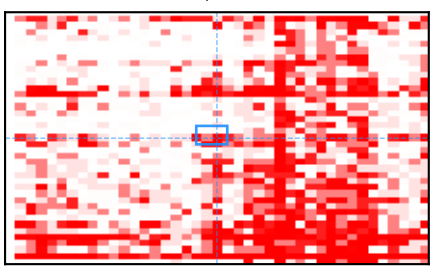
Unphased Σ=190
View: 82,951,877 – 83,151,877
Loop: 83,051,696 – 83,052,059



Ref (hap1) Σ=23

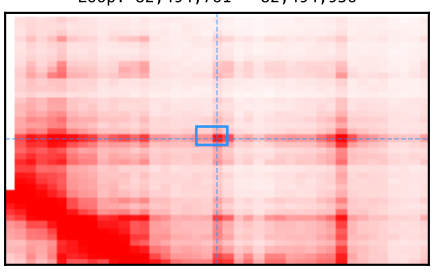


Alt (hap2) Σ=11

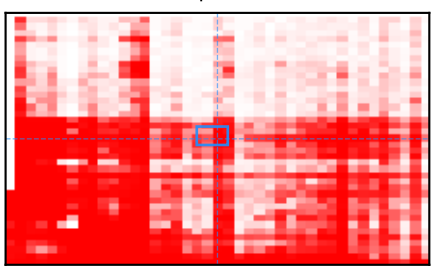


Loop 10
chr17:82,494,761-82,494,956
× chr17:83,051,696-83,052,059
dist: 557 kb
R/A: 2.09× log2FC: +1.06

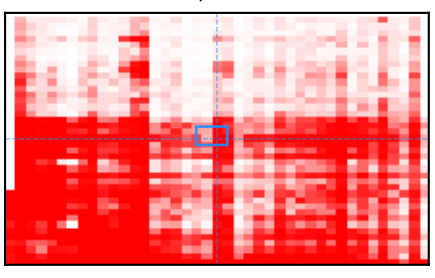
Unphased Σ=2,946
View: 82,394,858 – 82,594,858
Loop: 82,494,761 – 82,494,956



Ref (hap1) Σ=290



Alt (hap2) Σ=313



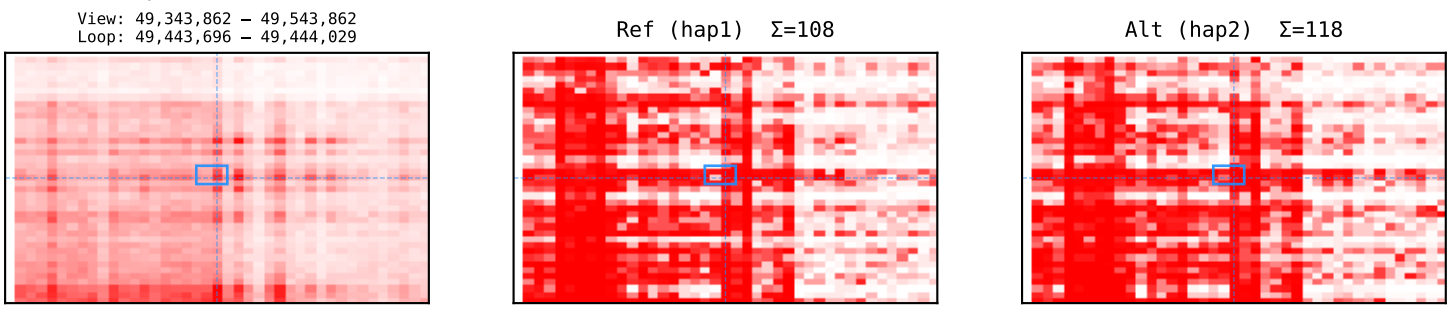
Loop 11
chr17:82,351,923-82,352,114
× chr17:82,494,761-82,494,956
dist: 143 kb
R/A: 0.93× log2FC: -0.11

Unphased Σ=718
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029

Ref (hap1) Σ=108

Alt (hap2) Σ=118

Loop 1
chr19:49,118,733-49,119,480
× chr19:49,443,696-49,444,029
dist: 325 kb
R/A: 0.92× log2FC: -0.13

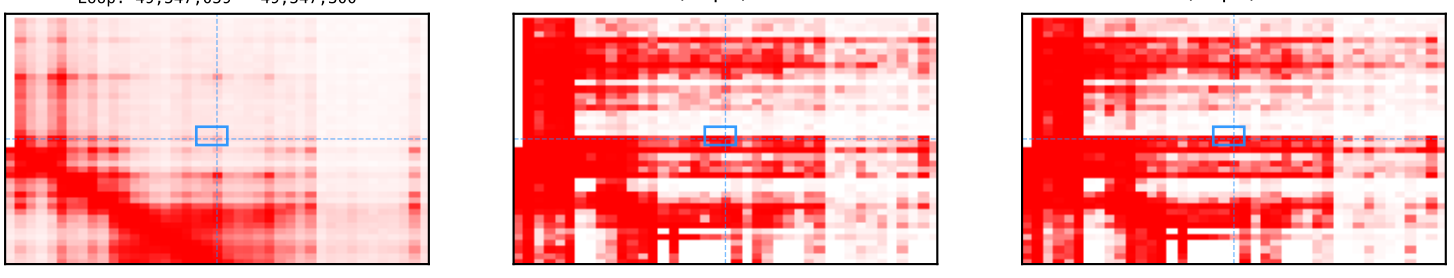


Unphased Σ=1,091
View: 49,447,179 – 49,647,179
Loop: 49,547,059 – 49,547,300

Ref (hap1) Σ=93

Alt (hap2) Σ=81

Loop 2
chr19:49,443,696-49,444,029
× chr19:49,547,059-49,547,300
dist: 103 kb
R/A: 1.15× log2FC: +0.20

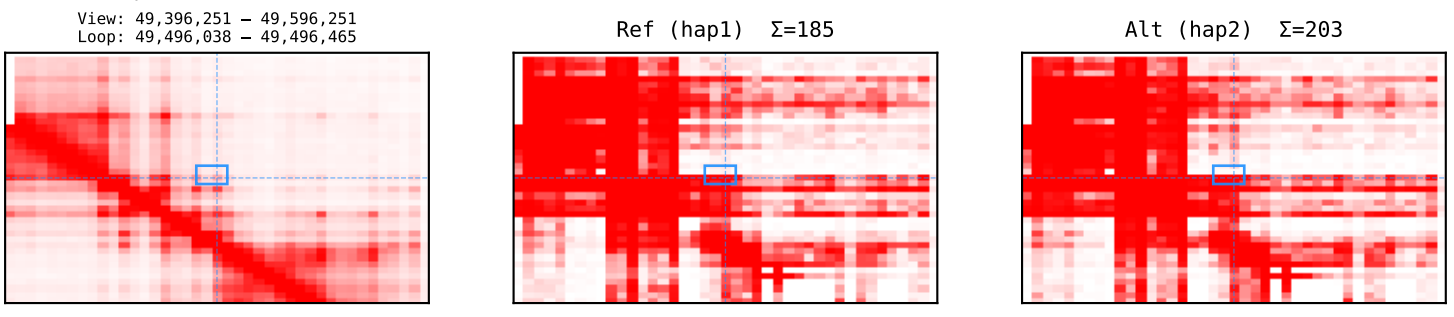


Unphased Σ=1,854
View: 49,396,251 – 49,596,251
Loop: 49,496,038 – 49,496,465

Ref (hap1) Σ=185

Alt (hap2) Σ=203

Loop 3
chr19:49,443,696-49,444,029
× chr19:49,496,038-49,496,465
dist: 52 kb
R/A: 0.91× log2FC: -0.13

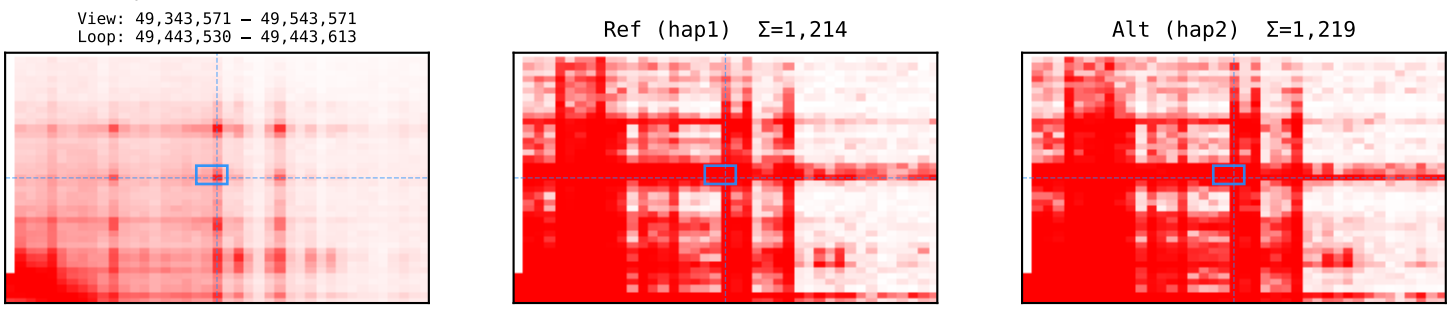


Unphased Σ=3,634
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613

Ref (hap1) Σ=1,214

Alt (hap2) Σ=1,219

Loop 4
chr19:49,266,508-49,266,589
× chr19:49,443,530-49,443,613
dist: 177 kb
R/A: 1.00× log2FC: -0.01

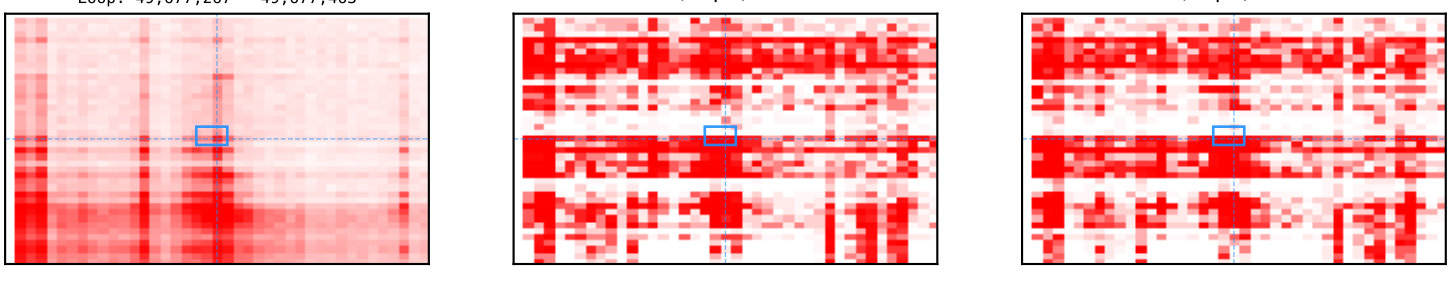


Unphased Σ=1,020
View: 49,577,365 – 49,777,365
Loop: 49,677,267 – 49,677,463

Ref (hap1) Σ=72

Alt (hap2) Σ=102

Loop 5
chr19:49,443,696-49,444,029
× chr19:49,677,267-49,677,463
dist: 234 kb
R/A: 0.71× log2FC: -0.50

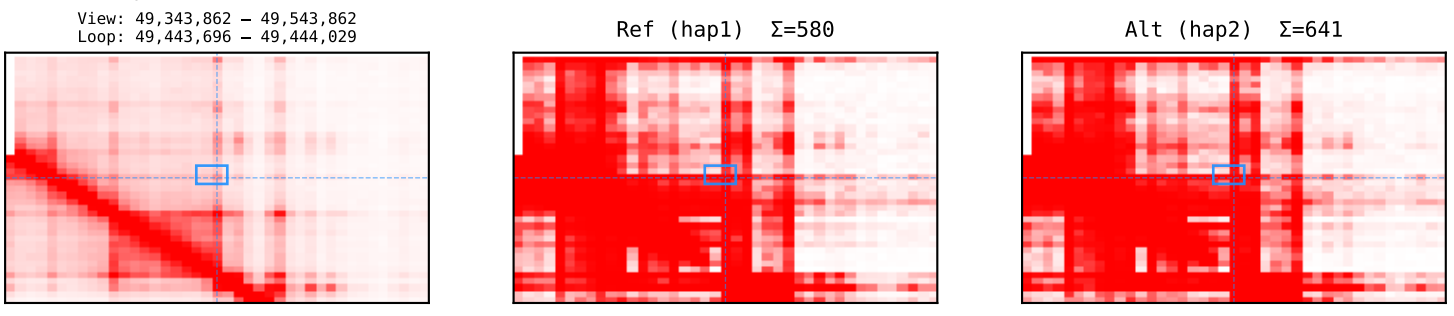


Unphased Σ=3,711
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029

Ref (hap1) Σ=580

Alt (hap2) Σ=641

Loop 6
chr19:49,363,371-49,363,800
× chr19:49,443,696-49,444,029
dist: 80 kb
R/A: 0.90× log2FC: -0.14

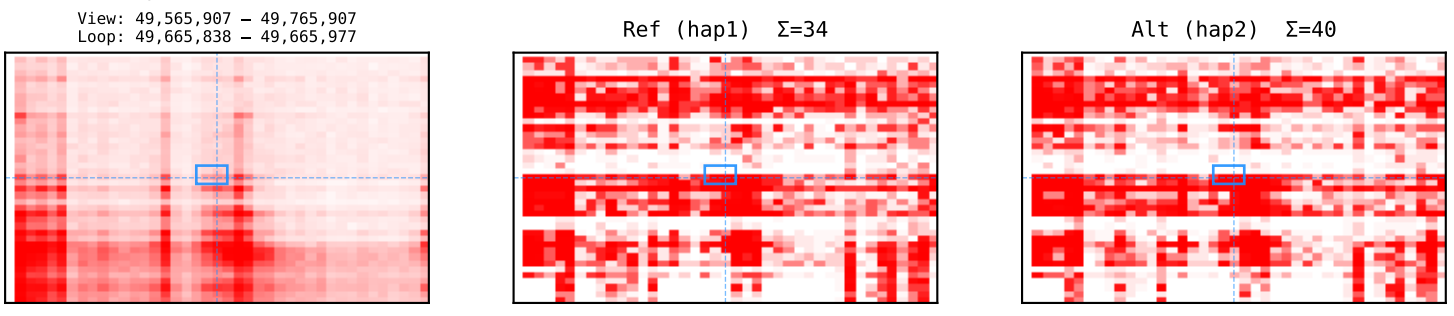


Unphased Σ=574
View: 49,565,907 – 49,765,907
Loop: 49,665,838 – 49,665,977

Ref (hap1) Σ=34

Alt (hap2) Σ=40

Loop 7
chr19:49,443,696-49,444,029
× chr19:49,665,838-49,665,977
dist: 222 kb
R/A: 0.85× log2FC: -0.23

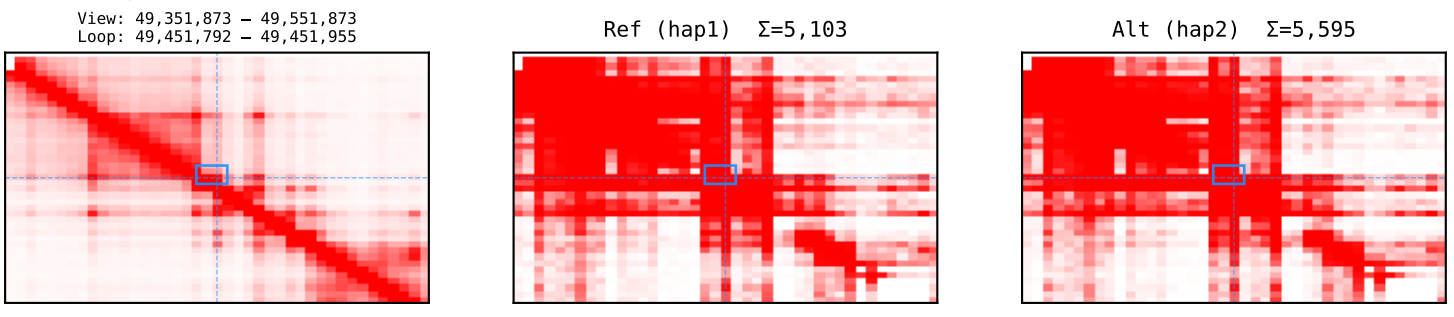


Unphased Σ=21,765
View: 49,351,873 – 49,551,873
Loop: 49,451,792 – 49,451,955

Ref (hap1) Σ=5,103

Alt (hap2) Σ=5,595

Loop 8
chr19:49,443,696-49,444,029
× chr19:49,451,792-49,451,955
dist: 8 kb
R/A: 0.91× log2FC: -0.13

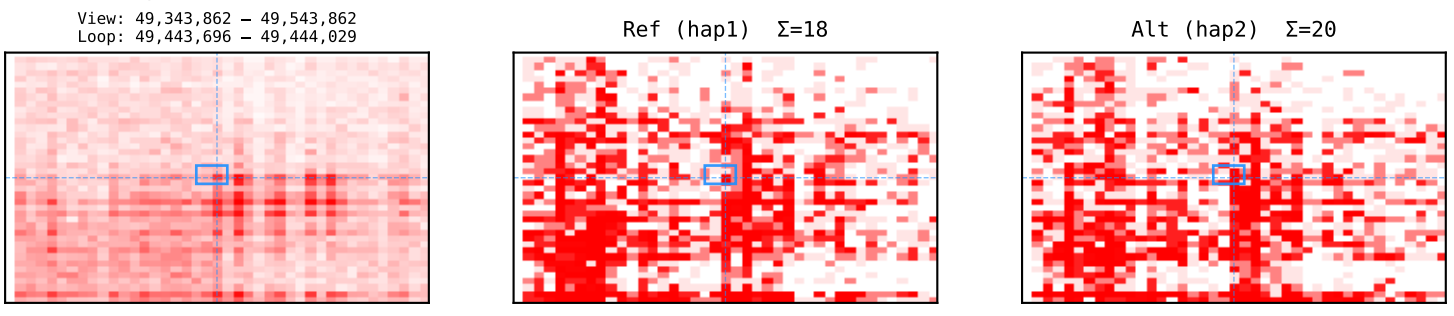


Unphased Σ=142
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029

Ref (hap1) Σ=18

Alt (hap2) Σ=20

Loop 9
chr19:48,619,150-48,619,533
× chr19:49,443,696-49,444,029
dist: 825 kb
R/A: 0.90× log2FC: -0.15

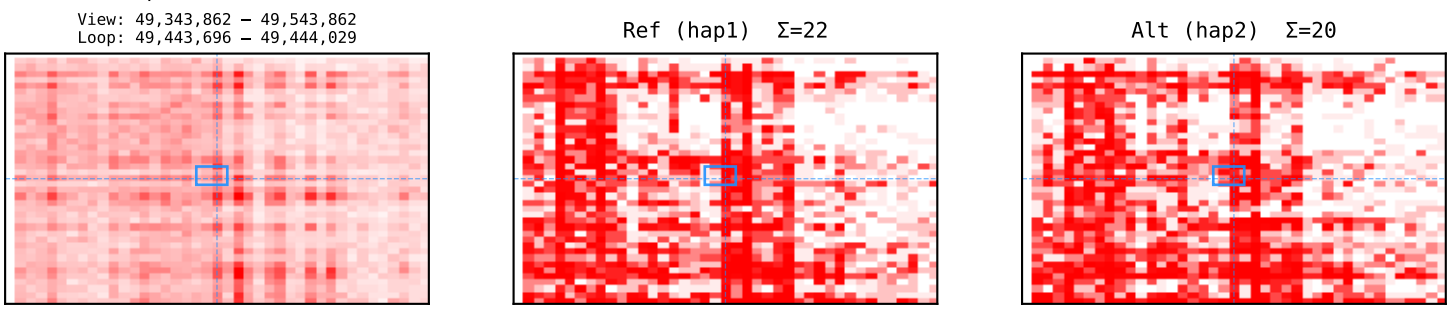


Unphased Σ=174
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029

Ref (hap1) Σ=22

Alt (hap2) Σ=20

Loop 10
chr19:48,795,489-48,795,628
× chr19:49,443,696-49,444,029
dist: 648 kb
R/A: 1.10× log2FC: +0.14

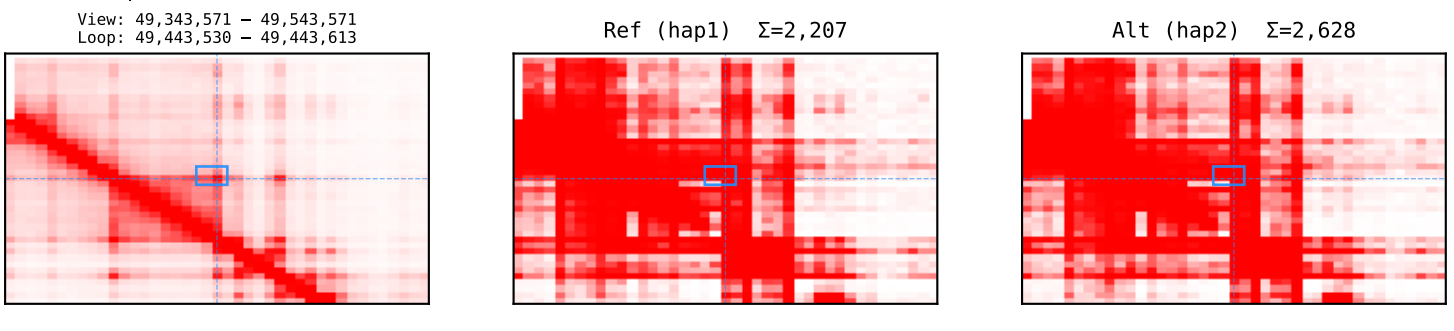


Unphased Σ=17,170
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613

Ref (hap1) Σ=2,207

Alt (hap2) Σ=2,628

Loop 11
chr19:49,391,977-49,392,116
× chr19:49,443,530-49,443,613
dist: 52 kb
R/A: 0.84× log2FC: -0.25

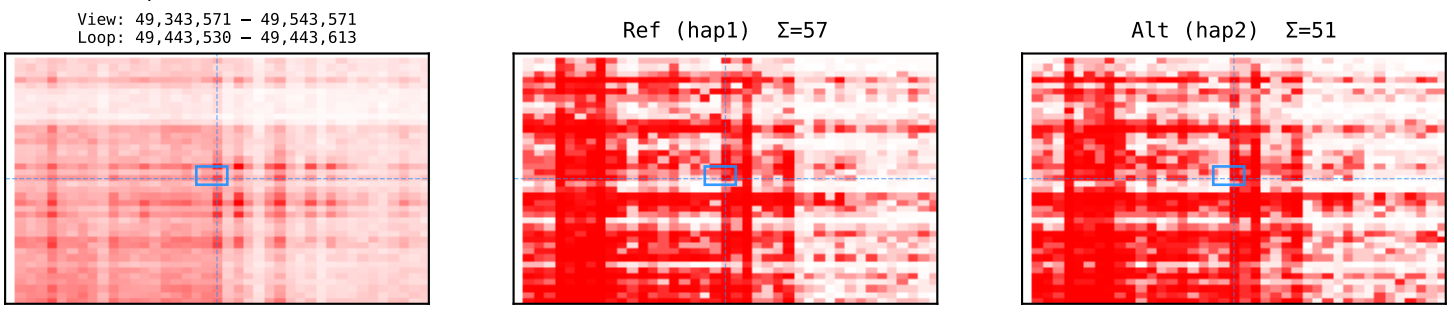


Unphased Σ=547
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613

Ref (hap1) Σ=57

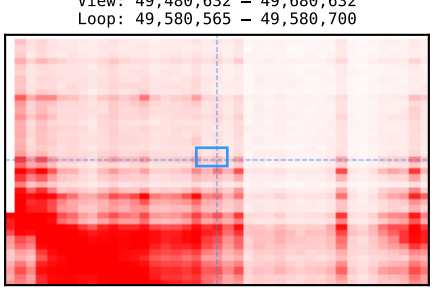
Alt (hap2) Σ=51

Loop 12
chr19:49,097,923-49,098,026
× chr19:49,443,530-49,443,613
dist: 346 kb
R/A: 1.12× log2FC: +0.16

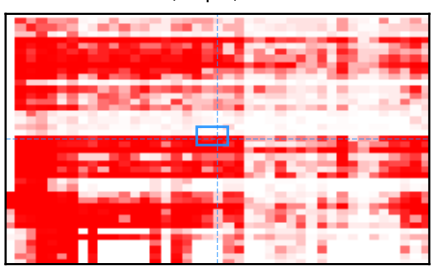


Loop 1
chr19:49,443,696-49,444,029
× chr19:49,580,565-49,580,700
dist: 137 kb
R/A: 0.85× log2FC: -0.23

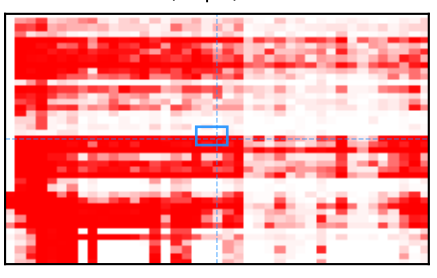
Unphased Σ=797
View: 49,480,632 – 49,680,632
Loop: 49,580,565 – 49,580,700



Ref (hap1) Σ=47

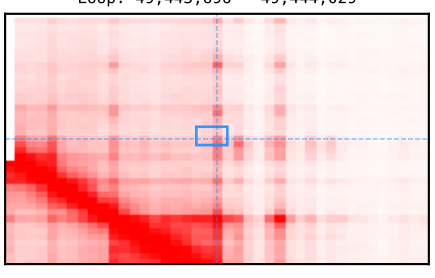


Alt (hap2) Σ=55

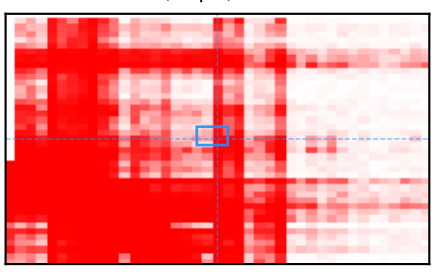


Loop 2
chr19:49,329,890-49,330,100
× chr19:49,443,696-49,444,029
dist: 114 kb
R/A: 0.77× log2FC: -0.38

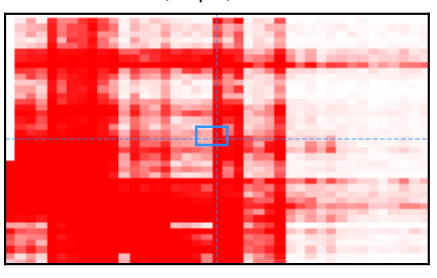
Unphased Σ=2,835
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=325

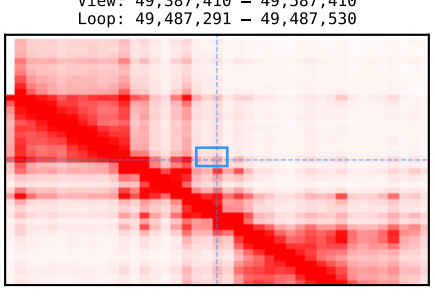


Alt (hap2) Σ=423

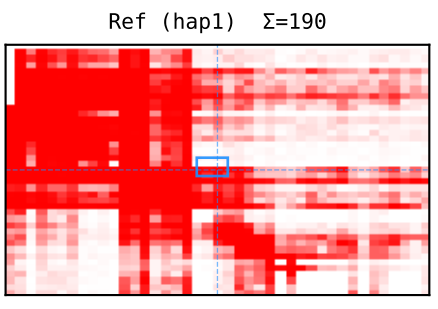


Loop 3
chr19:49,443,696-49,444,029
× chr19:49,487,291-49,487,530
dist: 44 kb
R/A: 0.97× log2FC: -0.04

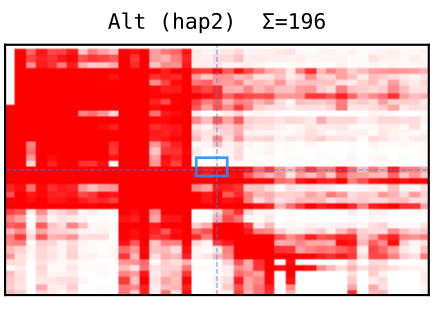
Unphased Σ=2,444
View: 49,387,410 – 49,587,410
Loop: 49,487,291 – 49,487,530



Ref (hap1) Σ=190

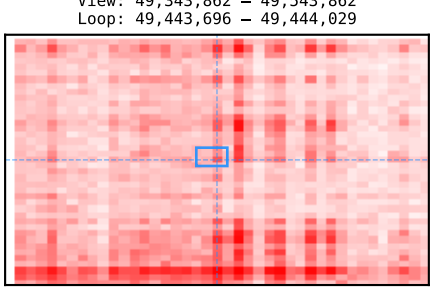


Alt (hap2) Σ=196

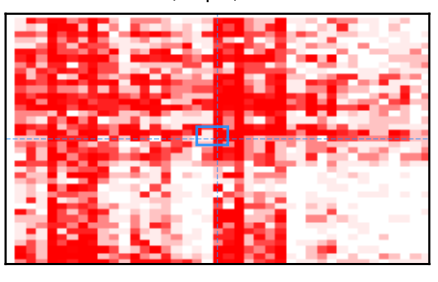


Loop 4
chr19:48,900,128-48,900,317
× chr19:49,443,696-49,444,029
dist: 544 kb
R/A: 0.66× log2FC: -0.60

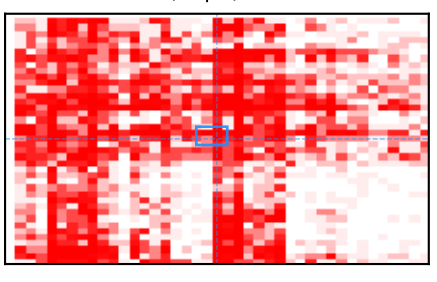
Unphased Σ=186
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=25

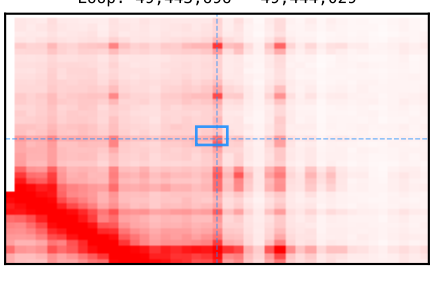


Alt (hap2) Σ=38

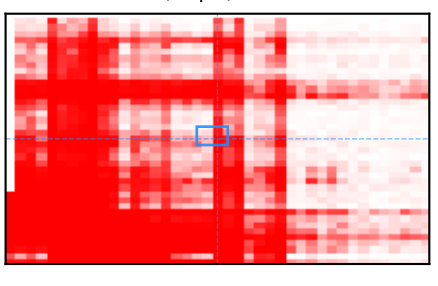


Loop 5
chr19:49,301,386-49,301,511
× chr19:49,443,696-49,444,029
dist: 142 kb
R/A: 0.75× log2FC: -0.42

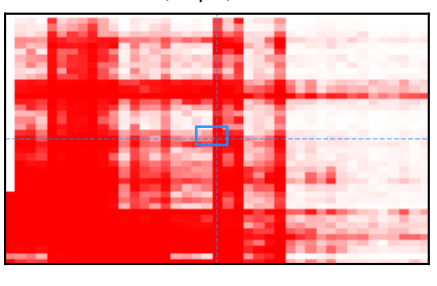
Unphased Σ=2,882
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=267

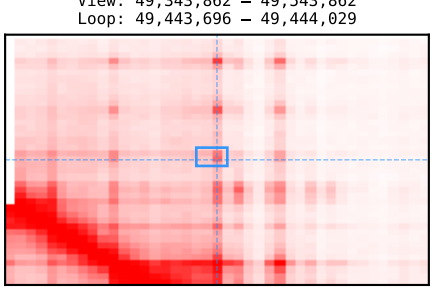


Alt (hap2) Σ=357

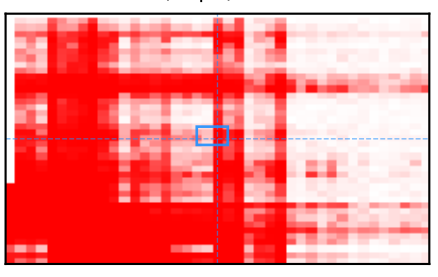


Loop 6
chr19:49,306,886-49,306,987
× chr19:49,443,696-49,444,029
dist: 137 kb
R/A: 0.62× log2FC: -0.68

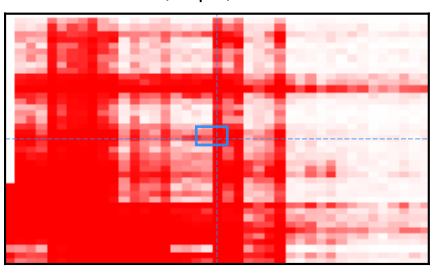
Unphased Σ=3,946
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=413

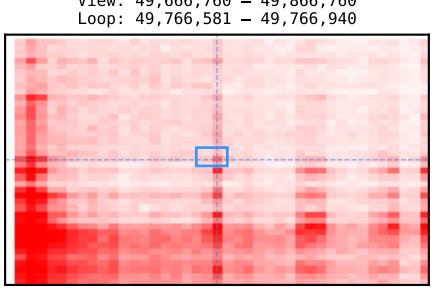


Alt (hap2) Σ=661

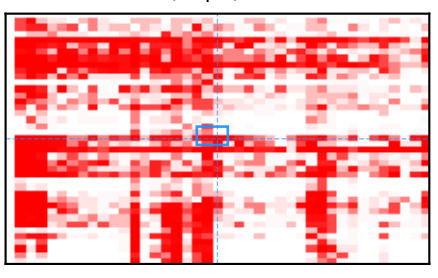


Loop 7
chr19:49,443,696-49,444,029
× chr19:49,766,581-49,766,940
dist: 323 kb
R/A: 0.89× log2FC: -0.16

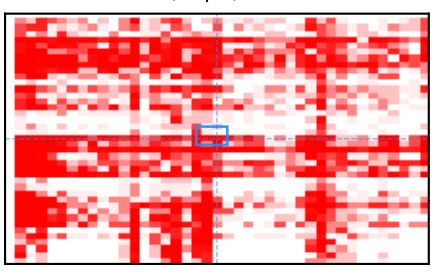
Unphased Σ=317
View: 49,666,760 – 49,866,760
Loop: 49,766,581 – 49,766,940



Ref (hap1) Σ=25

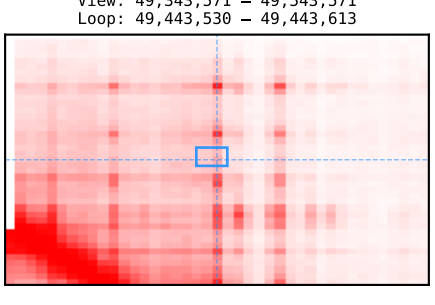


Alt (hap2) Σ=28

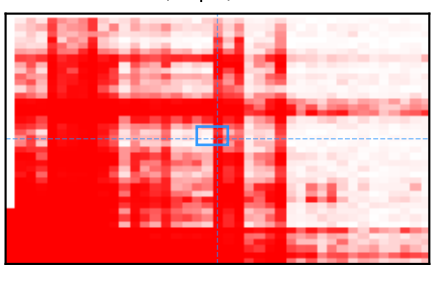


Loop 8
chr19:49,288,928-49,288,961
× chr19:49,443,530-49,443,613
dist: 155 kb
R/A: 0.80× log2FC: -0.32

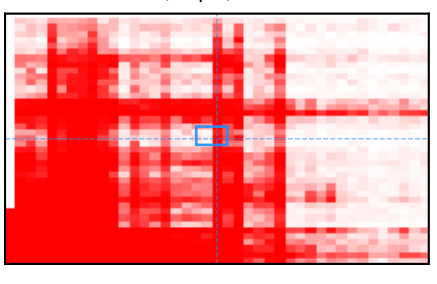
Unphased Σ=1,836
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613



Ref (hap1) Σ=121

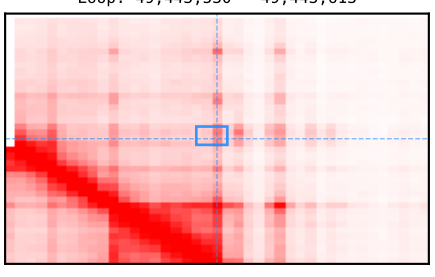


Alt (hap2) Σ=151

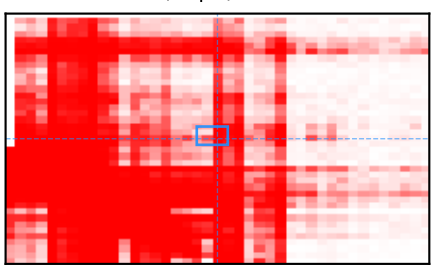


Loop 9
chr19:49,335,455-49,335,680
× chr19:49,443,530-49,443,613
dist: 108 kb
R/A: 0.89× log2FC: -0.17

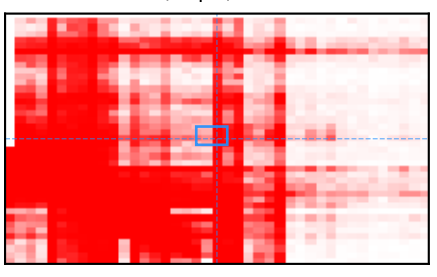
Unphased Σ=4,060
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613



Ref (hap1) Σ=510

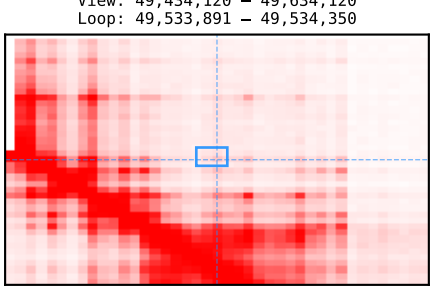


Alt (hap2) Σ=573

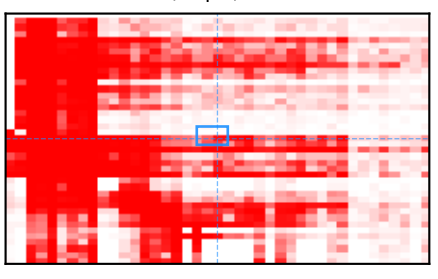


Loop 10
chr19:49,443,696-49,444,029
× chr19:49,533,891-49,534,350
dist: 90 kb
R/A: 0.76× log2FC: -0.39

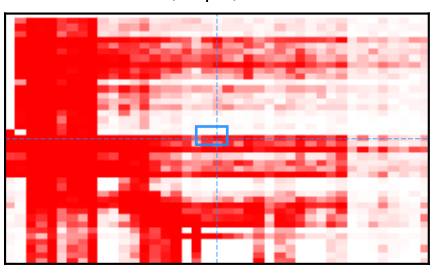
Unphased Σ=1,084
View: 49,434,120 – 49,634,120
Loop: 49,533,891 – 49,534,350



Ref (hap1) Σ=51

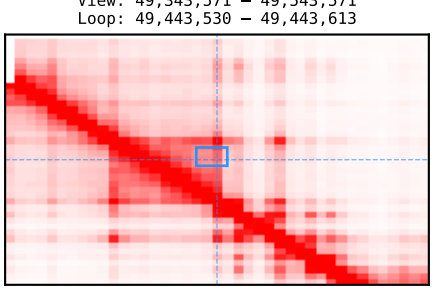


Alt (hap2) Σ=67

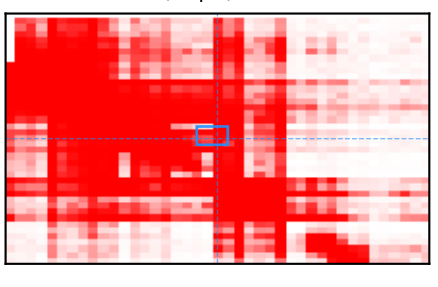


Loop 11
chr19:49,409,847-49,410,043
× chr19:49,443,530-49,443,613
dist: 34 kb
R/A: 0.80× log2FC: -0.33

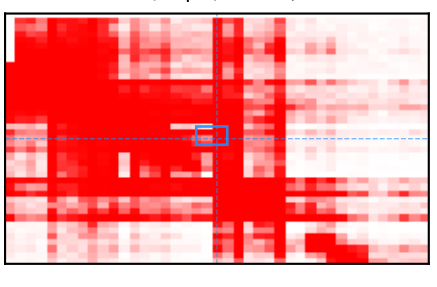
Unphased Σ=8,820
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613



Ref (hap1) Σ=826

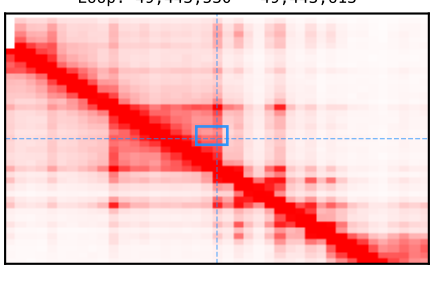


Alt (hap2) Σ=1,035

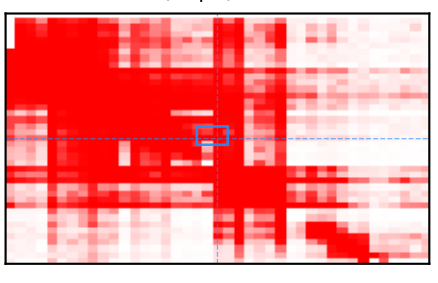


Loop 12
chr19:49,415,165-49,415,426
× chr19:49,443,530-49,443,613
dist: 28 kb
R/A: 0.89× log2FC: -0.17

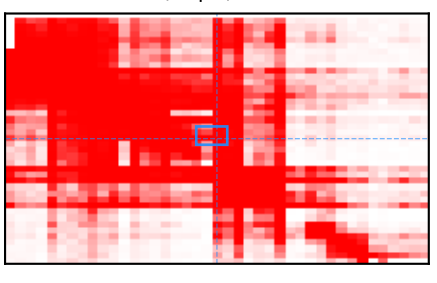
Unphased Σ=8,778
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613



Ref (hap1) Σ=839

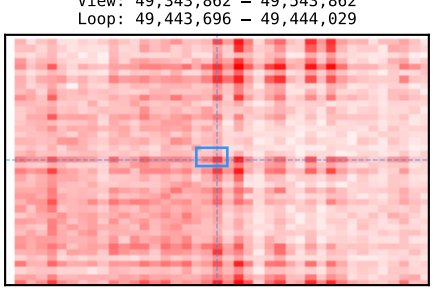


Alt (hap2) Σ=944

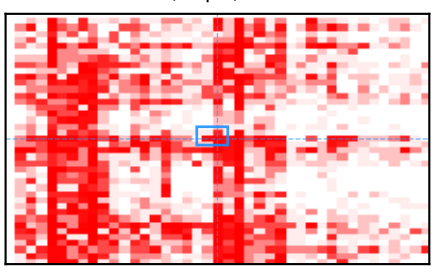


Loop 1
chr19:48,714,125-48,714,448
× chr19:49,443,696-49,444,029
dist: 730 kb
R/A: 1.12× log2FC: +0.17

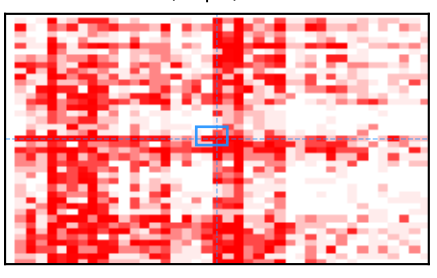
Unphased Σ=190
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=27

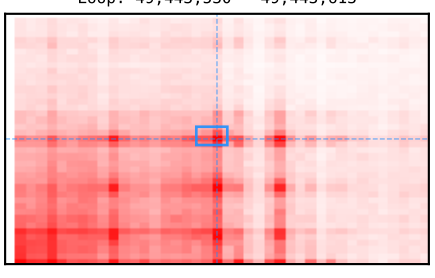


Alt (hap2) Σ=24

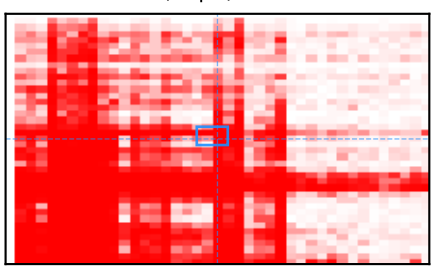


Loop 2
chr19:49,226,146-49,226,293
× chr19:49,443,530-49,443,613
dist: 217 kb
R/A: 1.00× log2FC: +0.00

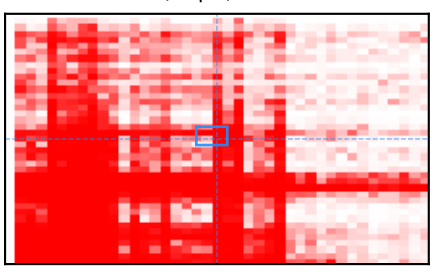
Unphased Σ=3,760
View: 49,343,571 – 49,543,571
Loop: 49,443,530 – 49,443,613



Ref (hap1) Σ=511

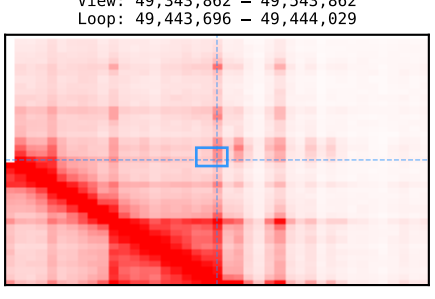


Alt (hap2) Σ=511

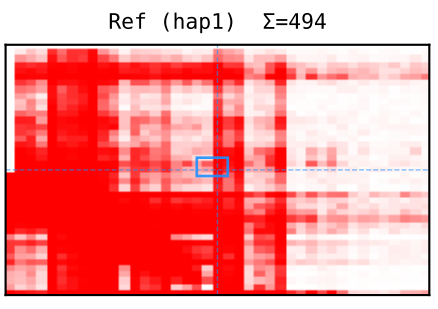


Loop 3
chr19:49,340,318-49,340,573
× chr19:49,443,696-49,444,029
dist: 103 kb
R/A: 0.90× log2FC: -0.15

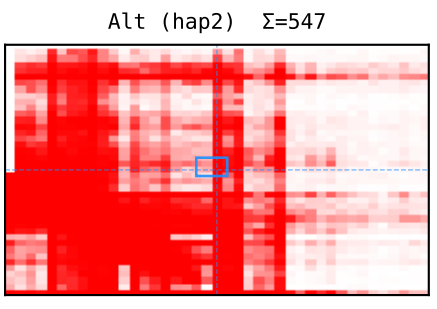
Unphased Σ=3,852
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=494

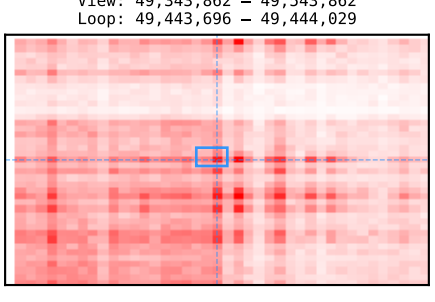


Alt (hap2) Σ=547

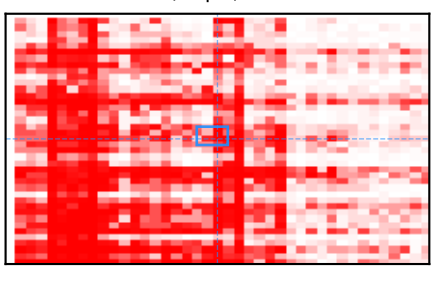


Loop 4
chr19:49,085,099-49,085,404
× chr19:49,443,696-49,444,029
dist: 359 kb
R/A: 1.17× log2FC: +0.23

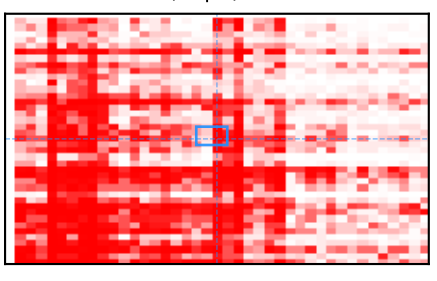
Unphased Σ=628
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=88

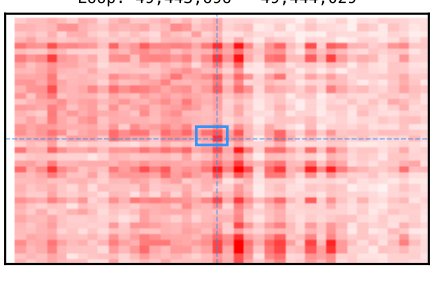


Alt (hap2) Σ=75

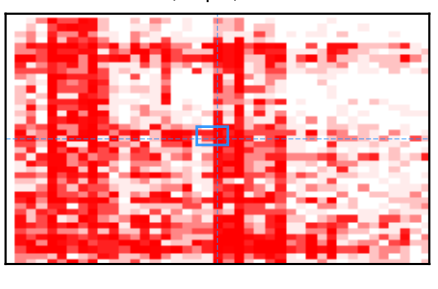


Loop 5
chr19:48,788,224-48,788,407
× chr19:49,443,696-49,444,029
dist: 656 kb
R/A: 1.20× log2FC: +0.26

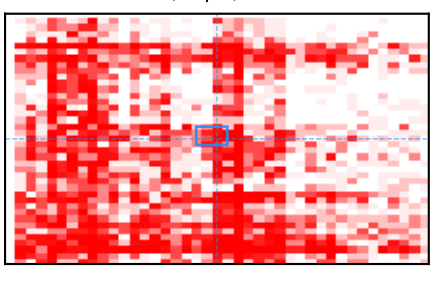
Unphased Σ=238
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=42

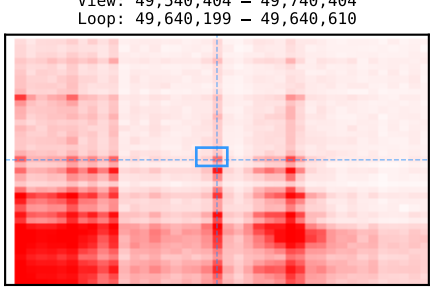


Alt (hap2) Σ=35

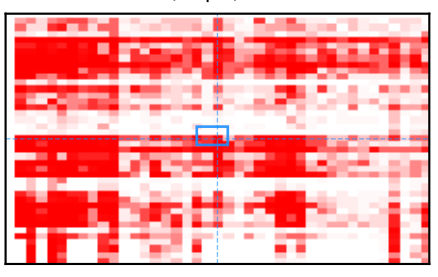


Loop 6
chr19:49,443,696-49,444,029
× chr19:49,640,199-49,640,610
dist: 197 kb
R/A: 1.24× log2FC: +0.31

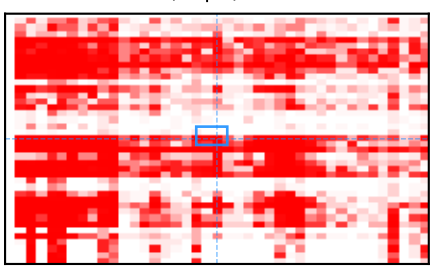
Unphased Σ=723
View: 49,540,404 – 49,740,404
Loop: 49,640,199 – 49,640,610



Ref (hap1) Σ=68

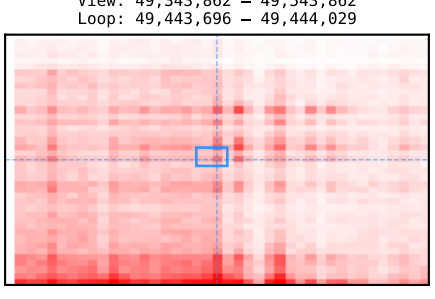


Alt (hap2) Σ=55

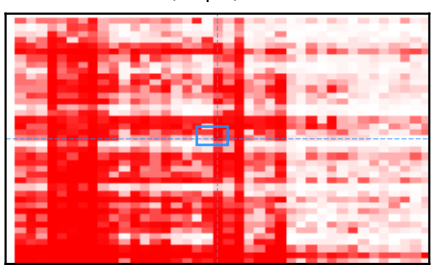


Loop 7
chr19:49,128,084-49,128,275
× chr19:49,443,696-49,444,029
dist: 316 kb
R/A: 0.76× log2FC: -0.39

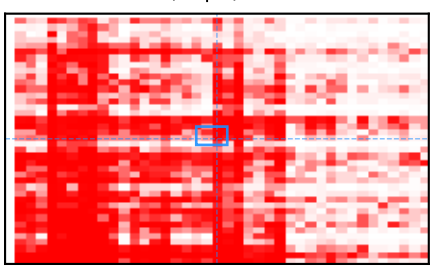
Unphased Σ=590
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=71

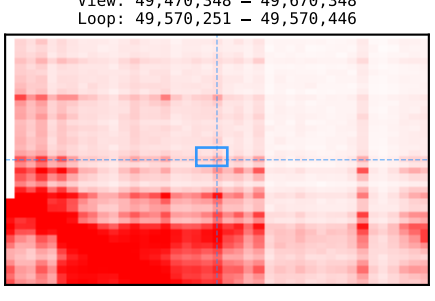


Alt (hap2) Σ=93

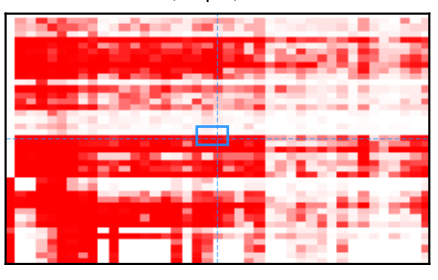


Loop 8
chr19:49,443,696-49,444,029
× chr19:49,570,251-49,570,446
dist: 126 kb
R/A: 1.04× log2FC: +0.06

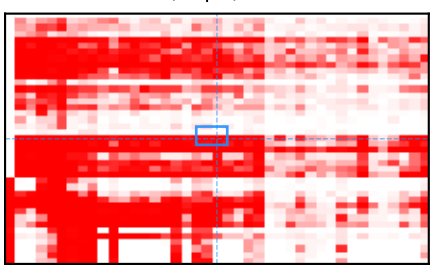
Unphased Σ=937
View: 49,470,348 – 49,670,348
Loop: 49,570,251 – 49,570,446



Ref (hap1) Σ=74

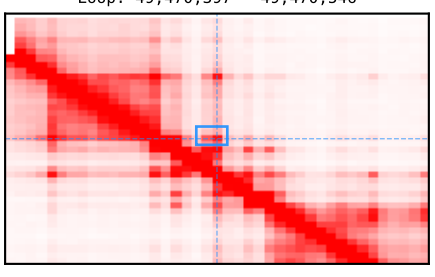


Alt (hap2) Σ=71

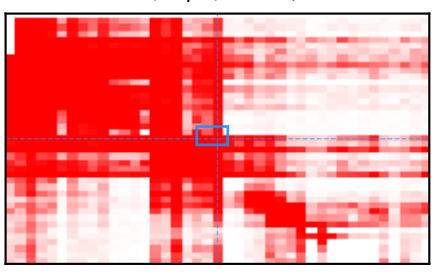


Loop 9
chr19:49,443,696-49,444,029
× chr19:49,470,397-49,470,546
dist: 27 kb
R/A: 0.78× log2FC: -0.36

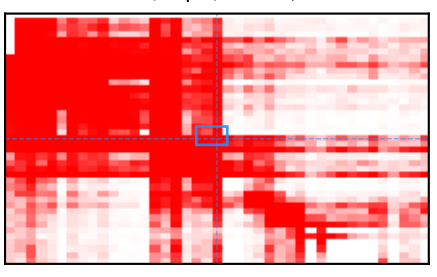
Unphased Σ=9,523
View: 49,370,471 – 49,570,471
Loop: 49,470,397 – 49,470,546



Ref (hap1) Σ=1,250

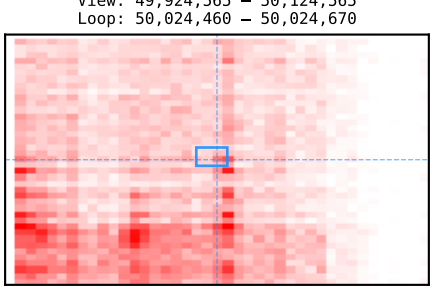


Alt (hap2) Σ=1,605

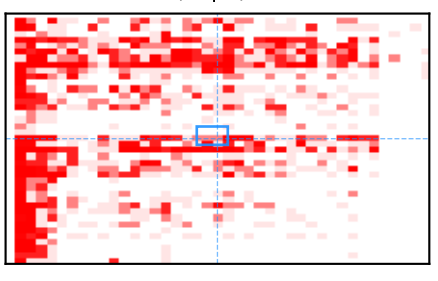


Loop 10
chr19:49,443,696-49,444,029
× chr19:50,024,460-50,024,670
dist: 581 kb
R/A: 0.36× log2FC: -1.46

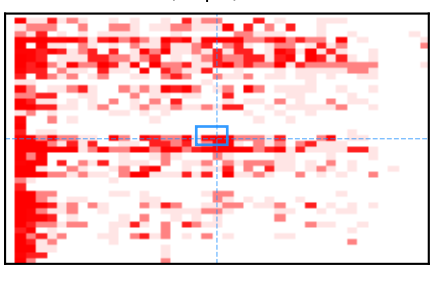
Unphased Σ=112
View: 49,924,565 – 50,124,565
Loop: 50,024,460 – 50,024,670



Ref (hap1) Σ=4

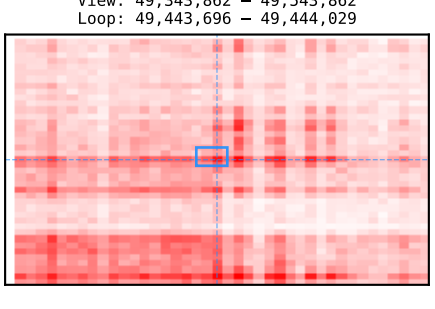


Alt (hap2) Σ=11

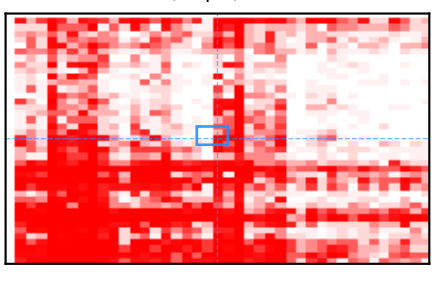


Loop 11
chr19:48,993,273-48,994,006
× chr19:49,443,696-49,444,029
dist: 450 kb
R/A: 1.03× log2FC: +0.04

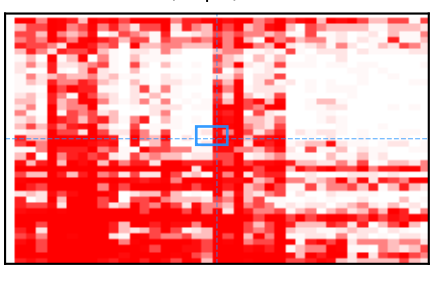
Unphased Σ=437
View: 49,343,862 – 49,543,862
Loop: 49,443,696 – 49,444,029



Ref (hap1) Σ=36

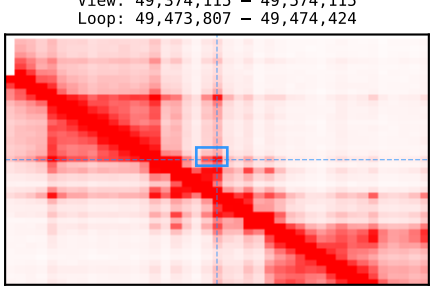


Alt (hap2) Σ=35

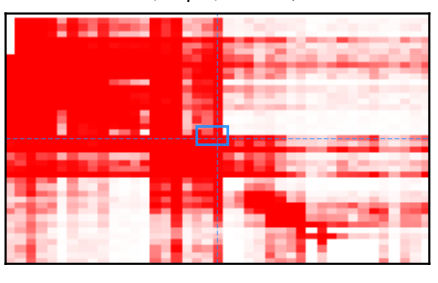


Loop 12
chr19:49,443,696-49,444,029
× chr19:49,473,807-49,474,424
dist: 30 kb
R/A: 0.78× log2FC: -0.36

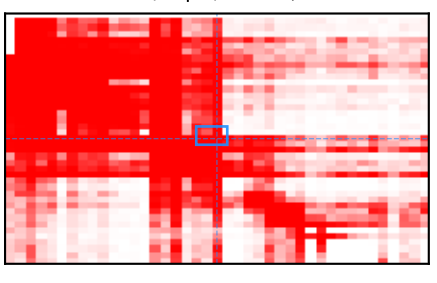
Unphased Σ=9,523
View: 49,374,115 – 49,574,115
Loop: 49,473,807 – 49,474,424



Ref (hap1) Σ=1,250



Alt (hap2) Σ=1,605

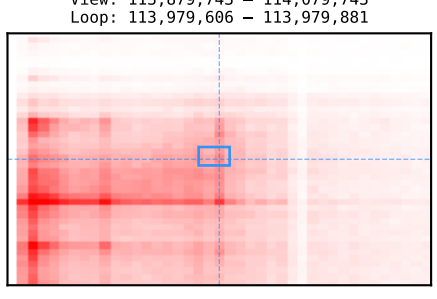


Anchor: chr2:113,979,627–113,979,899 | AlphaGenome pred: 0.0565 | empiric ChIP-seq: -2.68

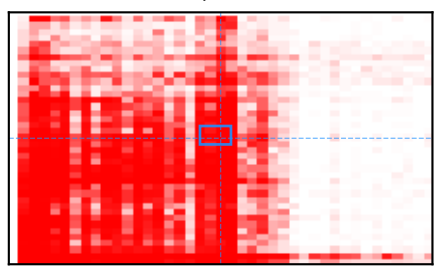
Unphased Σ=1,747

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

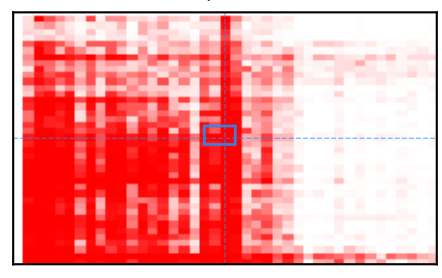
Loop 1
chr2:113,720,220-113,720,430
× chr2:113,979,606-113,979,881
dist: 259 kb
R/A: 0.85× log2FC: -0.23



Ref (hap1) Σ=223



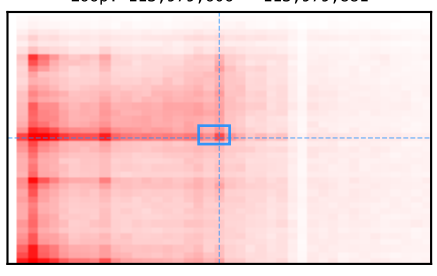
Alt (hap2) Σ=262



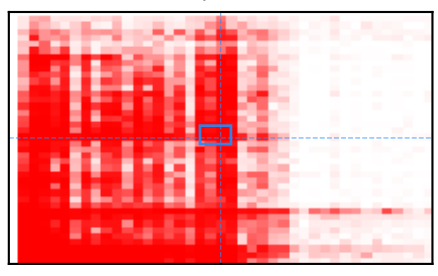
Unphased Σ=2,800

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

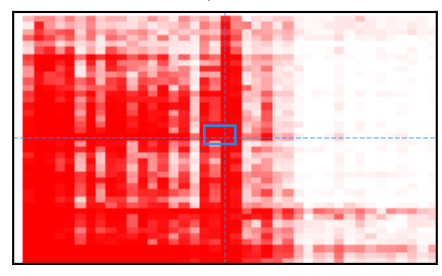
Loop 2
chr2:113,756,604-113,756,951
× chr2:113,979,606-113,979,881
dist: 223 kb
R/A: 0.64× log2FC: -0.63



Ref (hap1) Σ=348



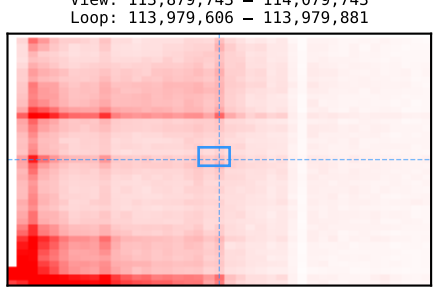
Alt (hap2) Σ=540



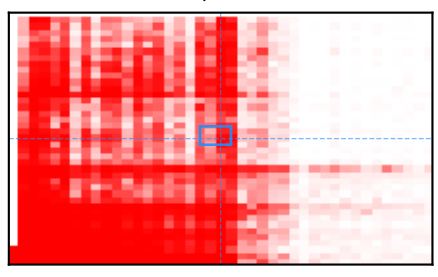
Unphased Σ=1,308

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

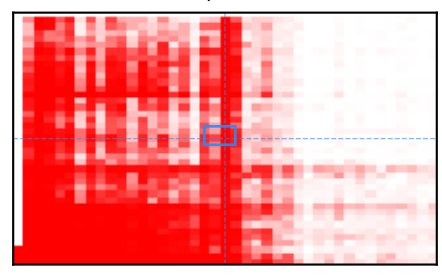
Loop 3
chr2:113,792,081-113,792,202
× chr2:113,979,606-113,979,881
dist: 188 kb
R/A: 0.63× log2FC: -0.67



Ref (hap1) Σ=137



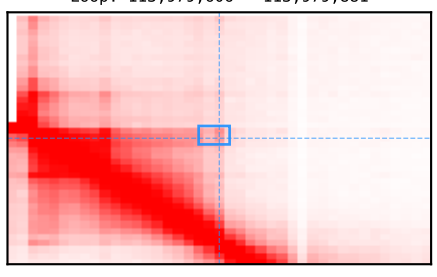
Alt (hap2) Σ=218



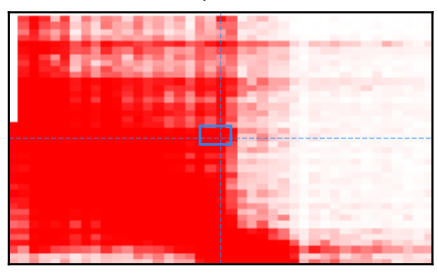
Unphased Σ=3,355

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

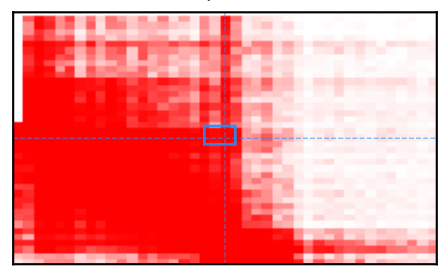
Loop 4
chr2:113,889,891-113,890,140
× chr2:113,979,606-113,979,881
dist: 90 kb
R/A: 0.81× log2FC: -0.30



Ref (hap1) Σ=644



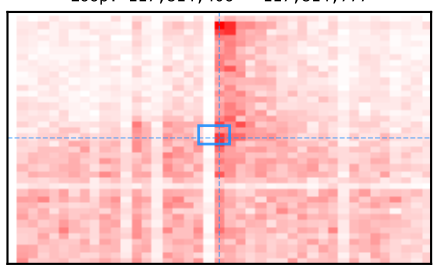
Alt (hap2) Σ=795



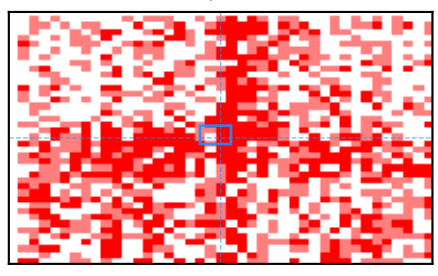
Unphased Σ=76

View: 117,714,591 – 117,914,591
Loop: 117,814,406 – 117,814,777

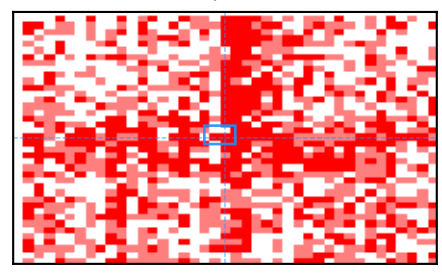
Loop 5
chr2:113,979,606-113,979,881
× chr2:117,814,406-117,814,777
dist: 3835 kb
R/A: 0.65× log2FC: -0.63



Ref (hap1) Σ=20



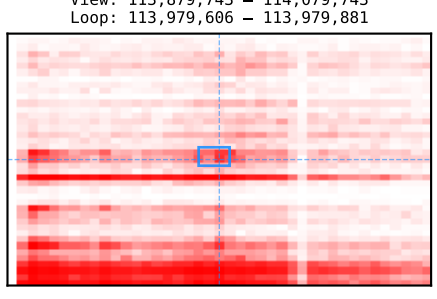
Alt (hap2) Σ=31



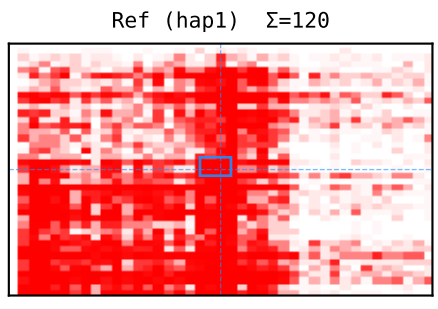
Unphased Σ=290

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

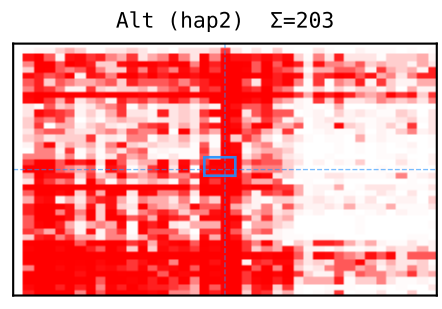
Loop 6
chr2:113,587,305-113,587,528
× chr2:113,979,606-113,979,881
dist: 392 kb
R/A: 0.59× log2FC: -0.76



Ref (hap1) Σ=120



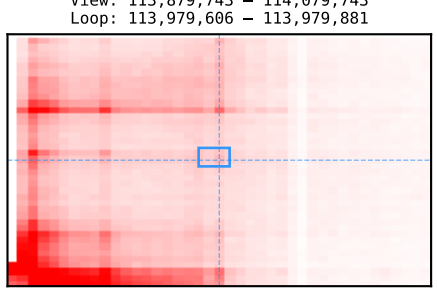
Alt (hap2) Σ=203



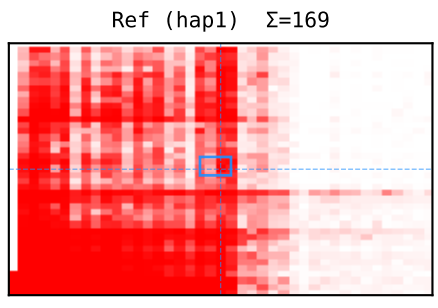
Unphased Σ=1,622

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

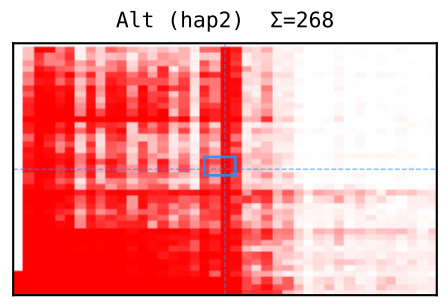
Loop 7
chr2:113,799,485-113,799,564
× chr2:113,979,606-113,979,881
dist: 180 kb
R/A: 0.63× log2FC: -0.67



Ref (hap1) Σ=169



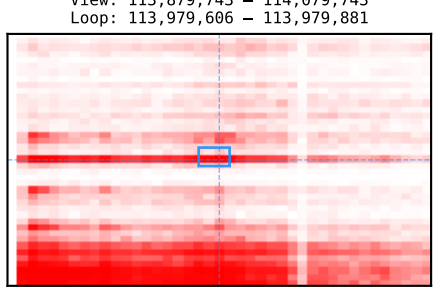
Alt (hap2) Σ=268



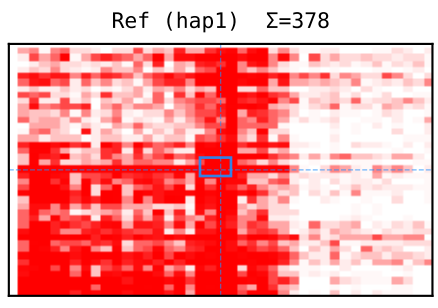
Unphased Σ=809

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

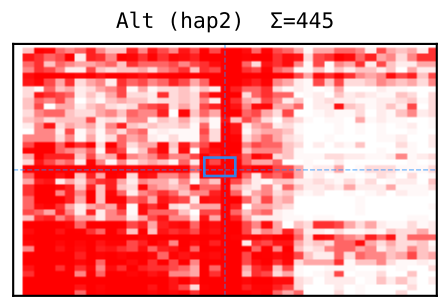
Loop 8
chr2:113,602,074-113,602,229
× chr2:113,979,606-113,979,881
dist: 378 kb
R/A: 0.85× log2FC: -0.24



Ref (hap1) Σ=378



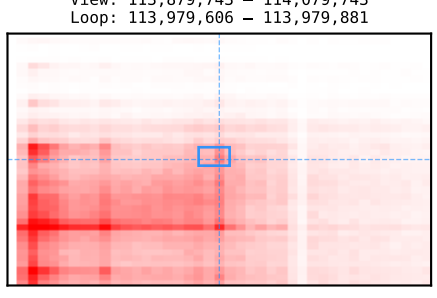
Alt (hap2) Σ=445



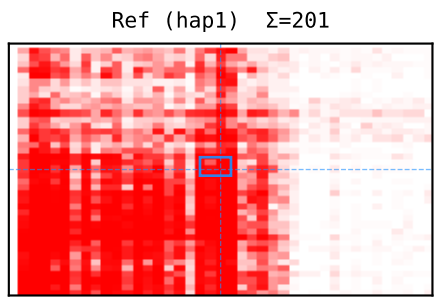
Unphased Σ=1,556

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

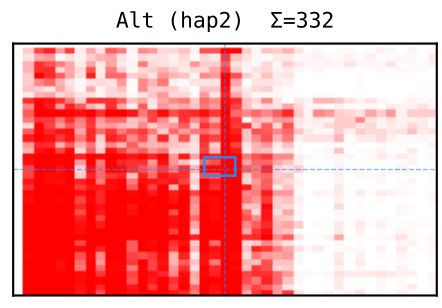
Loop 9
chr2:113,700,259-113,700,476
× chr2:113,979,606-113,979,881
dist: 279 kb
R/A: 0.61× log2FC: -0.72



Ref (hap1) Σ=201



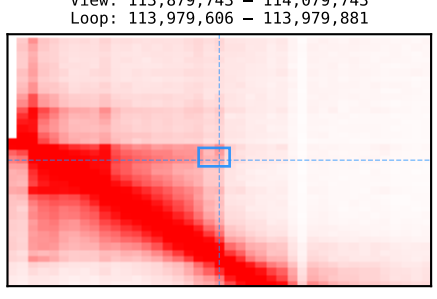
Alt (hap2) Σ=332



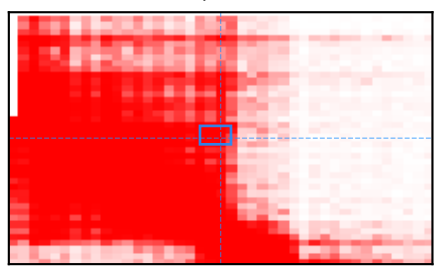
Unphased Σ=3,035

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

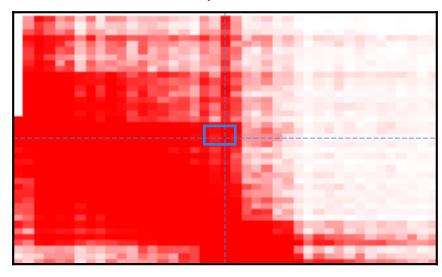
Loop 10
chr2:113,895,160-113,895,370
× chr2:113,979,606-113,979,881
dist: 84 kb
R/A: 0.88× log2FC: -0.18



Ref (hap1) Σ=565



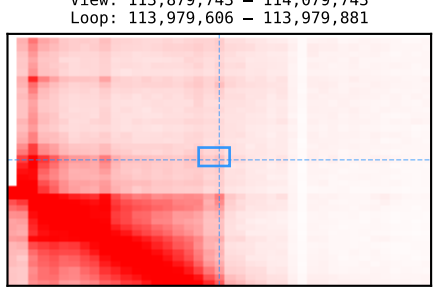
Alt (hap2) Σ=640



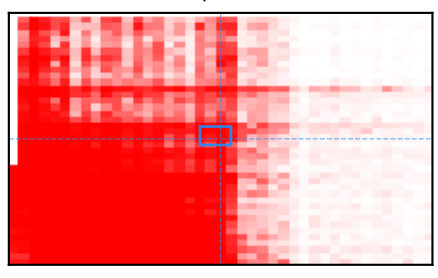
Unphased Σ=1,581

View: 113,879,743 – 114,079,743
Loop: 113,979,606 – 113,979,881

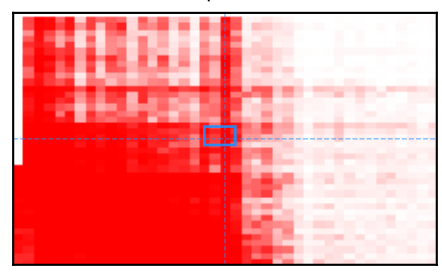
Loop 11
chr2:113,859,766-113,859,985
× chr2:113,979,606-113,979,881
dist: 120 kb
R/A: 0.99× log2FC: -0.01



Ref (hap1) Σ=311



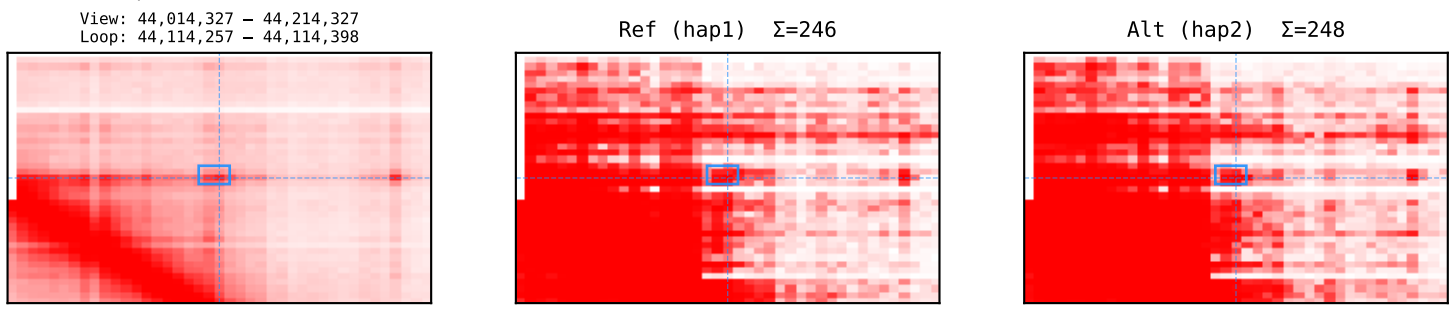
Alt (hap2) Σ=313



Unphased Σ=5,057
View: 44,014,327 – 44,214,327
Loop: 44,114,257 – 44,114,398

Ref (hap1) Σ=246

Alt (hap2) Σ=248

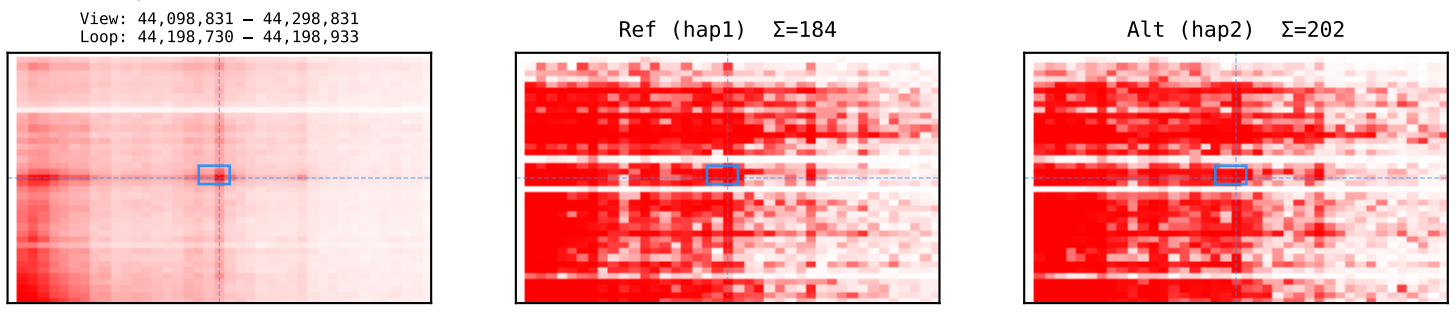


Loop 1
chr3:43,996,410-43,996,503
× chr3:44,114,257-44,114,398
dist: 118 kb
R/A: 0.99× log2FC: -0.01

Unphased Σ=3,587
View: 44,098,831 – 44,298,831
Loop: 44,198,730 – 44,198,933

Ref (hap1) Σ=184

Alt (hap2) Σ=202

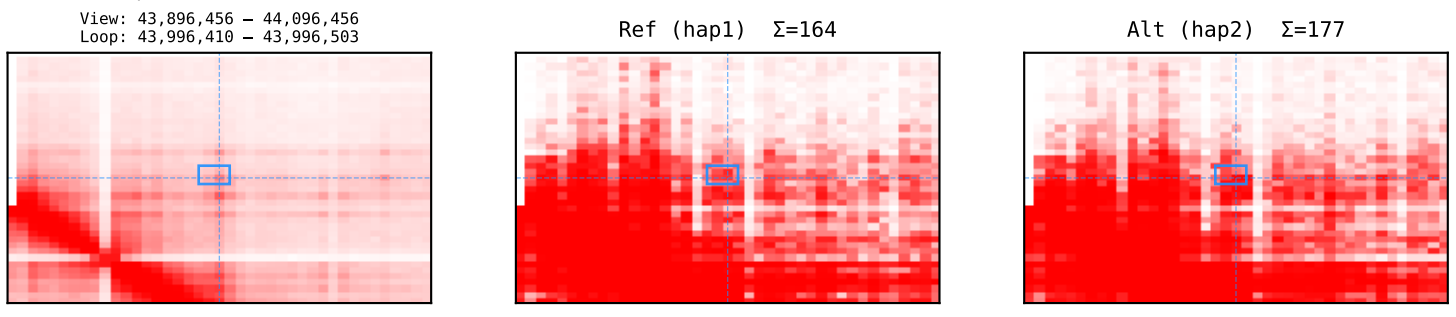


Loop 2
chr3:43,996,410-43,996,503
× chr3:44,198,730-44,198,933
dist: 202 kb
R/A: 0.91× log2FC: -0.13

Unphased Σ=2,384
View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

Ref (hap1) Σ=164

Alt (hap2) Σ=177

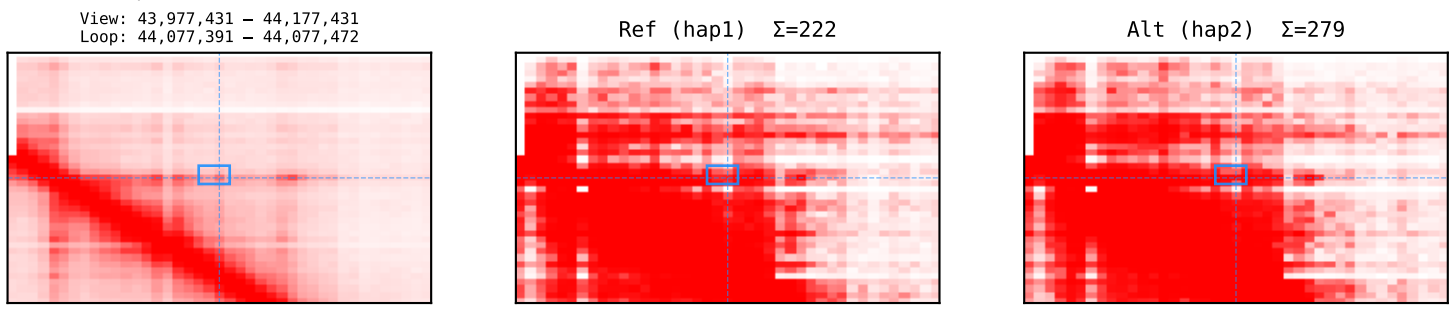


Loop 3
chr3:43,879,330-43,879,405
× chr3:43,996,410-43,996,503
dist: 117 kb
R/A: 0.93× log2FC: -0.11

Unphased Σ=3,834
View: 43,977,431 – 44,177,431
Loop: 44,077,391 – 44,077,472

Ref (hap1) Σ=222

Alt (hap2) Σ=279

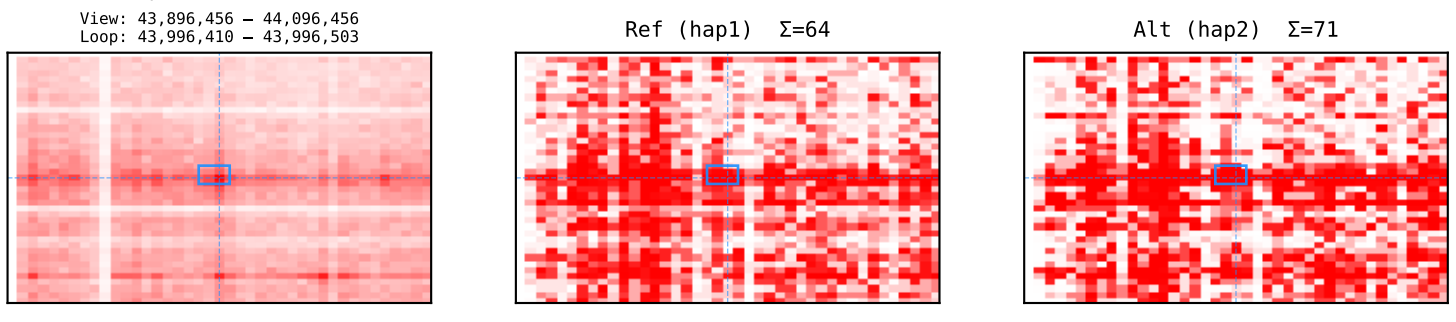


Loop 4
chr3:43,996,410-43,996,503
× chr3:44,077,391-44,077,472
dist: 81 kb
R/A: 0.80× log2FC: -0.33

Unphased Σ=743
View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

Ref (hap1) Σ=64

Alt (hap2) Σ=71

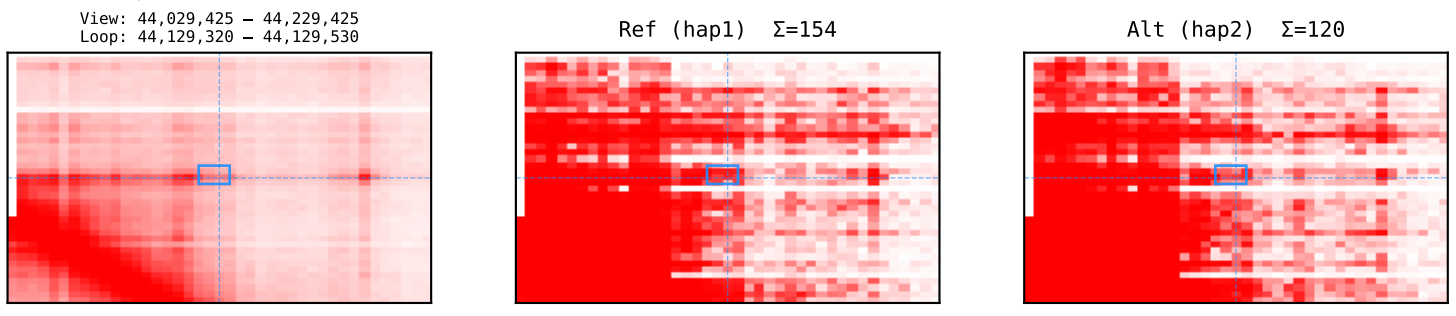


Loop 5
chr3:43,544,314-43,544,447
× chr3:43,996,410-43,996,503
dist: 452 kb
R/A: 0.90× log2FC: -0.15

Unphased Σ=2,502
View: 44,029,425 – 44,229,425
Loop: 44,129,320 – 44,129,530

Ref (hap1) Σ=154

Alt (hap2) Σ=120

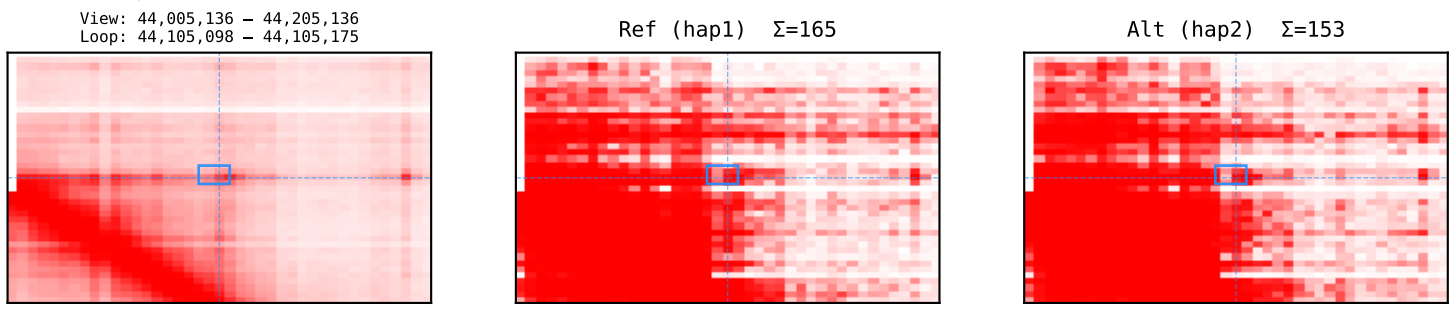


Loop 6
chr3:43,996,410-43,996,503
× chr3:44,129,320-44,129,530
dist: 133 kb
R/A: 1.28× log2FC: +0.36

Unphased Σ=3,742
View: 44,005,136 – 44,205,136
Loop: 44,105,098 – 44,105,175

Ref (hap1) Σ=165

Alt (hap2) Σ=153

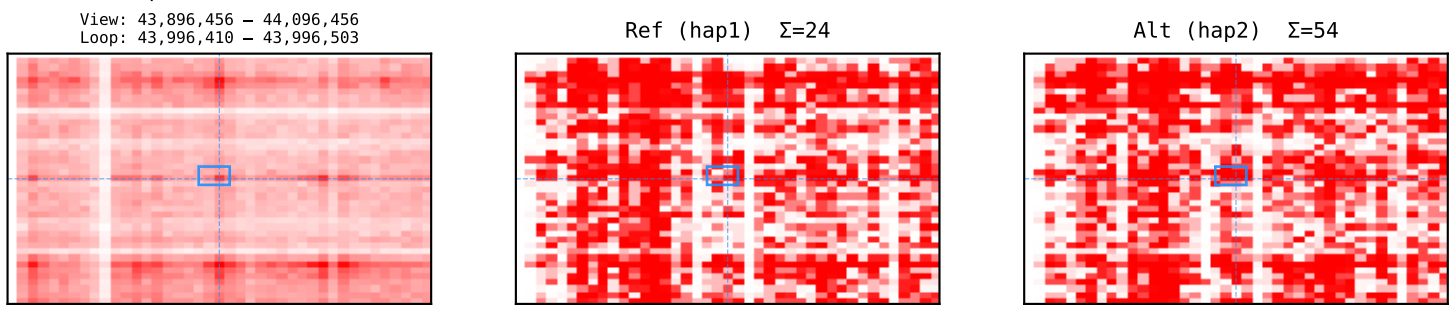


Loop 7
chr3:43,996,410-43,996,503
× chr3:44,105,098-44,105,175
dist: 109 kb
R/A: 1.08× log2FC: +0.11

Unphased Σ=504
View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

Ref (hap1) Σ=24

Alt (hap2) Σ=54

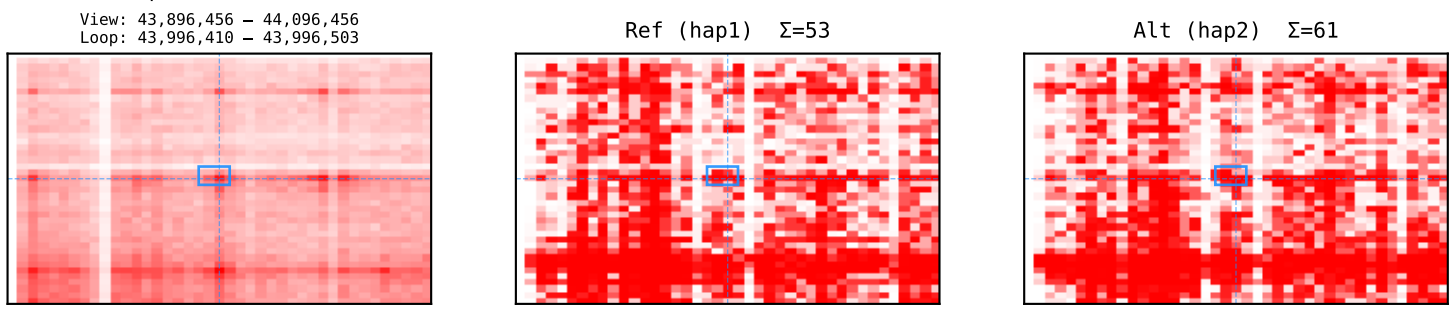


Loop 8
chr3:43,622,047-43,622,306
× chr3:43,996,410-43,996,503
dist: 374 kb
R/A: 0.44× log2FC: -1.17

Unphased Σ=781
View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

Ref (hap1) Σ=53

Alt (hap2) Σ=61

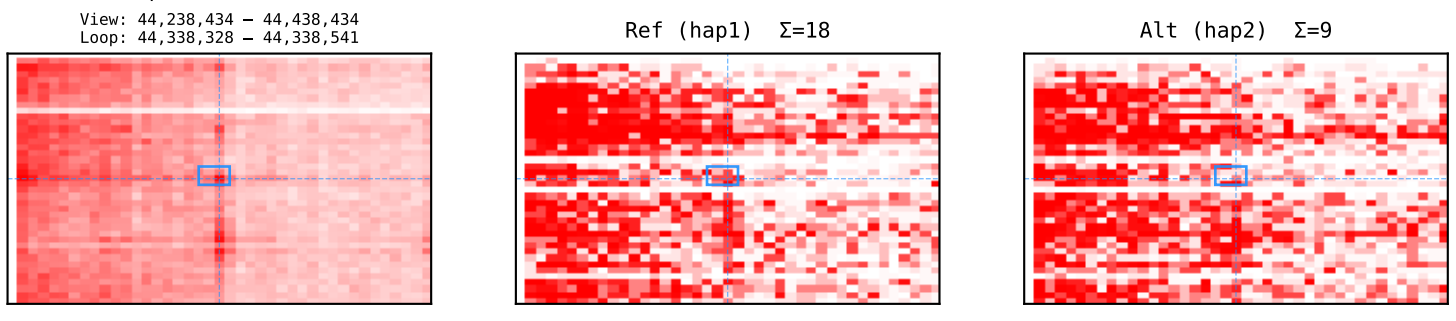


Loop 9
chr3:43,690,788-43,691,067
× chr3:43,996,410-43,996,503
dist: 306 kb
R/A: 0.87× log2FC: -0.20

Unphased Σ=437
View: 44,238,434 – 44,438,434
Loop: 44,338,328 – 44,338,541

Ref (hap1) Σ=18

Alt (hap2) Σ=9

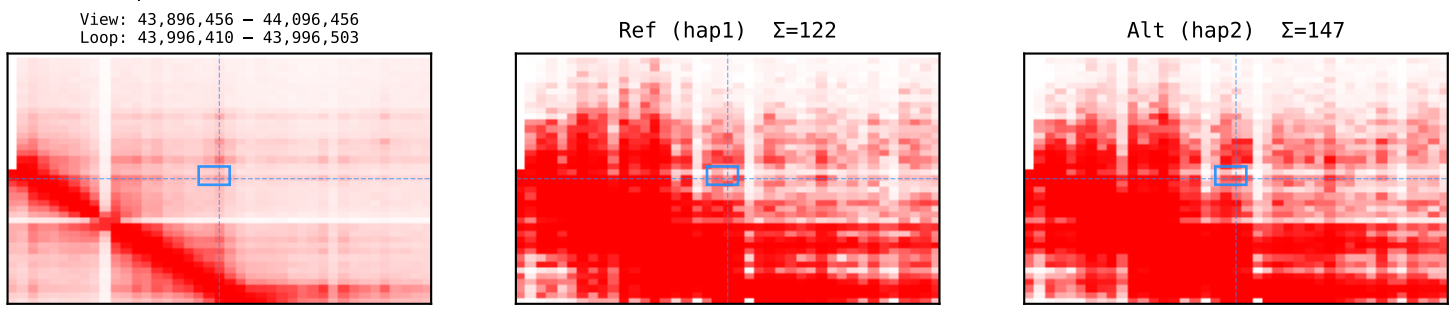


Loop 10
chr3:43,996,410-43,996,503
× chr3:44,338,328-44,338,541
dist: 342 kb
R/A: 2.00× log2FC: +1.00

Unphased Σ=2,500
View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

Ref (hap1) Σ=122

Alt (hap2) Σ=147

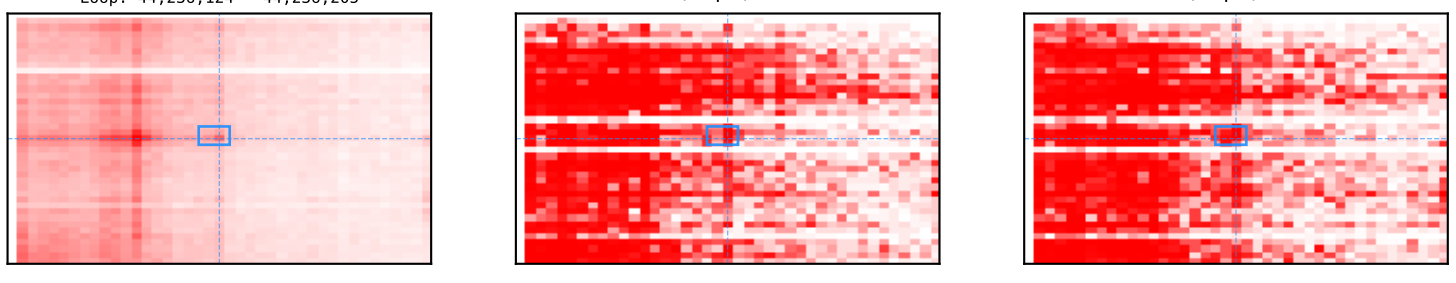


Loop 11
chr3:43,907,586-43,907,703
× chr3:43,996,410-43,996,503
dist: 89 kb
R/A: 0.83× log2FC: -0.27

Unphased Σ=1,260
View: 44,136,164 – 44,336,164
Loop: 44,236,124 – 44,236,205

Ref (hap1) Σ=65

Alt (hap2) Σ=84



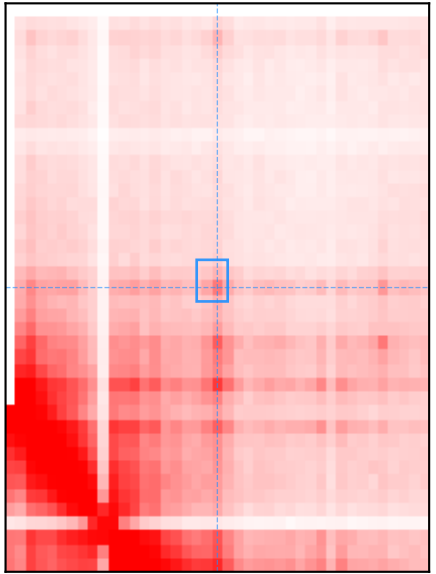
Loop 12
chr3:43,996,410-43,996,503
× chr3:44,236,124-44,236,205
dist: 240 kb
R/A: 0.77× log2FC: -0.37

Anchor: chr3:43,996,358–43,996,504 | AlphaGenome pred: -0.0872 | empiric ChIP-seq: -2 | page 2 of 2

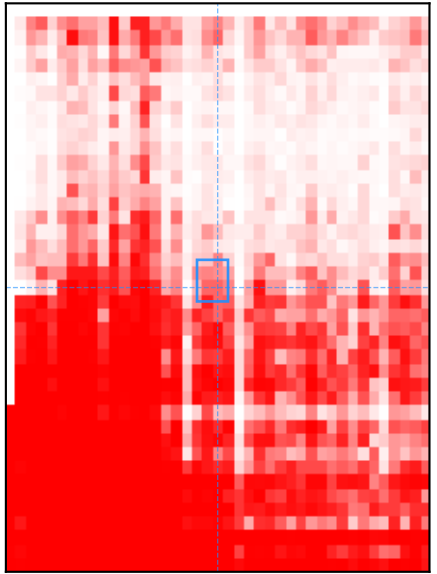
Unphased Σ=1,895

View: 43,896,456 – 44,096,456
Loop: 43,996,410 – 43,996,503

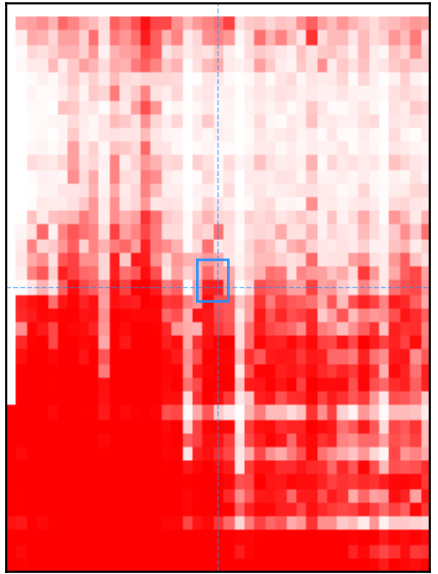
Loop 1
chr3:43,855,816-43,855,927
× chr3:43,996,410-43,996,503
dist: 141 kb
R/A: 0.82× log2FC: -0.29



Ref (hap1) Σ=97

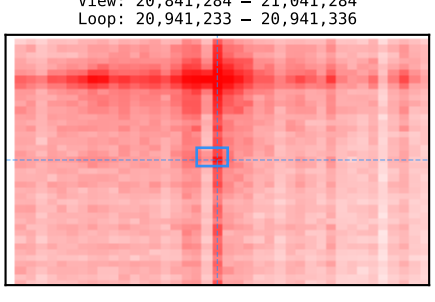


Alt (hap2) Σ=119

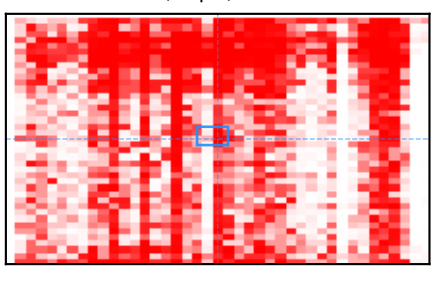


Unphased Σ=1,503

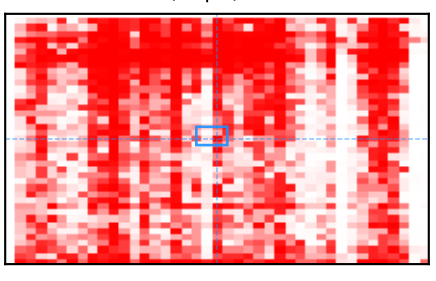
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=100



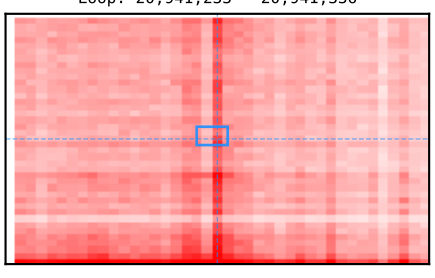
Alt (hap2) Σ=55



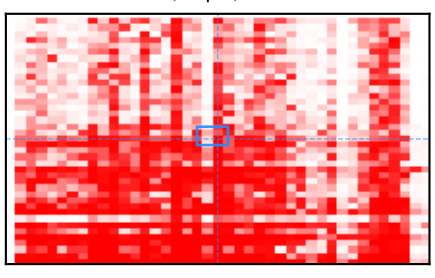
Loop 1
chr9:20,423,297-20,423,618
× chr9:20,941,233-20,941,336
dist: 518 kb
R/A: 1.82× log2FC: +0.86

Unphased Σ=873

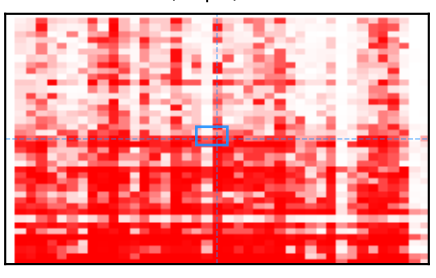
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=98



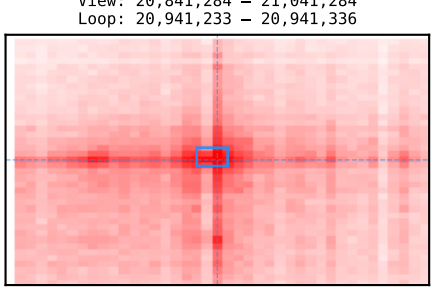
Alt (hap2) Σ=79



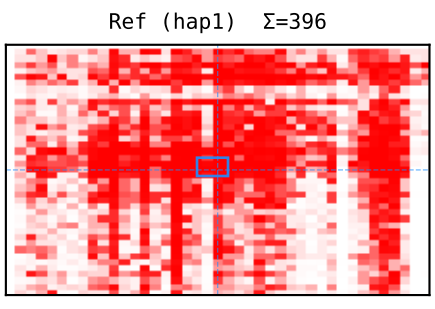
Loop 2
chr9:20,524,860-20,525,070
× chr9:20,941,233-20,941,336
dist: 416 kb
R/A: 1.24× log2FC: +0.31

Unphased Σ=6,083

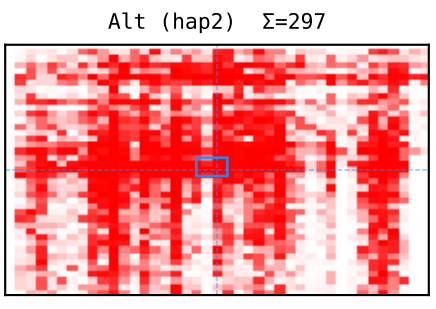
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=396



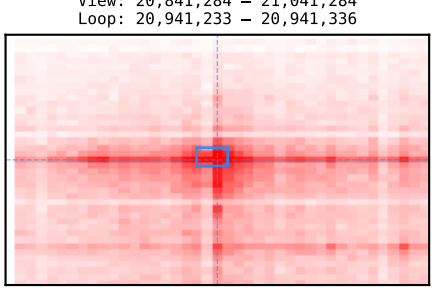
Alt (hap2) Σ=297



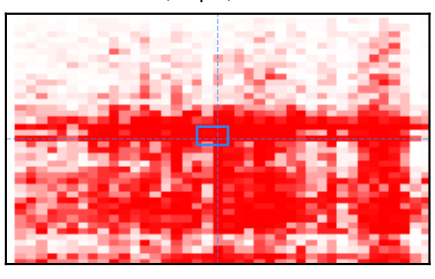
Loop 3
chr9:20,356,466-20,356,543
× chr9:20,941,233-20,941,336
dist: 585 kb
R/A: 1.33× log2FC: +0.42

Unphased Σ=4,450

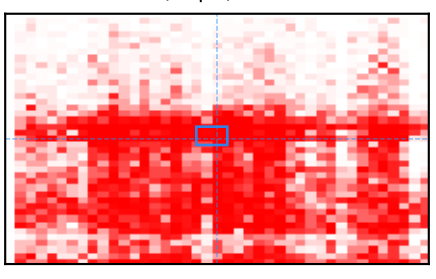
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=400



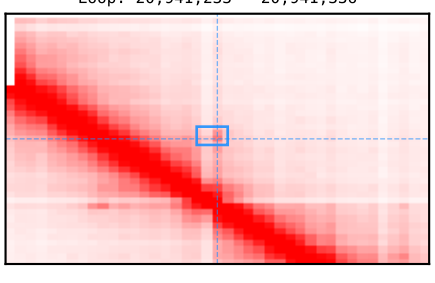
Alt (hap2) Σ=308



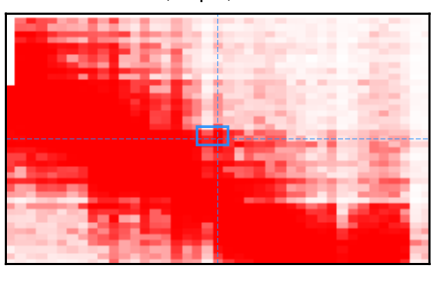
Loop 4
chr9:19,926,041-19,926,260
× chr9:20,941,233-20,941,336
dist: 1015 kb
R/A: 1.30× log2FC: +0.38

Unphased Σ=3,879

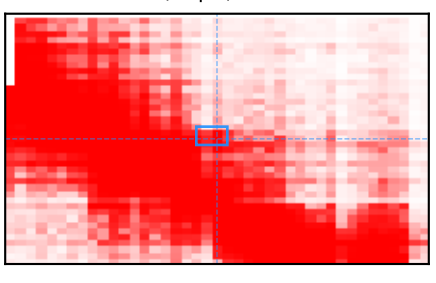
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=475



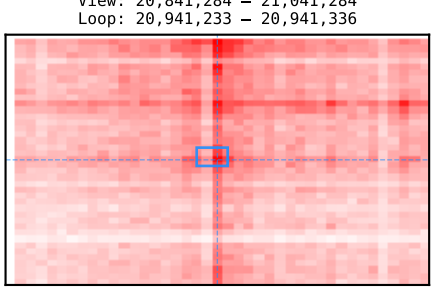
Alt (hap2) Σ=528



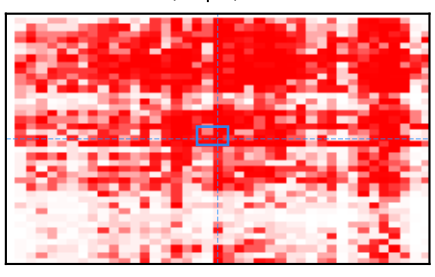
Loop 5
chr9:20,885,144-20,885,229
× chr9:20,941,233-20,941,336
dist: 56 kb
R/A: 0.90× log2FC: -0.15

Unphased Σ=726

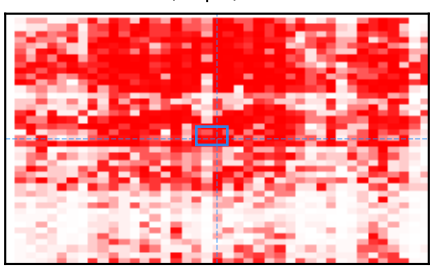
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=67



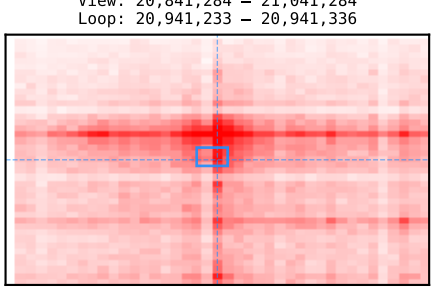
Alt (hap2) Σ=56



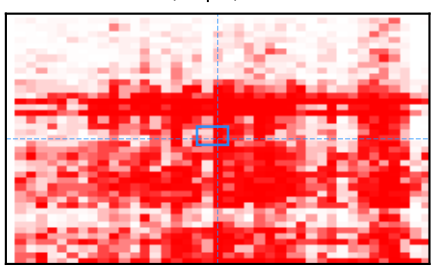
Loop 6
chr9:20,043,584-20,043,745
× chr9:20,941,233-20,941,336
dist: 898 kb
R/A: 1.20× log2FC: +0.26

Unphased Σ=1,212

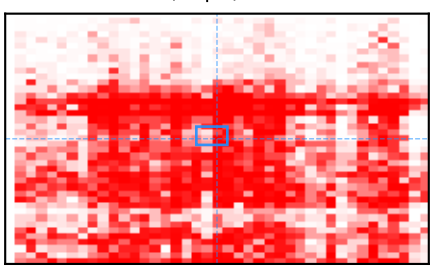
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=61



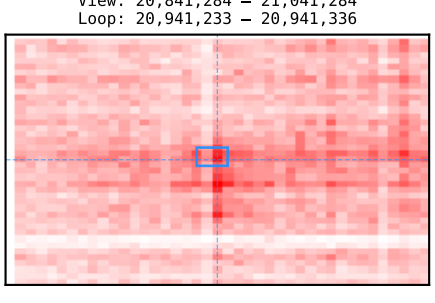
Alt (hap2) Σ=37



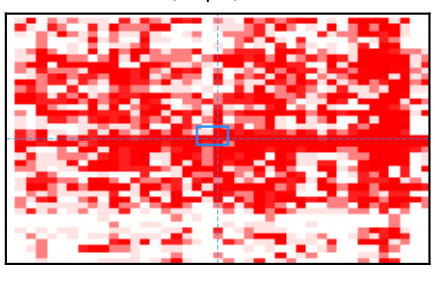
Loop 7
chr9:19,946,701-19,946,720
× chr9:20,941,233-20,941,336
dist: 995 kb
R/A: 1.65× log2FC: +0.72

Unphased Σ=352

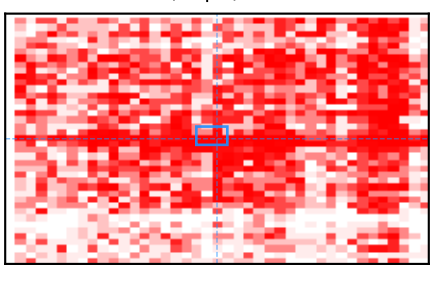
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=36



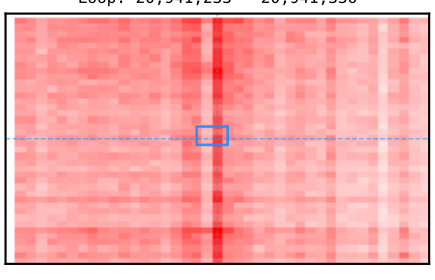
Alt (hap2) Σ=50



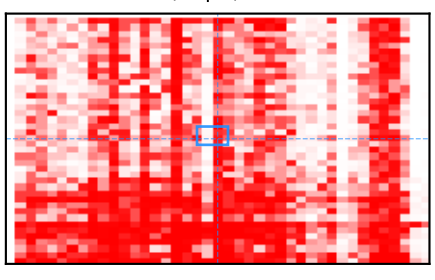
Loop 8
chr9:19,471,947-19,472,028
× chr9:20,941,233-20,941,336
dist: 1469 kb
R/A: 0.72× log2FC: -0.47

Unphased Σ=903

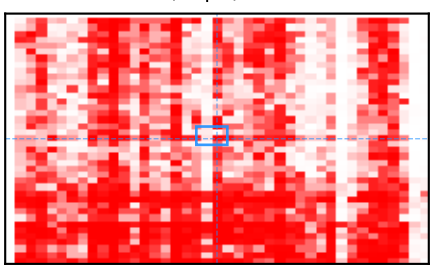
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=51



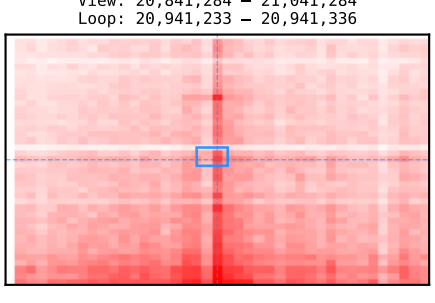
Alt (hap2) Σ=47



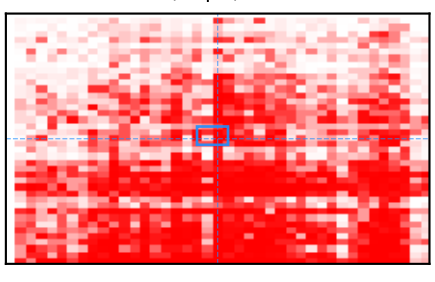
Loop 9
chr9:20,479,910-20,480,071
× chr9:20,941,233-20,941,336
dist: 461 kb
R/A: 1.09× log2FC: +0.12

Unphased Σ=718

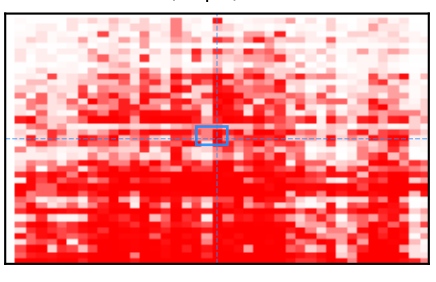
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=73



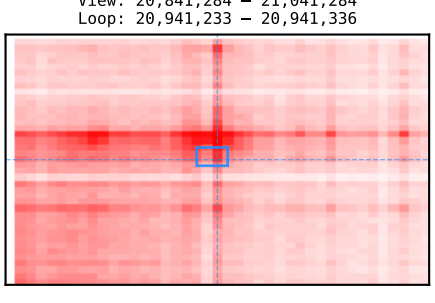
Alt (hap2) Σ=65



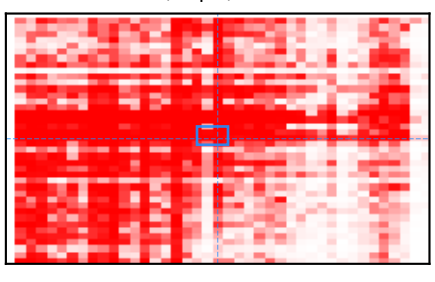
Loop 10
chr9:20,242,931-20,243,068
× chr9:20,941,233-20,941,336
dist: 698 kb
R/A: 1.12× log2FC: +0.17

Unphased Σ=2,601

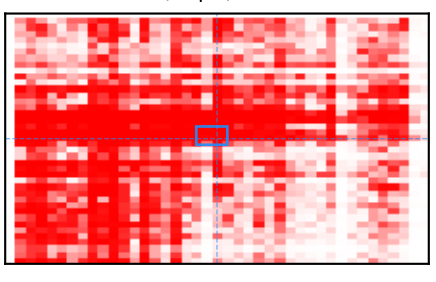
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=360



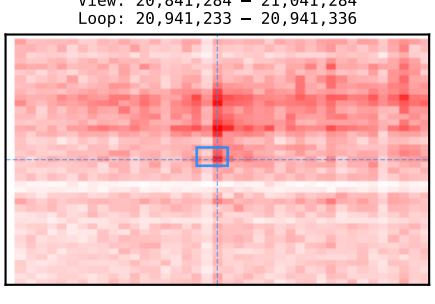
Alt (hap2) Σ=304



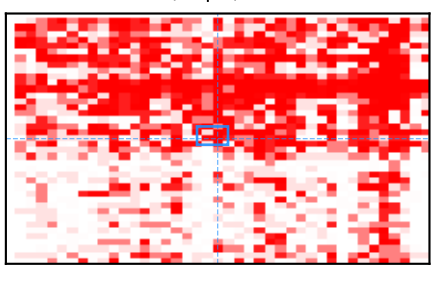
Loop 11
chr9:20,640,418-20,640,707
× chr9:20,941,233-20,941,336
dist: 301 kb
R/A: 1.18× log2FC: +0.24

Unphased Σ=251

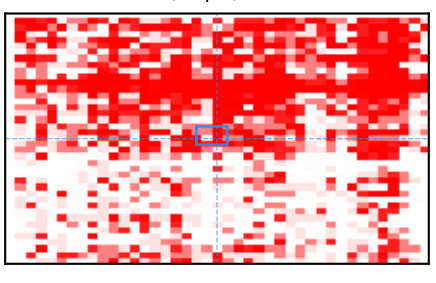
View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336



Ref (hap1) Σ=20



Alt (hap2) Σ=18

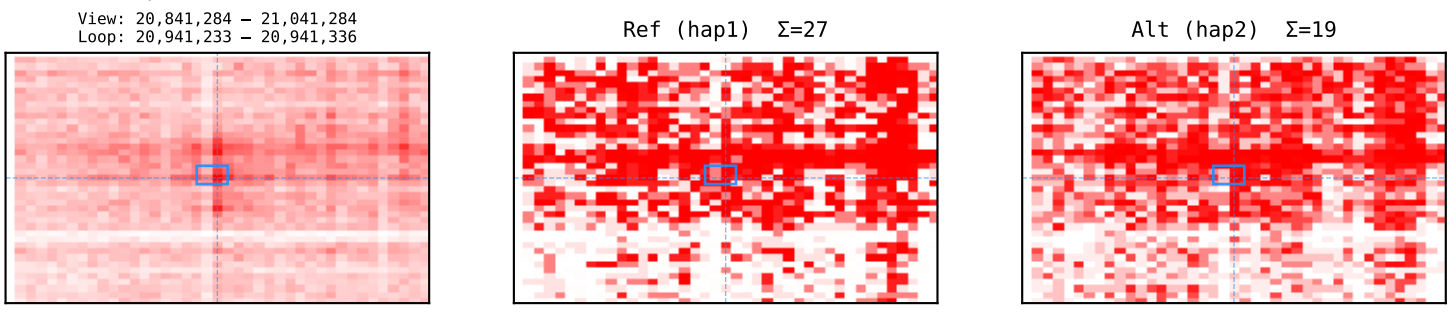


Loop 12
chr9:19,517,685-19,517,792
× chr9:20,941,233-20,941,336
dist: 1424 kb
R/A: 1.11× log2FC: +0.15

Unphased Σ=386

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

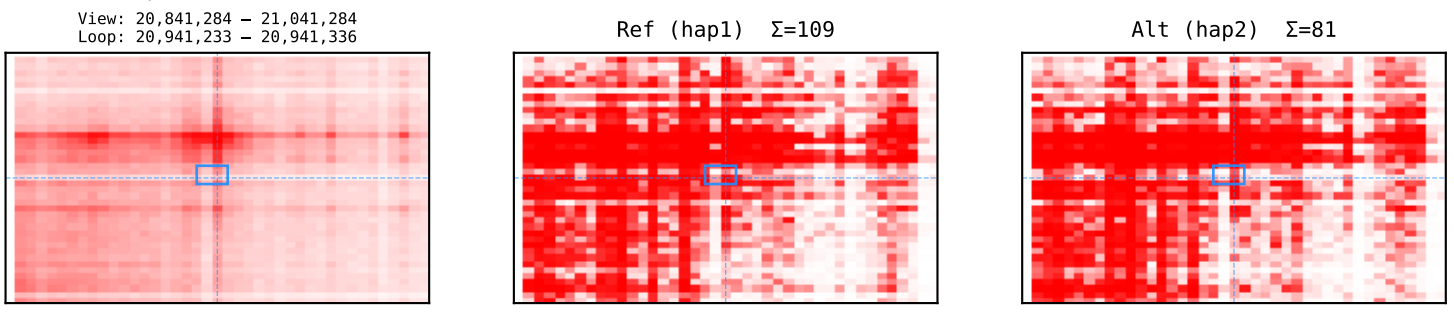
Loop 1
chr9:19,493,236-19,493,333
× chr9:20,941,233-20,941,336
dist: 1448 kb
R/A: 1.42× log2FC: +0.51



Unphased Σ=942

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

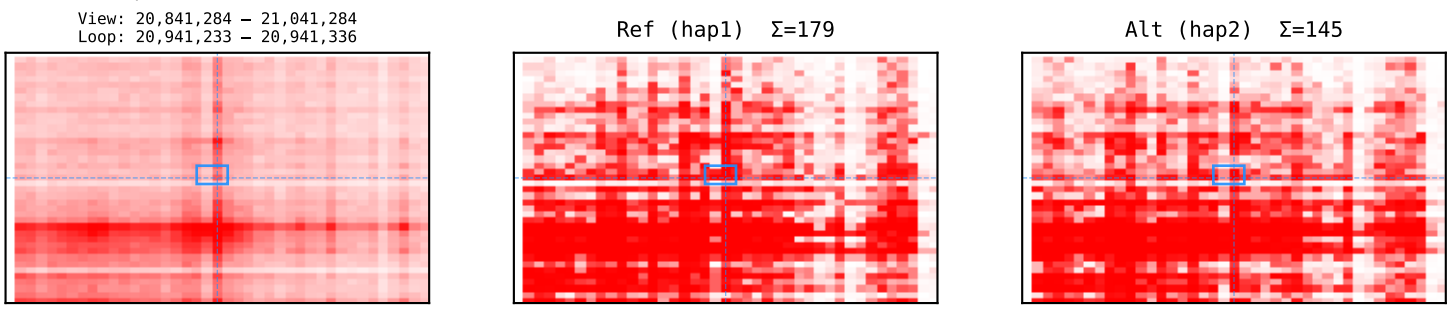
Loop 2
chr9:20,658,712-20,658,927
× chr9:20,941,233-20,941,336
dist: 282 kb
R/A: 1.35× log2FC: +0.43



Unphased Σ=1,048

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

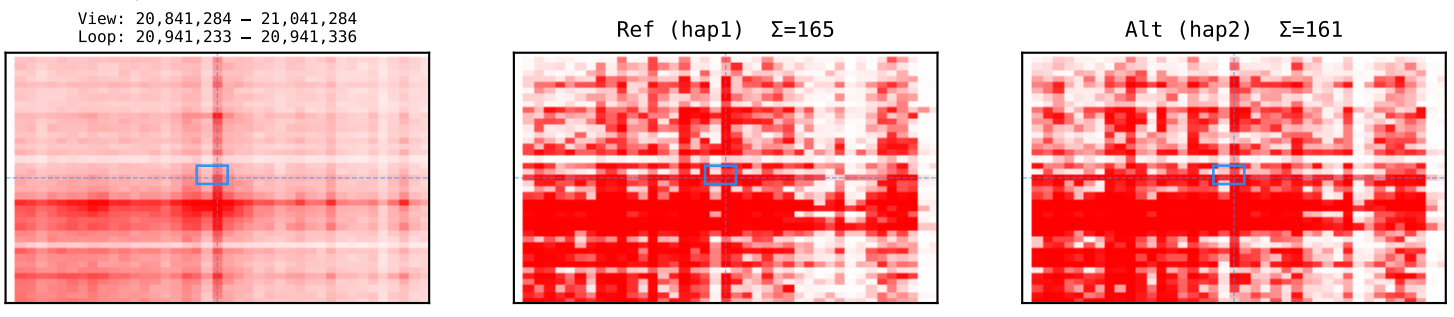
Loop 3
chr9:20,582,278-20,582,297
× chr9:20,941,233-20,941,336
dist: 359 kb
R/A: 1.23× log2FC: +0.30



Unphased Σ=1,281

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

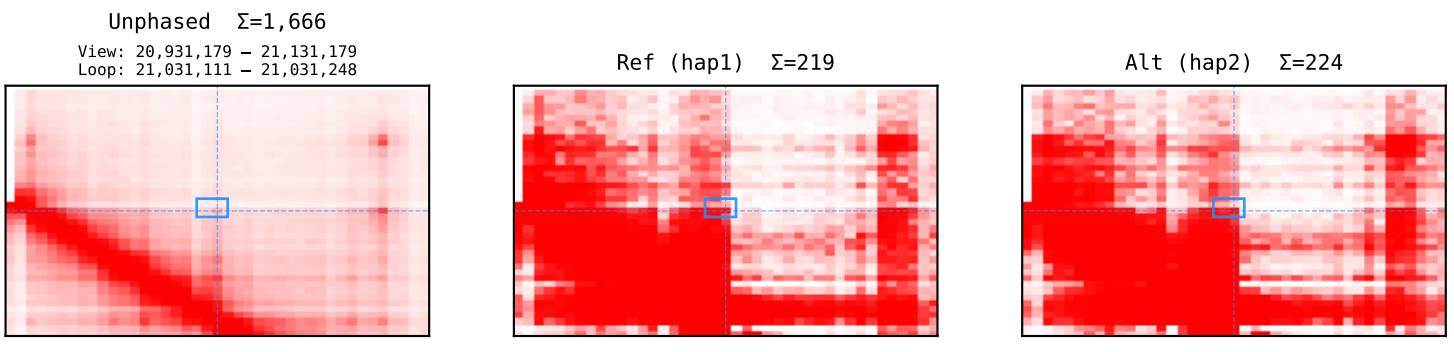
Loop 4
chr9:20,604,691-20,604,798
× chr9:20,941,233-20,941,336
dist: 337 kb
R/A: 1.02× log2FC: +0.04



Unphased Σ=1,666

View: 20,931,179 – 21,131,179
Loop: 21,031,111 – 21,031,248

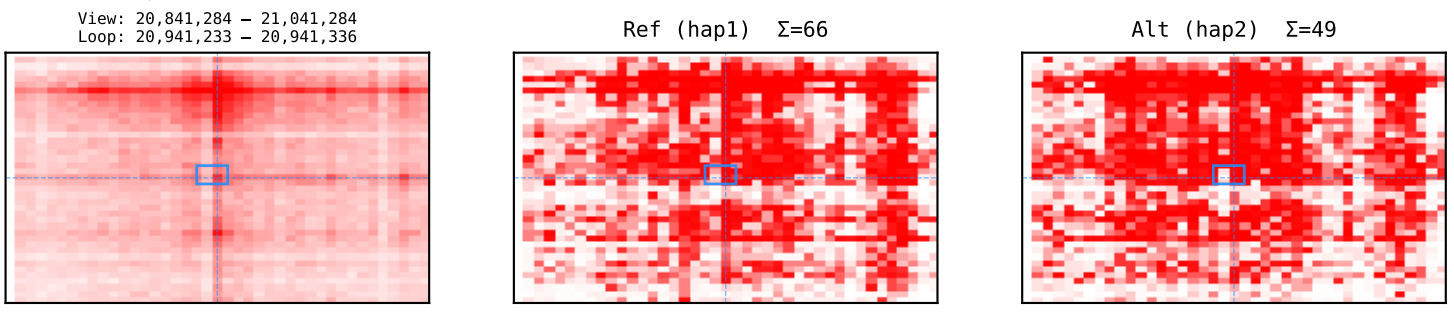
Loop 5
chr9:20,941,233-20,941,336
× chr9:21,031,111-21,031,248
dist: 90 kb
R/A: 0.98× log2FC: -0.03



Unphased Σ=585

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

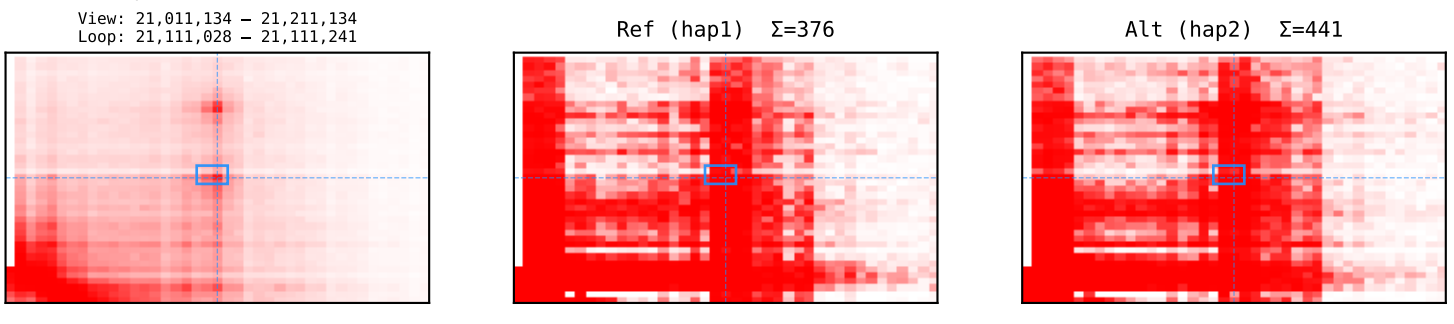
Loop 6
chr9:19,999,548-19,999,741
× chr9:20,941,233-20,941,336
dist: 942 kb
R/A: 1.35× log2FC: +0.43



Unphased Σ=4,255

View: 21,011,134 – 21,211,134
Loop: 21,111,028 – 21,111,241

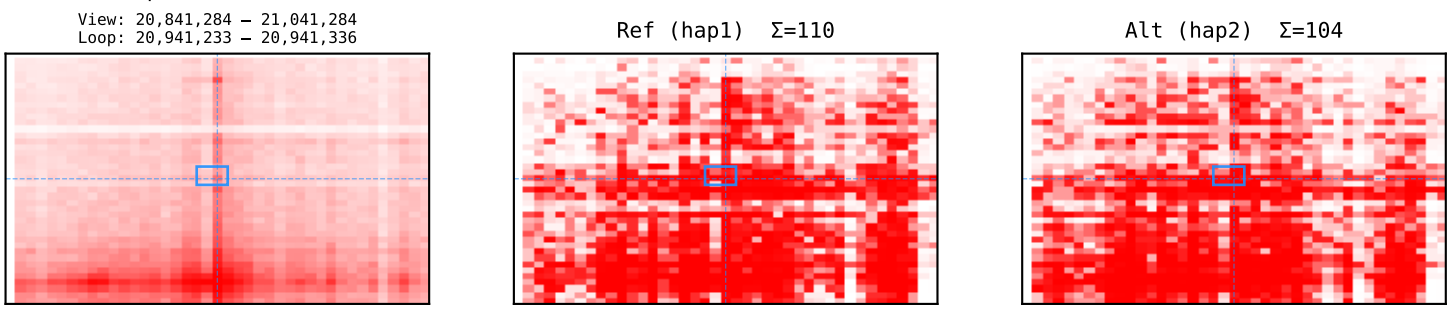
Loop 7
chr9:20,941,233-20,941,336
× chr9:21,111,028-21,111,241
dist: 170 kb
R/A: 0.85× log2FC: -0.23



Unphased Σ=653

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

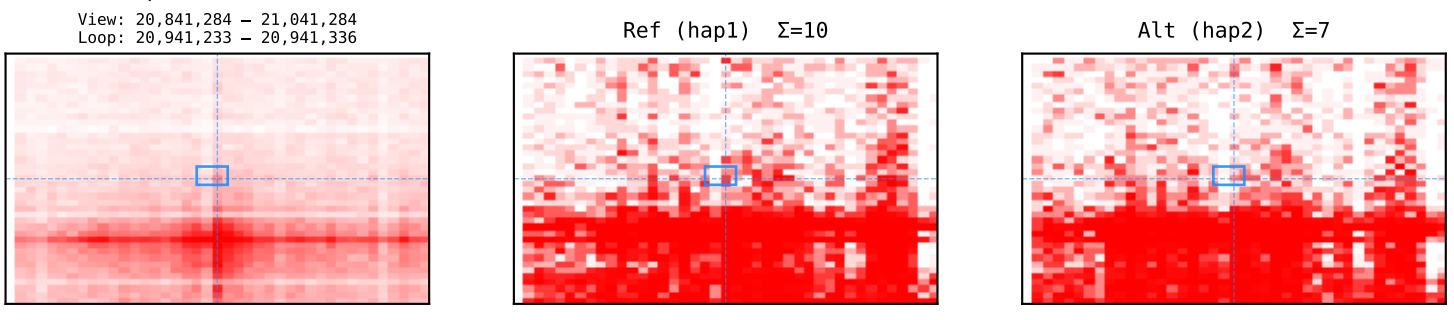
Loop 8
chr9:20,271,210-20,271,420
× chr9:20,941,233-20,941,336
dist: 670 kb
R/A: 1.06× log2FC: +0.08



Unphased Σ=265

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

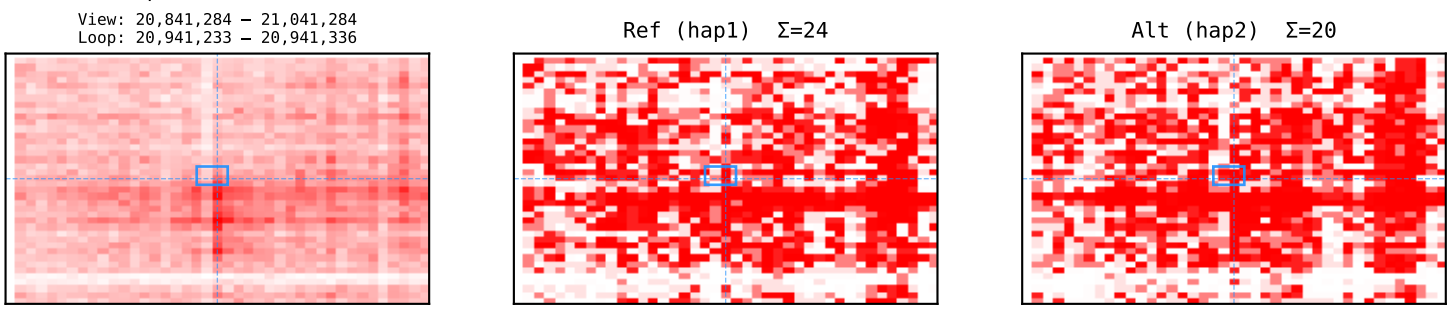
Loop 9
chr9:19,879,660-19,879,870
× chr9:20,941,233-20,941,336
dist: 1062 kb
R/A: 1.43× log2FC: +0.51



Unphased Σ=139

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

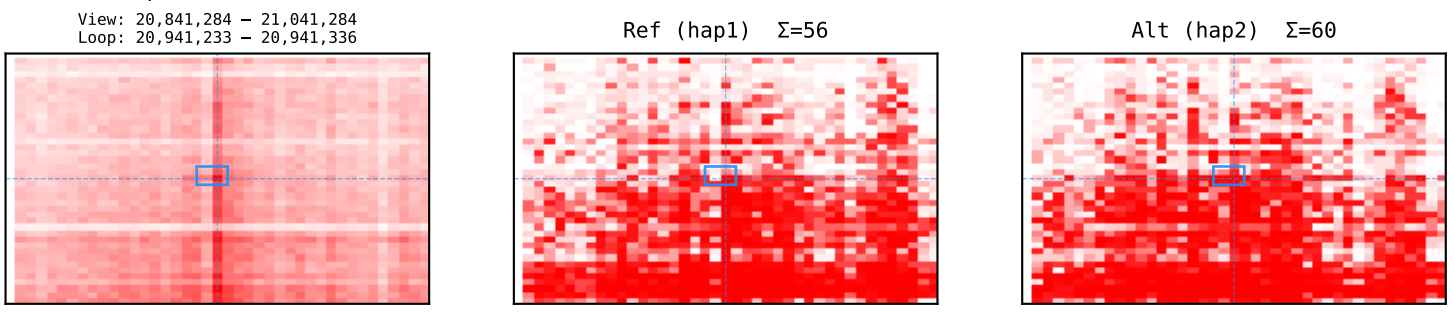
Loop 10
chr9:19,457,503-19,457,674
× chr9:20,941,233-20,941,336
dist: 1484 kb
R/A: 1.20× log2FC: +0.26



Unphased Σ=708

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

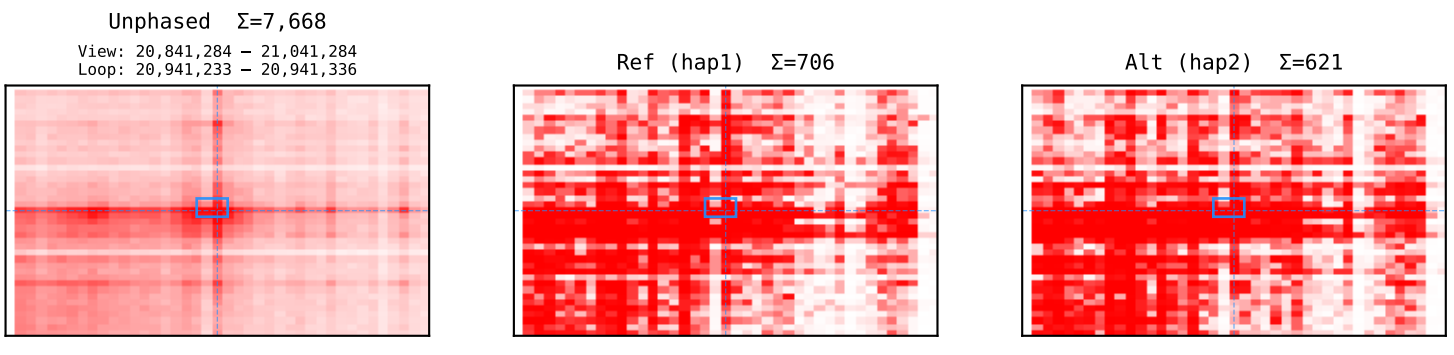
Loop 11
chr9:20,192,590-20,192,609
× chr9:20,941,233-20,941,336
dist: 749 kb
R/A: 0.93× log2FC: -0.10



Unphased Σ=7,668

View: 20,841,284 – 21,041,284
Loop: 20,941,233 – 20,941,336

Loop 12
chr9:20,624,899-20,624,994
× chr9:20,941,233-20,941,336
dist: 316 kb
R/A: 1.14× log2FC: +0.19

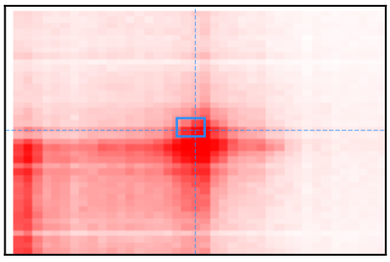


Anchor: chrX:110,145,938–110,146,092 | AlphaGenome pred: 0.0957 | empiric ChIP-seq: -1.25

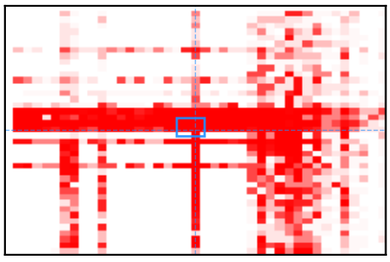
Unphased Σ=4,270

View: 110,308,336 – 110,508,336
Loop: 110,408,259 – 110,408,414

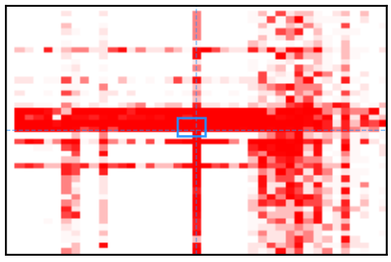
Loop 1
chrX:110,145,922-110,146,103
× chrX:110,408,259-110,408,414
dist: 262 kb
R/A: 1.12× log2FC: +0.17



Ref (hap1) Σ=273



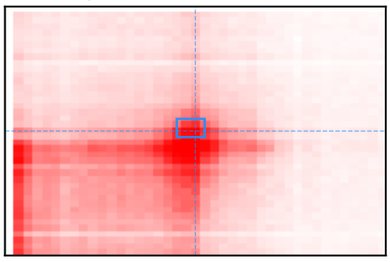
Alt (hap2) Σ=243



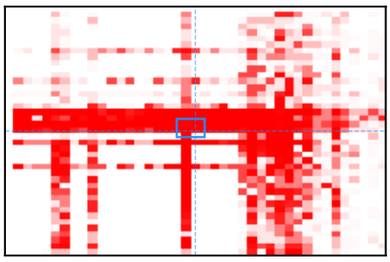
Unphased Σ=5,973

View: 110,313,303 – 110,513,303
Loop: 110,413,200 – 110,413,407

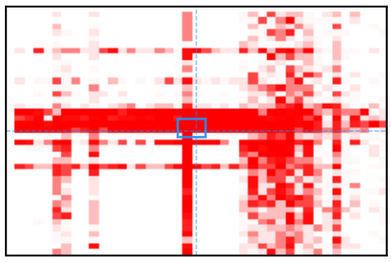
Loop 2
chrX:110,145,922-110,146,103
× chrX:110,413,200-110,413,407
dist: 267 kb
R/A: 1.17× log2FC: +0.23



Ref (hap1) Σ=391



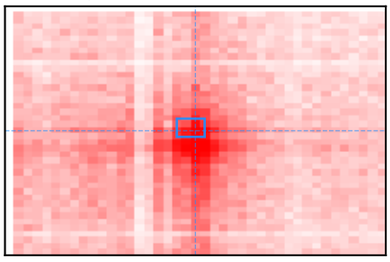
Alt (hap2) Σ=334



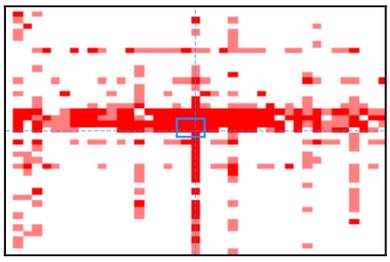
Unphased Σ=911

View: 110,747,484 – 110,947,484
Loop: 110,847,427 – 110,847,542

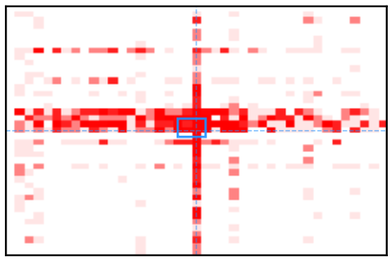
Loop 3
chrX:110,145,922-110,146,103
× chrX:110,847,427-110,847,542
dist: 701 kb
R/A: 0.88× log2FC: -0.19



Ref (hap1) Σ=65



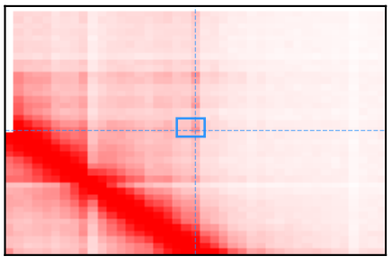
Alt (hap2) Σ=74



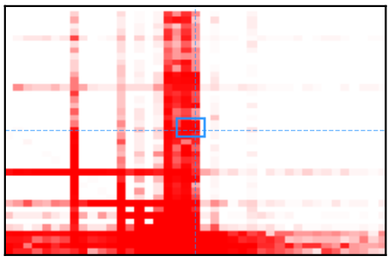
Unphased Σ=2,340

View: 110,046,012 – 110,246,012
Loop: 110,145,922 – 110,146,103

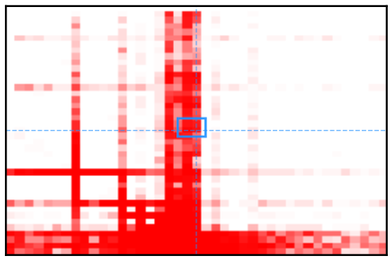
Loop 4
chrX:110,046,190-110,046,400
× chrX:110,145,922-110,146,103
dist: 100 kb
R/A: 1.35× log2FC: +0.43



Ref (hap1) Σ=128



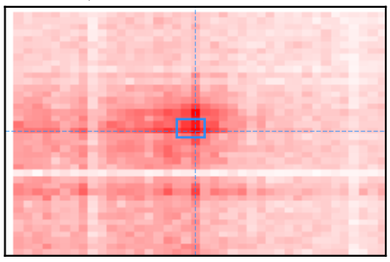
Alt (hap2) Σ=95



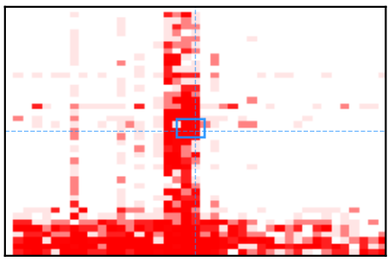
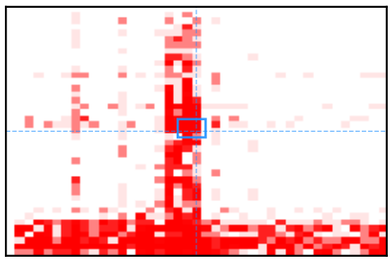
Unphased Σ=877

View: 110,046,012 – 110,246,012
Loop: 110,145,922 – 110,146,103

Loop 5
chrX:109,485,843-109,486,050
× chrX:110,145,922-110,146,103
dist: 660 kb
R/A: 1.52× log2FC: +0.60



Ref (hap1) Σ=44



Alt (hap2) Σ=29